Pollard Rho on NVIDIA Blackwell: ECC2K-130, Carry-less Multiplication and the NIST Curves

Anonymous

Abstract

We measure what Pollard rho costs on the current generation of NVIDIA hardware, using the Certicom challenge ECC2K-130 as the yardstick. ECC2K-130 is the Koblitz curve $y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{131}}$, open since 1997, with an expected cost of $2^{60.9}$ iterations of a Frobenius-class rho. In 2010 the best published GPU kernel ran at 0.064 billion iterations per second on a GTX 295 card. Rebuilt around the carry-less multiply that CUDA 13.3 exposes in PTX (`clmad`), three-input logic and a packed normal-basis representation, the same walk now runs at 14.64 billion iterations per second on one RTX PRO 6000 Blackwell in its audited campaign build and 15.88 in the best later build, about 230 times the 2010 card, so the whole budget is about five GPU-years. We trace that gain through sixteen paired optimisation steps and measure `clmad` on three dies: its issue rate per SM-clock differs by up to 18× between the workstation and data-centre parts, so the same change can win on one Blackwell part and lose on another. We report twelve GPUs, Turing to Blackwell, per card, per SM and per dollar, show the kernel and where its speedup came from (none of it from changing the algorithm), and set the work against the instruction sets that made it possible on GPUs and CPUs. We separate the σ-walk from the table walk: the table walk's 20.08 was measured under a cycle rule later found to leave cycles inconsistently, the exact rule runs at 5.10 today, and the σ-walk is the fastest correct walk we have. Applying the same optimisations to Pollard rho on NIST P-256 and P-384 on the same card gives 1.52× and 2.61×, to 1.632 and 0.548 billion iterations per second; the levers transfer, but their relative sizes do not. Upper bounds derived from measured pipe rates put any one-addition walk on one RTX PRO 6000 at about 24–26 billion iterations per second. In calendar terms, a fleet of five RTX PRO 6000s expects to solve ECC2K-130 in a year, 56 in a month and a thousand in about three days, where the 2010 kernel needed over a thousand cards for a one-year solve; at the same rates P-256 remains about $6 \cdot 10^{21}$ card-years away. The walk constant is measured rather than assumed, at $2^{60.91}$–$2^{60.92}$ iterations, and a floor derived from the carry-less unit before tuning rules out 28 billion iterations per second on one card. We also report an FPGA engine at 8.25 billion iterations per second on one AWS F2; a campaign that stored 486 million distinguished points and $2^{56.68}$ iterations (5.4% of the expected work) before its fleet was reclaimed; structural alternatives that land far above rho; and the open questions we would pursue next. Every number cites an evidence record; the code, data and a Docker image that starts the ECC2K-130 walk are being prepared for release with the paper. None of this improves on the generic-group bound, and we do not claim that it does.

Contents

1 Introduction **4**

2 Prior ECC2K-130 implementations **6**
2.1 Comparison with current hardware 7

3 Curve, field and reference model **8**
3.1 Type-II optimal normal basis 8
3.2 Reference model 9
3.3 ECC2K-95 as an end-to-end control 9

4 Iteration functions **9**
4.1 The σ-walk . . . 10
4.2 The r-adding table walk . . . 11
4.3 An x-only doubling-and-bridge map . . . 11
4.4 Measured walk constants . . . 11
4.5 Cost per solution . . . 12

5 Field arithmetic **12**
5.1 Bitsliced representation . . . 13
5.2 Packed representation . . . 13
5.3 Cross-validation . . . 14

6 Instruction-set support **14**

7 Carry-less multiplication on current NVIDIA GPUs **16**
7.1 The PTX `clmad` instruction . . . 16
7.2 `clmad` throughput across dies . . . 17
7.3 Three-input logic and the integer pipeline . . . 17

8 GPU implementation **18**
8.1 Performance floor and roofline . . . 18
8.2 Acceptance criteria . . . 20
8.3 Optimisation ledger . . . 21
8.4 Sources of the speedup . . . 22
8.5 Table-walk arithmetic . . . 23
8.6 σ-walk versus table walk . . . 24
8.7 RTX PRO 4500 . . . 25
8.8 Cross-GPU comparison . . . 25
8.9 Rejected optimisations . . . 26

9 CPU implementations **26**
9.1 Bitsliced vector implementations . . . 27
9.2 Carry-less multiplication on CPUs . . . 27
9.3 Prime-field arithmetic on CPUs . . . 28
9.4 CPU contribution to the campaign . . . 28

10 Application to NIST P-256 and P-384 **29**
10.1 Walk and field arithmetic . . . 29
10.2 Baseline measurement . . . 31
10.3 Optimisation steps . . . 31
10.4 Transfer from ECC2K-130 . . . 32

11 Other GPU kernels **33**
11.1 Implementation status . . . 33
11.2 256-bit prime-field arithmetic for secp256k1 . . . 34
11.3 Automorphism folding and baby-step giant-step . . . 35
11.4 Pollard kangaroo for interval discrete logarithms . . . 36
11.5 P-256 and P-384 . . . 36
11.6 Binary fields beyond ECC2K-130 . . . 36
11.7 F4 elimination on cloud GPUs . . . 38
11.8 Summary . . . 38

12 FPGA implementation **39**
12.1 Pre-implementation estimate . . . 39
12.2 Architecture . . . 39
12.3 Measurements on AWS F2 . . . 40
12.4 Comparison with GPUs . . . 41

13 The ECC2K-130 campaign **41**
13.1 Configuration . . . 41
13.2 Distinguished-point cutoff . . . 42
13.3 Fleet economics . . . 42
13.4 Storage, seeding and certification . . . 43
13.5 Current state . . . 44

14 Structural alternatives to rho **44**
14.1 Index calculus on the Koblitz curve . . . 44
14.2 Point decomposition . . . 46
14.3 Discussion . . . 46

15 Performance upper bounds **46**
15.1 RTX PRO 6000 . . . 46
15.2 Data-centre parts and the RTX PRO 4500 . . . 47
15.3 FPGA . . . 47
15.4 NIST walk . . . 47
15.5 Implications for solution cost . . . 48

16 Time and cost to solution **48**
16.1 Model . . . 48
16.2 Estimates by device and fleet size . . . 49
16.3 NIST P-256 and P-384 . . . 51

17 Open problems and future work **51**
17.1 A fast exact table walk . . . 51
17.2 Remaining σ-walk optimisations . . . 52
17.3 Blackwell pipeline behaviour . . . 52
17.4 Data-centre parts . . . 52
17.5 NIST walk . . . 53
17.6 CPUs . . . 53
17.7 Methodology . . . 53

18 Related work **54**

19 Conclusion **55**

A Reproduction **56**

B Evidence records **57**

1 Introduction

In 1997 Certicom published a list of elliptic-curve discrete logarithm challenges [21]. Three of them, ECC2K-95, ECC2K-108 and ECC2K-130, use the Koblitz curve $y^2 + xy = x^3 + 1$ over $\mathbb{F}_{2^{97}}$, $\mathbb{F}_{2^{109}}$ and $\mathbb{F}_{2^{131}}$ and ask for the k with $Q = [k]P$. Harley's group solved ECC2K-95 in 1998 [32] and ECC2K-108 in 2000. ECC2K-130 is still open. In 2009 a consortium of twenty authors published the arithmetic, the iteration function and per-platform throughput for an attack on it [11, 12, 19, 25], and estimated the work at $2^{60.9}$ iterations. That attack was never completed.

This paper is about what Pollard rho costs on the current generation of NVIDIA hardware, and ECC2K-130 is the yardstick. In 2009 the fastest GPU in the consortium's plan had no carry-less multiplier and a binary-field product had to be built bit by bit from AND and XOR (Section 2). Today CUDA 13.3 exposes a carry-less multiply-add, `clmad`, in PTX; on this walk a Blackwell SM retires integer logic at about twice an Ada SM's per-clock rate; and three-input logic, funnel shifts and byte permutes all issue at full rate. We rebuilt the ECC2K-130 attack from scratch around that hardware and measured, step by step, how far each change moves the rate. On one RTX PRO 6000 Blackwell the campaign's walk runs at 14.64 billion iterations per second in its audited build and 15.88 in the best later build, about 230 times the best published GTX 295 kernel of 2010 [15], so the whole $2^{60.9}$ budget is now about five GPU-years (Section 2.1).

We also wanted to know which of those gains belong to binary fields and which belong to rho on GPUs in general. So we carried the same optimisations to Pollard rho on NIST P-256 and P-384, on the same card. Most of the levers transferred; their relative sizes did not.

We tried to hold the work to the standard the first attempt's reports set, and in places to a stricter one: every rate is paired against an adjacent control on the same machine, every target was written down before we tuned toward it, and every claim cites the evidence record that backs it. None of this improves on the generic-group bound, and we do not pretend otherwise. What we report is the cost of one step on GPUs, CPUs and an FPGA; the number of steps, measured on small curves of the same family and extrapolated; and a campaign that has walked $2^{56.7}$ of the $2^{60.9}$ iterations, most of it before its cloud fleet was taken away.

All rates in this paper are in billions of iterations per second on one device unless a row says otherwise. An iteration is one complete step of one walk, including its share of the batched inversion, the distinguished-point test and the restart path.

Contributions.

- A side-by-side account of ECC2K-130 rates on 2009–2010 hardware and on current GPUs, in one unit (Section 2).
- Background on the instructions this arithmetic relies on, from Fermi's `PRMT` to the PTX 9.3 `clmad`, and on x86, Arm and RISC-V from `PCLMULQDQ` to AVX-512, SVE2 and Zvbc (Section 6).
- A measured description of what current NVIDIA hardware changes for rho: the PTX 9.3 `clmad` instruction, its rate on three dies (from 1.69 lanes per SM-clock on the RTX PRO 6000 to 29.9 on the H100), and why the same source change can win on one Blackwell part and lose on another (Section 7).
- Diagrams of the kernel itself (thread layout, the step's data flow, the carry-less product, the memory layout) and a breakdown of where the speedup came from by kind of change. None of it changed the algorithm (Section 8.4).
- A GPU ledger of sixteen paired optimisation steps, taking one RTX PRO 6000 from 0.85 to 14.64 billion iterations per second for the campaign build, with later σ-walk builds at 15.44

and 15.88, a roofline, a floor derived from the carry-less unit before tuning that rules out 28 on one card, and every candidate we built and rejected (Section 8).

- A clear separation of the σ-walk from the table walk. The table walk's 20.08 was measured under a cycle rule later found to exit cycles inconsistently; the exact rule runs at 5.10 today, so the σ-walk is the fastest correct walk we have (Section 8.6).
- Measurements on twelve GPUs from Turing to Blackwell, per card, per SM and per dollar (Section 8.8).
- The same ladder applied to Pollard rho on NIST P-256 and P-384 on the RTX PRO 6000: 1.52× and 2.61× over the first working build, to 1.632 and 0.548 billion iterations per second (Section 10).
- Supporting work: a code generator that builds every field routine from one checked model of $\mathbb{F}_{2^{131}}$ (Sections 3 and 5); walk constants measured rather than assumed, $2^{60.91}$–$2^{60.92}$ iterations for the campaign's walk (Section 4); bitsliced CPU clients (Section 9); an inventory of the repositories' other GPU kernels (Section 11); an FPGA engine at 8.25 billion iterations per second on one AWS F2 (Section 12); the campaign at 486 M points and $2^{56.68}$ iterations (Section 13); and structural alternatives priced in the same unit, none of which come close (Section 14).
- CPU measurements by instruction set: AVX2, AVX-512's ternary logic, NEON with `BSL` and `EOR3`, and a packed client on the CPU's carry-less multiply at 2.9× its software product (Section 9).
- Upper bounds per platform: about 24–26 billion iterations per second for any one-addition walk on one RTX PRO 6000, a 1.55× logic cut for 30 on a B200, 9.3 for the F2 FPGA, and 4.6–12.2 for the NIST walk, with what each would mean for the cost of a solve (Section 15).
- The time to a solve on every measured device and fleet size, including rho's spread and the detection delay of the distinguished-point cutoff. Five RTX PRO 6000s solve ECC2K-130 in about a year, 56 in a month, 290 in a week, and a thousand in about three days. In 2010, a one-year solve needed 1,066 GTX 295 cards. The same model puts P-256 at about $6 \cdot 10^{21}$ card-years (Section 16).
- A list of open questions and next optimisations, each with what would settle it (Section 17).

Reporting methodology. One rule from the repository governs every section, so we state it once here. A change is measured as a ratio to a boundary that was written down before the change was attempted. That boundary is either a *floor*, a counting, generic-group or hardware argument the method cannot cross, or a *reference*, the best existing method on the same instance in the same unit. Every cost the method incurs goes inside the ratio: precomputation, verification, the inversion a batch shares, and the work still in flight when a collision lands. A number that prices only one phase is a stage diagnostic, never a speed. We then classify each change by what moved. An *advance* lowers the ratio to the floor. *Engineering* lowers the cost while the ratio to the floor stays flat. *Relabelling* lowers a headline while the total cost rises. *Accounting* changes a number without changing the algorithm. Every row in this paper falls into one of the last three. The same rule, applied across Pollard rho, baby-step giant-step, kangaroo and prime-field index calculus, is the subject of a companion draft in the `cryptanalysis` repository [7], though this paper does not depend on it. The mistake we tried hardest to avoid is quoting a rate without its paired control, its hardware and its launch geometry, and Sections 8.3 and 13.2 each record a case where an earlier draft in the repository made exactly that mistake.

Table 1: ECC2K-130 rates reported in 2009–2012, all for the Bailey et al. iteration function, converted to billions of iterations per second per chip. "Reported" is the source's own figure; the converted value is ours where the source gives cycles per iteration (cores × clock / cycles). GTX 295 figures are per card, which carries two GPU dies. The last column is the number of chips the source says (or its rate implies) would perform $2^{60.9}$ iterations in one year.

year	platform	reported	billion it/s	chips, 1 yr	source
2009	Core 2 Q6850, 4 cores, 3.0 GHz	534 cycles/it per core	0.0225	3,039	[12]
2009	Phenom 9550, 4 cores, 2.2 GHz	637 cycles/it per core	0.0138	—	[12]
2009	Cell (PS3, 6 SPEs + PPE)	27.67 M it/s	0.0277	2,466	[12]
2009	GTX 295 (two GT200b dies)	25.12 M it/s	0.0251	2,716	[12]
2009	Spartan-3 XC3S5000, 20 engines	33.67 M it/s	0.0337	2,026	[12]
2009	ASIC, 90 nm (synthesis est.)	800 M it/s	0.800	85	[12]
2010	Spartan-3 XC3S5000	111 M it/s	0.111	—	[25]
2010	GTX 295, final kernel	1,164 cycles/it per MP	0.064	1,066	[15, 41]
2012	Cell (PS3, 6 SPEs)	741.79 cycles/it per SPE	0.0259	2,636	[41]

Availability. We are preparing one companion repository, `rho-blackwell` [8], to hold everything needed to read, check and rerun this work: the paper source, the ECC2K-130 GPU and CPU clients, the NIST P-256/P-384 rho kernel, the benchmark data behind every figure, a copy of every evidence record, and a Docker image that builds the client and starts walking ECC2K-130 by default (Appendix A). At the time of writing it is not yet public, and the Docker image has not yet been built and run on a GPU; until it is, the records are available in the development repositories below. Every measured number in the paper cites an evidence record, written [E*n*], and Appendix B lists them. The records were produced in three development repositories (`crypto`, `cryptanalysis` and the research ledger `crypto-autoresearcher`), and the appendix names which one each came from. Live campaign aggregates are published at `https://aburan28.github.io/crypto/status/`.

Scope and limitations. We have not recovered a logarithm on the challenge curve, and we do not expect one from the corpus collected so far; collection is currently paused. We quote no rate for hardware we did not measure. The L40S collection rate, the RTX PRO 4500 collection rate under the bridge map and the G7e instance family are unmeasured, and we say so where they come up. The table walk's 20.08 and 19.40 were measured under superseded cycle rules and are rates for the arithmetic, not for a correct walk. Projections, such as the 7.2–7.4 billion iterations/s fused build on the 4500, the table walk's 0.81–0.86 relative cost per solve, and the 136- and 144-engine F2 images, are labelled as projections wherever they appear. The NIST rates in Section 10 are throughput on a walk that will not finish, and say nothing about the security of those curves.

2 Prior ECC2K-130 implementations

The 2009 attack is the baseline every number in this paper should be read against. Bailey et al. fixed the iteration function, the distinguished-point rule and the $2^{60.9}$ budget, and measured one step on every platform they could get: Core 2 and Phenom CPUs, the Cell processor in a PlayStation 3, the NVIDIA GTX 295, Spartan-3 FPGAs, and a synthesised ASIC [12]. Follow-up papers then tuned the Cell [19], the GPU [15] and the FPGA [25] clients, and Niederhagen's thesis collects the final figures [41]. Table 1 restates them in this paper's unit.

A few details matter when comparing. The GTX 295 is two GPUs on one card, so its 25.12 and 64 million iterations per second are per card; per die they are half. The INDOCRYPT paper prints "63 million" for the final kernel, while its own cycle count gives 64.0, which is also what the thesis and the two-year device count use [15, 41]. The 2009 FPGA figure was partly

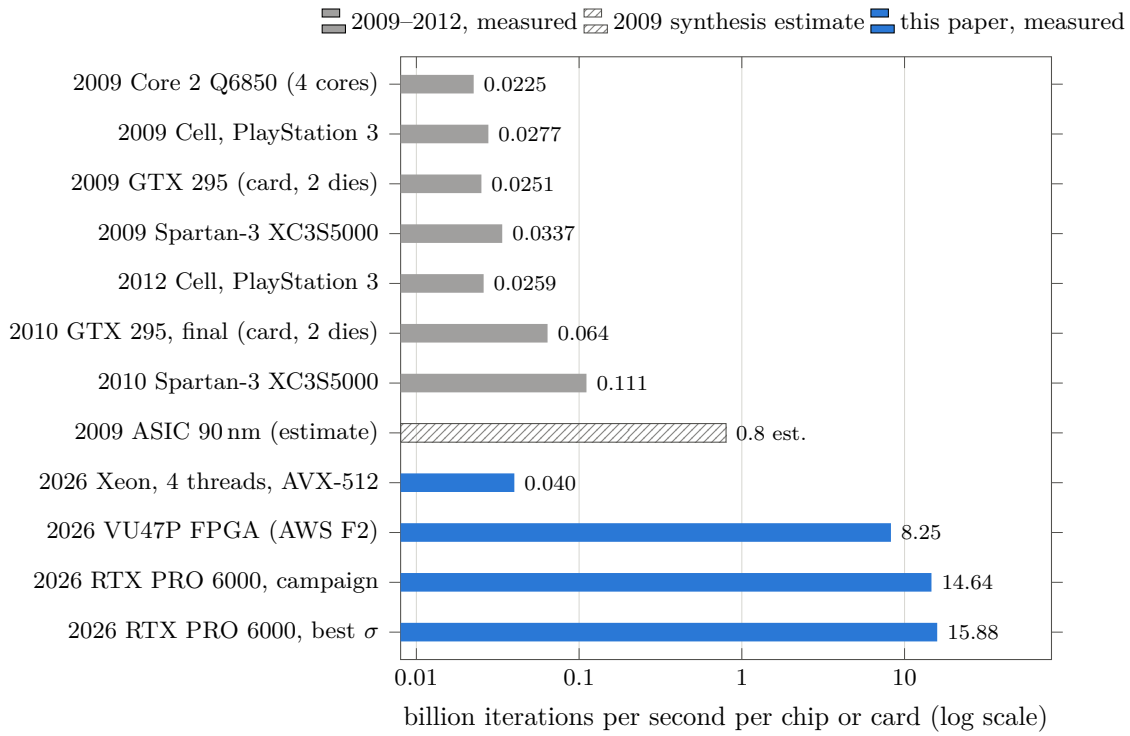


Figure 1: ECC2K-130 rates, 2009–2012 against today, on one log scale (Tables 1 and 2). Each bar is one chip except the two GTX 295 bars, which are one card carrying two GPU dies, as their sources report them; per die they are half. One RTX PRO 6000 does the work of about 229 of the 2010 cards.

estimated (only the multiplier of the chosen design was implemented), the ASIC is a synthesis result rather than silicon, and the 111 million figure for the 2010 FPGA design is as cited by later work, since we could not obtain the primary text. The 85-ASIC count comes from the project's status page, which Bailey et al.'s "less than 100 dedicated ASICs" matches.

2.1 Comparison with current hardware

Table 2 sets the best 2009–2012 figure for each platform class next to what we measured. All rows walk the same classes of the same curve with the same $2^{60.9}$ budget; the 2009 walk and our campaign walk are the same map, and only the arithmetic and the hardware differ.

The GPU row is the headline. One RTX PRO 6000 running the campaign build does the work of about 229 GTX 295 cards, and at that rate the whole $2^{60.9}$ budget is about 4.7 GPU-years instead of the 1,066 card-years of 2012. Of that factor, part is simply a bigger and faster chip: 188 SMs at about 2.4 GHz against 60 multiprocessors at 1.24 GHz is about 6× more SM-clocks per second. The rest, roughly 38× per SM-clock, is the arithmetic: a native carry-less product instead of a bitsliced one, three-input logic, a packed representation with the Frobenius as a permutation network, and a batch shape tuned against measured pipe rates. Figure 8 splits that factor, and Sections 5 through 8 take it apart step by step.

The FPGA gain is smaller, about 74× from a 2010 Spartan-3 to a current VU47P, and on cost per solve the F2 is level with the GPU (Section 12.4). The CPU gain per core is only about 2×: the CPU client is the same bitsliced design as in 2009, and it is the GPU that changed. The one platform class we have not revisited is the ASIC. A 2009 synthesis estimate of 0.8 billion iterations per second per chip at 90 nm already needed fewer than a hundred chips for a one-year solve; a modern node would need far fewer, but that is a manufacturing question and outside this paper.

Table 2: Best published rate per platform class against this paper's measurements, billions of iterations per second per chip. Ratios are ours. One year of $2^{60.9}$ iterations needs about 68.2 billion iterations per second.

class	then	billion it/s	now	billion it/s	ratio
GPU, per card	GTX 295 (2010)	0.064	RTX PRO 6000, campaign	14.64	229×
			RTX PRO 6000, best σ	15.88	248×
GPU, per die	GT200b (2010)	0.032	RTX PRO 6000	14.64	458×
FPGA	Spartan-3 XC3S5000 (2010)	0.111	VU47P (AWS F2)	8.25	74×
ASIC	90 nm estimate (2009)	0.800	—	—	—
CPU, per core	Core 2, 3.0 GHz (2009)	0.0056	Xeon 2.1 GHz, AVX-512	0.0126	2.2×

3 Curve, field and reference model

ECC2K-130 is the Certicom challenge [21]

$$E:\ y^2 + xy = x^3 + 1 \qquad \text{over } \mathbb{F}_{2^{131}},$$

a Koblitz curve [39] defined over $\mathbb{F}_2$, whose group order is $4 \cdot \ell$ with

$$\ell = 680564733841876926932320129493409985129,$$

a 129-bit prime [E1, "The problems"]. The challenge gives points P and Q of order ℓ and asks for k with $Q = [k]P$. It has been open since 1997. The 2009 consortium estimated the work at $2^{60.9}$ iterations of a Frobenius-equivariant walk [12], and Section 4.4 measures our walk at $2^{60.91}$–$2^{60.92}$.

We did not take the curve on trust. $E(\mathbb{F}_2)$ has four points, so the Frobenius trace is $t = -1$, and the Koblitz recursion $s_m = t\,s_{m-1} - 2s_{m-2}$ gives $\#E(\mathbb{F}_{2^{131}}) = 2^{131} + 1 - s_{131} = 4\ell$. The `cryptanalysis` library's test suite checks the two facts that pin down *this* curve, on random points multiplied by the cofactor: $\sigma^2 + \sigma + 2 = 0$, where $\sigma(x, y) = (x^2, y^2)$ is the Frobenius, and $\ell \cdot R = \mathcal{O}$ [E2; E2]. Converting the published challenge point out of the normal basis gives back Certicom's polynomial-basis hex, `05 1C99BFA6 F18DE467 C80C23B9 8C7994AA` [E1, "Validation"].

3.1 Type-II optimal normal basis

Because $2 \cdot 131 + 1 = 263$ is prime and 2 has order 131 modulo 263, $\mathbb{F}_{2^{131}}$ has a type-II optimal normal basis. Taking ζ to be a primitive 263rd root of unity, the basis is $\gamma_i = \zeta^i + \zeta^{-i}$ for $i = 1, \ldots, 131$, and an element is a 131-bit vector. Every design in this paper, on every platform, leans on three consequences of that choice [E1, "Field representation"].

fact	consequence
$\gamma_i\gamma_j = \gamma_{i+j} + \gamma_{i-j}$ (indices folded by $\gamma_{263-k} = \gamma_k$, $\gamma_0 = 0$)	a product is a cyclic convolution of two symmetric 263-bit vectors or, after a Bernstein–Lange change of basis, a polynomial product with no reduction step
$\gamma_i^2 = \gamma_{2i \bmod 263}$	squaring is the index permutation $i \mapsto \text{fold}(2i)$, which is free in unrolled code and just wiring on an FPGA, so every Frobenius power σ^k costs nothing
the Hamming weight of the vector is invariant under σ and under negation $-(x, y) = (x, x + y)$	the iteration's branch selector and the distinguished-point test are functions on $\langle\sigma, -1\rangle$-classes that read the vector the hardware already holds

3.2 Reference model

The code generator models the field as symmetric vectors over $\mathbb{Z}/263\mathbb{Z}$ modulo the all-ones vector, with multiplication as cyclic convolution. That model is complete and simple enough to verify by inspection, and every generated routine is checked against it on 64 random inputs before it is written out [E3; E3]. The bitsliced multiply, square, inverse, σ^7 and Hamming weight are also tested differentially against a reference implementation that shares no code with them. We check that the iteration function commutes with Frobenius and with negation, that the collision solver recovers a planted logarithm, and that the distinguished-point comparison is right for every weight and cutoff. In all, `make test` runs 52 such checks across GF(2^{23}), GF(2^{41}), GF(2^{83}) and GF(2^{131}) [E1, "Validation"].

The same model is the oracle for every other platform. The VHDL engine's reference script imports it directly and uses it to prove the basis-change maps, the multiplier route, σ^j, the inversion chain and a full step before it emits test vectors (Section 12) [E4]. The `cryptanalysis` library's C model of the same field and curve [E2] was written separately and re-walks every GPU report that library imports. When the two models agree on a record, they agree on the exact 131-bit vector in the same permuted basis, not merely up to a change of basis [7].

Longer runs exercise the reporting path rather than the algebra. On GF(2^{83}), 12.6 M iterations produced 172 distinguished points, within 30% of the predicted rate, and the reference recomputed 60 of them from their seeds without a mismatch, including walks that had been restarted. On the challenge curve itself, with the cutoff loosened from 34 to 50 so that points actually appear, 3.1 M iterations produced 12,368 reports, and the 25 we recomputed all matched exactly [E1, "Validation"].

3.3 ECC2K-95 as an end-to-end control

ECC2K-95 is the retired Certicom challenge on the same curve equation over $\mathbb{F}_{2^{97}} = \mathbb{F}_2[t]/(t^{97} + t^6 + 1)$, with group order 4ℓ for the 95-bit prime $\ell = 39614081257132074233778707191$. Harley and collaborators solved it in 1998 [32]. Since $2 \cdot 97 + 1 = 195$ is composite there is no type-II optimal normal basis, so the client runs a polynomial-basis backend and computes the orbit-invariant weight through one generated linear map into a normal basis, which is how Harley's own client did it [E1, "ECC2K-130 and ECC2K-95"].

We recovered the parameters from Harley's surviving client source. The generator refuses to emit the curve unless both points lie on it, both have order ℓ, and the published answer

$$k = 37837308472231540269443981458$$

satisfies $[k]P = Q$. The expected work is about 2^{44} iterations. Harley's group measured $2.16 \cdot 10^{13}$, against the $1.79 \cdot 10^{13}$ that this walk predicts from $\sqrt{\pi\ell/(4m)}$ [E1, "The problems"]. Per iteration the polynomial-basis field is about 1.9× cheaper than GF(2^{131}) in the normal basis (roughly 32,000 against 60,000 bitsliced instructions). That makes ECC2K-95 a job of a few GPU-hours where ECC2K-130 takes GPU-centuries, which is exactly what an end-to-end test of the whole pipeline needs: the same machinery at a scale that finishes [E1, "ECC2K-95"]. The repository's separate ECC2K-95 CUDA kernel has so far run only under CPU emulation, where it recovers 21-bit and 40-bit toy logarithms at 0.94× and 1.32× its own prediction; we do not use it for any throughput claim [E5].

4 Iteration functions

Pollard's rho [47], in the parallel collision-search form of van Oorschot and Wiener [52], runs many pseudo-random trails through the group, reports the distinguished points they pass, and solves the logarithm when two trails meet. On a Koblitz curve the Frobenius σ and negation

generate a group of $2m = 262$ automorphisms acting on $\langle P \rangle$. A walk defined on the orbits of that group needs $\sqrt{262} \approx 16.2$ times fewer steps [24, 28, 56]. With $N = (\ell - 1)/262$ classes, the expected number of classes visited before a collision is $\sqrt{\pi N/2}$, which here is $2^{60.809}$ [E6, §1]. Every walk in this paper is a map on those classes. They differ in how they respect the class action and in what each step costs.

4.1 The σ-walk

The campaign uses the map of Bailey et al. [12]:

$$\begin{aligned} j &= 3 + ((\mathrm{HW}(x_R) \gg 1) \bmod 8) \in [3, 10], \\ R' &= \sigma^j(R) + R, \\ \text{distinguished} &\iff \mathrm{HW}(x_R) \le w, \end{aligned}$$

with $w = 34$ by default in the client and $w = 32$ in the live campaign. Because the weight is a class invariant, $f(\sigma R) = \sigma f(R)$ and $f(-R) = -f(R)$, so the map descends to classes without ever computing a canonical representative inside the loop. The division by two in the selector matters more than it looks: x-coordinates of subgroup points always have even weight, so a selector on the raw weight would only ever use four of the eight branches [E5].

One step is one affine addition, with the inversion shared across a batch (Section 5). In the normal basis σ^j is an index permutation and costs nothing. As a scalar, the step is $R \mapsto [1+s^j]R$, where s is the root of $s^2 + s + 2 = 0$ that σ realises on $\langle P \rangle$. Selecting σ^j needs no branch: it is three conditional squarings, each a single `LOP3` per word, and j itself is never materialised [E1, "Iteration function"].

The walk tracks no coefficients. A walk is identified by its seed, and a report is the pair (seed, canonical endpoint), 32 bytes in all. The start point is $R_0 = Q + \sum_i c_i\, \sigma^i(P)$, where c is a 128-bit string from a PRF of the seed. It is computed with the same arithmetic and merged into finished lanes under a mask. Tracking a linear combination inside the loop would cost a 129-bit modular multiplication per step, which this walk cannot afford. Instead, collision resolution re-walks both trails and counts how often each branch $1 + s^j$ was applied. That gives endpoint $= [\mu](\alpha_0 P + Q)$ with $\mu = \prod_j (1 + s^j)^{n_j}$; the solver matches the two endpoints up to Frobenius and negation and solves for k modulo ℓ [E1, "Distinguished points and restarts"]. The σ-walk has no fruitless cycles: no product $\prod_j (1 + s^j)^{n_j}$ equals $\pm s^a$, so a trail can never come back to its own class [E6, §5].

The client can optionally carry the eight branch counters as it goes (`WITNESS=1`). Because μ is a commuting product, that is just eight counters and nothing is multiplied per step. The payoff is that a third party can check a report with one scalar multiplication $[\mu]R_0$, 196 point operations, instead of re-walking the $2^{25.27}$ steps that a weight-34 report represents, an asymmetry of about $2^{17.7}$. On the bitsliced CPU walk this costs 6.0%, the median of three paired rounds (40.026 to 37.762 million iterations/s, a 5.7% drop in rate, against 33.8% predicted; an earlier revision measured 4.2% on a pre-merge tree). On the packed GPU walk the rate fell 34.7%, from 14.003 to 9.150 billion iterations/s. That figure is stated in the note, but only its job script was frozen with the record. Either way it is far above the 2% the note had set as its threshold for abandoning a 16-bit counter layout that was never built, so the campaign builds pin `WITNESS=0` and the record stays at 32 bytes [E7; E8]. A later fused kernel narrows the packed cost (9.347867 to 11.800903 billion iterations/s in witness-carrying mode) but does not close it [E9].

4.2 The r-adding table walk

On a GPU, the σ-walk pays for σ^j in the basis conversions around it (Section 5.2). An r-adding walk on the same classes avoids that cost:

$$\begin{aligned} h(R) &= (\mathrm{HW}(x)/2) \bmod 8, \\ k(R) &= \left(\textstyle\sum_e L(e)\, x_e\right) \cdot \mathrm{HW}(x)^{-1} \bmod 131, \\ \varepsilon(R) &= \text{bit } p(R) \text{ of } y, \\ R' &= R + (-1)^{\varepsilon}\, \sigma^k(T_h), \end{aligned}$$

where $T_h = [a_h]P + [b_h]Q$ and the 131×8-point table lives in shared memory (37.7 KB, or 48.5 KB per block with the selection tables). Since k shifts by one under σ and ε flips under negation, the walk descends to classes. Each step is one affine addition with a table operand whose Frobenius power is a lookup rather than a conversion [E10, §4 and §6.1].

Unlike the σ-walk, an adding walk on classes does have fruitless cycles [16]. Two kinds show up here: four-step patterns in which the table points' Frobenius powers satisfy a relation coming from $\tau^2 + \tau + 2 = 0$, and six-step pairwise patterns. The first cycle rule let 24 four-step and 4 six-step patterns through. Observed against predicted counts were 1,145 against 1,312 (0.87 ± 0.03) for the τ-relation class and 126 against 166 for the pairwise class [E6, §5]. At the campaign's trail length of $2^{28.41}$ steps, a model built on those measured rates predicts that the cycles trap 53.7% of trails, a loss of 7.1–8.6× with the 2^{30} iteration guard [E6, §6]. A revised rule (v2) brings residual trapping down to $2.0 \cdot 10^{-5}$ of trails, at a 1.0048× iteration cost from merge parting. A third revision (v3, Section 4.5) replaces the history-dependent exits with a raw-cycle anchor [E6, §11.1].

4.3 An x-only doubling-and-bridge map

On the power-capped RTX PRO 4500 a third map turns out to pay. Its common transition is $R' = R + \sigma(R)$, which is the scalar $[1 + s]$. On this curve $x(R + \sigma R) = x + 1/x$, the square root of $x([2]R)$, so the walk can keep an x-only state. That scalar on its own generates a subgroup of index 12, so the map takes a sparse bridge $R' = R + \sigma^3(R)$ (index 109 on its own) whenever $\mathrm{HW}(x) \bmod 72 = 14$. Together the two transitions generate the full group, which the earlier [2]/[3] map had failed to do by an index of two (we called it "the scalar subgroup defect") [E1, "ECC2K-130 and ECC2K-95"]. We froze a study of 2,000 trials on $\mathrm{GF}(2^{23})$, where the modulus-72 bridge predicate selects exactly what the earlier modulus-32 one did (weight 14 only). It gives mean charged work of 1.006696× that of the map selected at the time, so the new map's throughput gain (Section 8.7) is not given back in extra iterations. The map also passes an independent 2,048-point affine checkpoint [E11].

4.4 Measured walk constants

Throughput is only half of what determines how long a solve takes. The other half is how many iterations the walk needs, which for a non-random map differs from the random-mapping $\sqrt{\pi N/2}$ by a constant c. We measure c as iterations divided by a random mapping's iterations on the class set, on small Koblitz curves of the same family. The device walk and a host emulation are measured separately, with an r-adding control using 256 branches alongside [E6, §4].

The σ-walk is one-to-one on each branch, so it pays the injective-branch penalty $1/\sqrt{1 - \sum_j p_j^2}$. With the weight distribution at $n = 131$ ($\sum p_j^2 = 0.127$) that penalty is 1.070. The measurements in Table 3 drift from 1.0995 at $n = 23$ toward it, and extrapolate to $[1.070, 1.082]$ at $n = 131$. The table walk is not injective per branch and sits at 1.0035 ± 0.0006

Table 3: Walk constant c on toy Koblitz curves, ECC2K-130 branch distribution ($H = 8$). Floor $c = 1$. The adding-walk control with 256 branches measures 1.0002 ± 0.0012 at $n = 23$. The injective-branch model $1/\sqrt{1 - \sum p^2}$ predicts 1.070 for the σ-walk at $n = 131$.

n	measured on	table walk	σ-walk
23	host emulation	1.0037 ± 0.0012	1.0995 ± 0.0013
23	device	1.0076 ± 0.0037	1.0948 ± 0.0040
31	host emulation	1.0035	1.0964
37	host emulation ($W = 16$)	1.0036	1.0949
41	host emulation	1.0125 ± 0.0059	1.0910 ± 0.0064
41	device	1.0051 ± 0.0299	1.0997 ± 0.0336
59	host emulation	0.9949	1.0817
pooled / extrapolated to $n = 131$		1.0035 ± 0.0006	[1.070, 1.082]

pooled, so the ratio c_t/c_σ is 0.928–0.938. The expected work for the campaign's σ-walk is therefore

$$2^{60.809} \times [1.070,\ 1.082] = 2^{60.91}\text{–}2^{60.92}$$

iterations. Under the original 2^{30} restart guard it would be a further 1.185× ($2^{61.15}$–$2^{61.17}$): a guard at three trail lengths cuts the longest 4.9% of trails and so throws away 15.6% of the steps (again a model on the measured rates). Bailey et al.'s $2^{60.9}$ holds up [E6, §6]. The guard has since been raised to 2^{32}, twelve trail lengths at weight 32, which discards only 0.007% of steps (1.00007×) [E12].

As a direct check, on GF(2^{41}) with 128 walks, cutoff 13 and 64 planted logarithms, the σ-walk needed $164{,}200 \pm 9{,}900$ iterations per solve and the table walk $157{,}800 \pm 7{,}500$, a ratio of 0.96 ± 0.07. The bare prediction is 102,600; the difference is roughly 1.5× of harness overhead from the distinguished-point tail [E10, §6.2].

4.5 Cost per solution

Put rate and constant together and the table walk looks very different. Its throughput advantage (Section 8.5) is 1.09–1.15× and its walk constant is 0.93×. But its as-built cycle rule is modelled to trap half of all campaign-length trails, so the as-built kernel's cost per solve was 4.85–6.26× that of the σ-walk. With rule v2 the projection improves to 0.81–0.86×, and the switch deadline, the fraction of the campaign after which switching no longer pays, is 15–22%. That projection is untested: the note declares the paired rate measurement for the v2 kernel that would test it, and it has not been run [E6, §11]. So the campaign still walks the σ-map, and the 20 billion iterations/s table-walk rows in Section 8.5 are rates, not costs per solve.

Rule v2 itself refused all 19,200 τ-cycles in a dedicated test, passed a 4,096-point host probe without a mismatch, solved 8/8 planted logarithms, and its scatter check against the predicted cycle counts gives $\chi^2 = 8.79$ on 15 degrees of freedom (the v1 rule gives 23.30) [E6, §11.1 and §11.6.1]. Rule v3 then replaced the history-dependent exits with a raw-cycle probe of at most eight steps and a least-key anchor exit. It is checked by regression tests on a raw two-cycle found on the challenge curve and on synthetic entry-phase cycles, with 300 of 300 device reports re-walked and identical corpora across arms. V2's walk-constant and throughput figures do not carry over to it, and the exact v3 kernel currently runs at about a third of the σ-walk's rate (Section 8.6) [E13]. V3 needs a new table corpus; the campaign's σ-walk corpus is unaffected.

5 Field arithmetic

Field multiplication dominates a step. The addition itself needs two products. With W walks sharing one inversion through Montgomery's trick [40], the product tree adds $3(W - 1)/W$

more, plus $1/W$ of an Itoh–Tsujii inversion [37], which costs only eight multiplications in a basis where squaring is free (the client's addition chain is $1, 2, 4, 8, 16, 32, 64, 65, 130$; the FPGA's is $1, 2, 4, \ldots, 128, 130$) [E14; E4]. Going from $W = 4$ to $W = 32$ is worth 60% on the CPU client, because it shrinks the amortised inversion from 25% to 3% of the multiplication budget [E1, "Throughput"]. How a multiplication is represented is therefore the central design decision, and the repository carries two answers.

5.1 Bitsliced representation

The original client is bitsliced in Biham's sense [18]: one word operation advances W walks at once, one per bit lane, and a field operation is a straight-line program of word-wide Boolean instructions generated from the field model. The GPU uses 32-bit words. The host uses the widest it has (512 lanes with AVX-512, 256 with AVX2, 128 with NEON), and the arithmetic source is identical at every width [E1, "Word width"].

Multiplication follows Bernstein and Lange [14]. Both operands are converted from the normal basis to the optimal polynomial basis $\{1, c, c^2, \ldots\}$ with $c = \gamma_1$, multiplied as polynomials, and the 261-coefficient product is converted back. We generate both conversions from the substitution $c = z + 1/z$, which splits by parity as $H(c) = H_{\text{even}}(c^2) + c\,H_{\text{odd}}(c^2)$; because squaring is an index permutation, this yields an $O(K \log K)$ conversion. The generated `multPrep` is 331 instructions and the double-size inverse conversion 903, against 325 and 909 for the hand-designed chains in the literature [E1, "Findings"].

This backend taught us two things that change the usual accounting. The first is that three-input logic fusion (`LOP3` on NVIDIA, `vpternlogd` on AVX-512, `BSL/EOR3` on Arm with `FEAT_SHA3`) is worth 1.65×. A 131-bit multiplication is 14,554 two-input bit operations but only 8,859 instructions once $x \oplus y \oplus z$ and $x \oplus (y \wedge z)$ are single instructions. Since the fused form makes the XOR that accompanies an AND free, schoolbook leaves beat Karatsuba leaves [38]: full recursion costs 18,819 bit operations and 14,770 instructions, while stopping at a 17-bit schoolbook leaf costs more bit operations but fewer instructions. A search over balanced and unbalanced splits found nothing below 8,859. Toom-3 over $\mathbb{F}_2$, the source of Bernstein's 11,961-bit-operation chain [13], is not implemented [E1, "What is not done"]. The second lesson is that code size, not arithmetic, is the real wall. Inlining everything produced 285,096 PTX instructions for one kernel, and `ptxas` had still not finished register allocation after five minutes. Turning the multiplication, inversion, conversions, leaf, weight tree and σ^j into real subroutines brought it down to 36,553 lines and a 2.4 s assembly [E1, "Findings"].

The Hamming-weight tree is 268 instructions against 654 bit operations, because a full adder is two `LOP3`s. The whole bitsliced iteration comes to about 60,000 instructions per 32 walks on $\mathrm{GF}(2^{131})$, as a static count. On the GPU that is the 0.85 billion iterations/s control of Section 8.3. For comparison, the instruction-count ceiling of the later packed kernel on the RTX PRO 6000 is 17.6 billion iterations/s (27.1 T lane-operations/s over about 1,540 lane-operations per update, a different count) [E15].

5.2 Packed representation

The packed backend holds one walk's coordinates in five 32-bit words per thread and multiplies with carry-less instructions [E16]. Coordinates live in $\mathbb{F}_2[c]/(m)$ with $c = \gamma_1$, so a 131-bit product is a Karatsuba over carry-less multiplies (six `CLMAD`) followed by a reduction [E10, §1]. The normal basis, where the weight, branch selection and distinguished-point test live, is reached through the same Bernstein–Lange conversions, implemented as shift-and-mask networks and, for the Frobenius powers, as Beneš permutation networks [E17]. Before CUDA 13.3 the carry-less product had to be built from integer multiplies. The native `clmad.lo/hi` instructions that PTX 9.3 exposes for `sm_80` and later were worth +22.4% on the RTX PRO 6000 (Section 8.3), and they are why the campaign preset requires that compiler; Section 7.1 describes

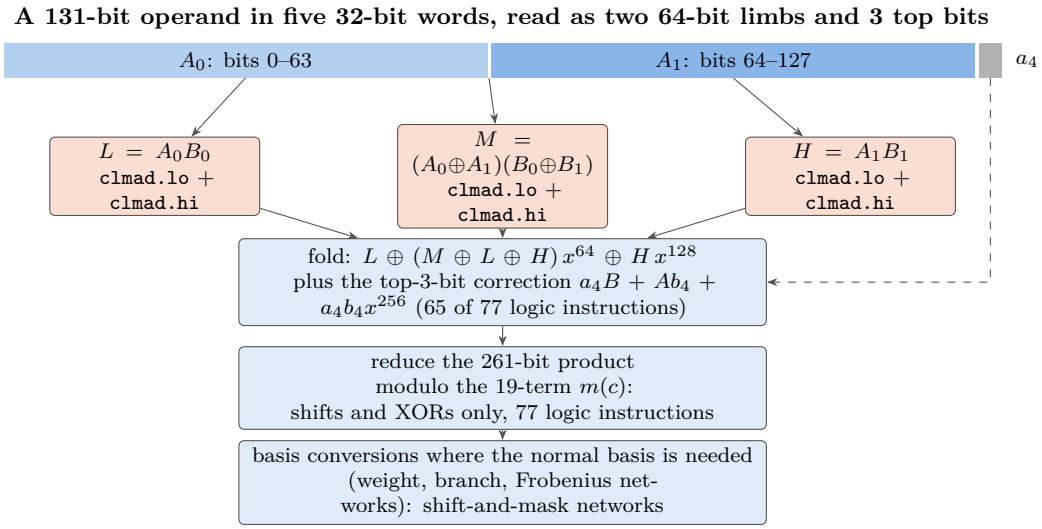


Figure 2: The packed 131-bit product. Three 64×64 carry-less leaves (six `CLMAD` instructions) form a one-level Karatsuba; the three top bits, the fold, the reduction and every basis change run on the logic pipe. On the RTX PRO 6000 a `CLMAD` costs the issue time of 37.8 logic instructions, so the split keeps only the six leaf products on the carry-less unit; on a B200 or H100, where it costs 2–3.8, moving the correction and even the reduction onto `CLMAD` pays instead (Section 7.2).

the instruction and its very different rates across parts [E18].

The natural unit of account changes with the representation. In the packed kernel the shipping walk issues 4.81 products per update at batch 16 and about 2,324 static ALU slots [E10, §1 and §3]. The count that matters is `CLMAD`s, and Section 8.1 prices it. The reduction `reducePolynomial131` alone is 440.6 lane-instructions per update, 29% of the integer work of the 20 billion iterations/s build [E19].

5.3 Cross-validation

The two backends, the CPU client at every word width, the VHDL engine and the `cryptanalysis` library's imported kernel all stay bit-for-bit compatible. A distinguished-point record written by any of them re-walks under any other, and every GPU report is re-walked on the host reference before we record a rate. The host check re-slices packed words into the same permuted type-II basis and compares vectors, not classes [E1, "Validation"]. One corner case is worth mentioning. When $\sigma^j(R) = \pm R$, the addition's denominator is zero; this happens with probability about 2^{-131} per step. The σ-walk client and the VHDL engine do not special-case it. A zero denominator inverts to zero, the batch's points become garbage, and the next distinguished-point check or the iteration guard throws them away. The `cryptanalysis` C model and the table walk abort the step instead [E20; E4].

6 Instruction-set support

Pollard rho on these curves needs two kinds of arithmetic: products of binary polynomials (for $\mathbb{F}_{2^{131}}$) and multi-word integer products with long carry chains (for the NIST primes). Both have gained dedicated instructions over the last fifteen years, on GPUs and CPUs alike. Table 4 and Figure 3 summarise the ones this paper uses or prices; the rest of this section says why each matters.

Table 4: Instructions relevant to rho on $\mathbb{F}_{2^{131}}$ and on prime fields, with the first generation that has each. "Use here" says how this paper's code uses it: in the measured clients, priced, or not yet tried.

instruction	what it does	first available	use here
NVIDIA GPUs			
`PRMT`, `POPC`, `FLO`	byte permute; population count; find leading one	Fermi (2010)	every client
`SHF`	funnel shift of a register pair	Kepler (2012–13)	every client
`LOP3`	any 3-input Boolean function	Maxwell (2014)	every client
`IADD3`, native `IMAD`	3-input add; 32-bit multiply-add	Maxwell; Volta (2017)	NIST walk
`mad.lo.cc`, `madc.hi.cc`	multiply-add with carry in/out (PTX)	all CUDA GPUs	NIST walk
`clmad.lo/.hi`	64×64 carry-less multiply-add (PTX 9.3)	CUDA 13.3 (2026), `sm_80+`	ECC2K-130 client
x86 (Intel, AMD)			
`PCLMULQDQ`	64×64 carry-less multiply	Westmere (2010)	packed CPU client
`MULX`, `ADCX/ADOX`	flag-free multiply; two carry chains	Haswell (2013); Broadwell (2014)	not tried
`VPTERNLOGD`	any 3-input Boolean function, 512-bit	Skylake-SP (2017)	bitsliced client
`VPMADD52` (IFMA)	52×52-bit multiply-add	Cannon Lake (2018)	not tried
`VPCLMULQDQ`, GFNI	4 carry-less products per 512 bits; GF(2^8) affine	Ice Lake (2019); AMD Zen 4 (2022)	priced
Arm			
`PMULL/PMULL2`	64×64 carry-less multiply	Armv8.0 Crypto	packed CPU client
`EOR3`, `BCAX`	3-way XOR; AND-NOT then XOR	Armv8.2 SHA3	bitsliced client
SVE2 `PMULLB/T`, `EOR3`	vector carry-less and 3-input logic	Armv9.0 (SVE2-AES)	not tried
RISC-V			
Zbc, Zvbc	scalar and vector carry-less multiply	ratified 2021, 2023	not tried

NVIDIA GPUs. The integer pipe has grown three instructions that a binary field uses on every step. `PRMT` and `POPC` date from Fermi, the funnel shift `SHF` from Kepler, and `LOP3`, which evaluates any Boolean function of three inputs from an 8-bit truth table, from Maxwell [42]. `LOP3` alone is worth 1.65× on the bitsliced product (Section 5.1). `POPC` still issues at a quarter of the logic rate on every current part, 16 per clock per SM, so each Hamming weight costs about four logic slots [43]. The prime-field side relies on PTX's carry-chain forms, which every CUDA GPU has; the extended-precision forms issue at half the rate of a plain multiply-add [43].

The change that matters most for ECC2K-130 is recent. PTX ISA 9.3, shipped with CUDA 13.3 in May 2026, added `clmad.lo` and `clmad.hi`: the low and high halves of a 64×64 carry-less product, XORed into an addend, for `sm_80` and later [44]. NVIDIA describes it as accelerated in hardware since Ampere and reports GHASH at about 6.3 TB/s on a B200 and 1.3 TB/s on an RTX 5090 [53]. NVIDIA documents neither its SASS mnemonic nor its issue rate, which is why we measured both (Section 7.2). Nsight attributes it to the FP64 pipe, and it is slow exactly where FP64 is slow: about two lanes per SM-clock on the RTX PRO 6000, ten to eighteen times that on the B200 and H100. That matters because NVIDIA's FP64 ratio varies by product rather than by generation: 2:1 against FP32 on the A100, H100 and B200, but 64:1 on the GA10x and Ada consumer parts, on the RTX PRO 6000 and on the B300 [1–3, 5, 10].

Two other generational changes show up in our rates. From Ampere's GA10x parts onward each SM partition pairs 16 FP32-only lanes with 16 lanes that do FP32 or INT32, and Ada kept that split; the workstation Blackwell chips (`sm_120`) unify them, so every lane can issue integer work [1, 3, 5]. That is consistent with the factor of two we measure per SM between Ada and `sm_120` on this integer-bound walk (Section 7.3). Hopper's DPX instructions (fused add with

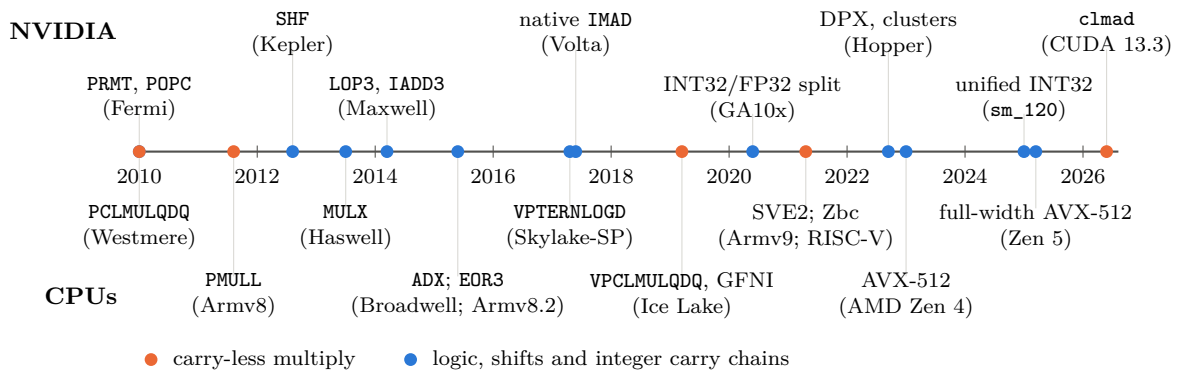


Figure 3: When the instructions this arithmetic relies on arrived. Dates are first availability in a shipping product (Armv8 and Armv8.2 are architecture versions; Zbc is the ratification date). The GPU carry-less multiply arrived last of all, in 2026, which is why the ECC2K-130 GPU client is new work rather than a port.

min or max for dynamic programming), its tensor memory accelerator and Blackwell's `tcgen05` tensor-memory instructions do not help this walk; the products here have no operand shared across lanes, which tensor cores need (Section 17.7).

x86. Intel added `PCLMULQDQ` with Westmere in 2010 [30], `MULX` with Haswell and the dual carry chains `ADCX`/`ADOX` with Broadwell [45], AVX-512 with its ternary-logic `VPTERNLOGD` in Skylake-SP, the 52-bit integer multiply-add IFMA in Cannon Lake [31], and the 512-bit `VPCLMULQDQ` (four carry-less products per instruction) with GFNI in Ice Lake [23, 36]. AMD implemented AVX-512, including `VPCLMULQDQ` and GFNI, in Zen 4 on a 256-bit datapath that issues each 512-bit operation in two halves, and Zen 5's server parts run it at full width [4, 6]. Intel's AVX10.2 makes 512-bit vectors universal and APX doubles the general-purpose registers to 32 [34, 35]; both would help a register-starved bitsliced leaf (Section 9.1).

Arm and RISC-V. Armv8.0's optional Crypto extension has the 64×64 carry-less `PMULL`; the Armv8.2 SHA3 extension adds the three-input `EOR3` and `BCAX`; SVE2 brings vector forms of both, with 128-bit-result `PMULLB`/`PMULLT` in the SVE2-AES extension [9]. Graviton2 (Neoverse N1) has PMULL but not SHA3, Graviton3 (V1) adds SHA3 and SVE, and Graviton4 (V2) has SVE2 with the 128-bit PMULL; Apple's M4 has SHA3 and SME but no non-streaming SVE. RISC-V ratified scalar carry-less multiply (Zbc) in 2021 and its vector form (Zvbc) in 2023 [48, 49].

7 Carry-less multiplication on current NVIDIA GPUs

The 2009 GPU client ran on hardware with no carry-less multiplier at all: a binary-field product had to be assembled from AND and XOR, one bit-slice at a time. Three things are different on current NVIDIA parts, and between them they explain most of the gap between the rates in Section 2 and the ones in this paper.

7.1 The PTX `clmad` instruction

PTX ISA 9.3, shipped with CUDA 13.3, added a carry-less multiply-add, `clmad.lo.u64` and `clmad.hi.u64`, which returns the low and high halves of a 64×64-bit product over $\mathbb{F}_2$. NVIDIA documents it for `sm_80` and later, so it covers Ampere, Ada, Hopper and Blackwell but not Turing [E18]. Our client uses both halves with zero addends and feeds them into a three-leaf Karatsuba, so a 131-bit product is six `CLMAD` instructions. The native binary for `sm_120`

Table 5: Issue rates in lanes per SM-clock, one probe, three parts, one session (best of three runs per stream). The last column prices one `CLMAD.lo` in `LOP3` slots, which is what decides whether moving work onto the carry-less unit pays.

GPU	`LOP3`	`CLMAD.lo`	`CLMAD` lo+hi	`IMAD.WIDE`	`POPC`	`LOP3/CLMAD.lo`
RTX PRO 6000 (`sm_120`)	63.9	1.69	1.99	31.3	16.1	37.8
B200 (`sm_100`)	63.7	16.9	29.1	22.9	16.0	3.78
H100 (`sm_90`)	59.8	29.9	30.1	22.6	16.0	2.00

contains real `CLMAD` opcodes in its SASS, and the build refuses to compile the native path with anything older than CUDA 13.3 while keeping the software product for `sm_75` slices of a mixed binary [E14; E21].

The gain depends on the part. On the RTX PRO 6000 the native product lifted the packed walk by 22.4% (7.110 to 8.704 billion iterations/s) and roughly halved the instructions issued per update, which is what later let the batch drop to 16 walks per inversion (Section 8.3) [E18]. On Ada the effect is larger, because the software product it replaces is more expensive there: 1.81× on the L40S and 1.88× on the L4 [E22, §3]. On the ECC2K-95 kernel the same port gave 1.19× on the walk. The T4, at `sm_75`, has no `clmad` and runs the software product, which is why it sits at the bottom of every table in this paper [E23].

7.2 `clmad` throughput across dies

Having the instruction is not the same as having it fast. We measured issue rates with one probe on three parts in one session (Table 5) [E24, §6.1].

The logic pipe, `POPC` and shared-memory loads run at the same per-clock rate on all three. The carry-less unit does not. Nsight attributes `CLMAD` to the FP64 pipe, and the workstation Blackwell die executes FP64 at 1/64 of its FP32 rate, about two lanes per SM-clock; the data-centre dies run it ten to eighteen times faster per clock. On the RTX PRO 6000 one `CLMAD.lo` costs as much issue time as 37.8 logic instructions, on the B200 3.78, on the H100 2.00 [E24, §6]. The RTX PRO 4500 matches the 6000 to within 1% [E10, §3.1 and §4.5].

This one number reshapes the optimisation problem. On the workstation die the carry-less unit is scarce, so the right move is to keep products on it and everything else off it. On the data-centre dies it is plentiful and the logic pipe binds, so the right move is the opposite: push reductions, corrections and even basis changes onto `CLMAD`. The same source change can therefore win on one Blackwell part and lose on another. Moving the top-bit correction onto `CLMAD` costs 15% on the RTX PRO 6000 and gains 13.4% on the B200; a three-limb Karatsuba loses 30% on the 6000 and gains 4.9% on the B200 [E25; E26, §3.2 and §4]. Kernel tuning does not travel between dies, and each part in this paper was tuned on its own.

7.3 Three-input logic and the integer pipeline

The other instructions that matter are older but just as important. `LOP3` evaluates any three-input Boolean function in one instruction, which is worth 1.65× on the bitsliced product (Section 5.1) and makes a full adder two instructions [E1, "Findings"]. On the RTX PRO 6000, `LOP3`, `IADD3`, the funnel shift `SHF`, `PRMT` and `IMAD` all issue at about 62–63 lanes per SM-clock, while `POPC` and `FLO` run at quarter rate (15.8), so each Hamming weight costs about four logic slots; the static cost model prices them that way [E15; E10, §3.1]. The prime-field walk of Section 10 depends instead on the carry-chain forms `mad.lo.cc`, `madc.hi.cc` and `addc`, which exist only at 32-bit granularity; writing them by hand was worth 1.19–1.38× there.

Finally, the integer pipe itself got wider. Blackwell's SM lanes are unified FP32/INT32, where Ada splits them into an FP32-only half and an FP32/INT32 half, so the note predicted

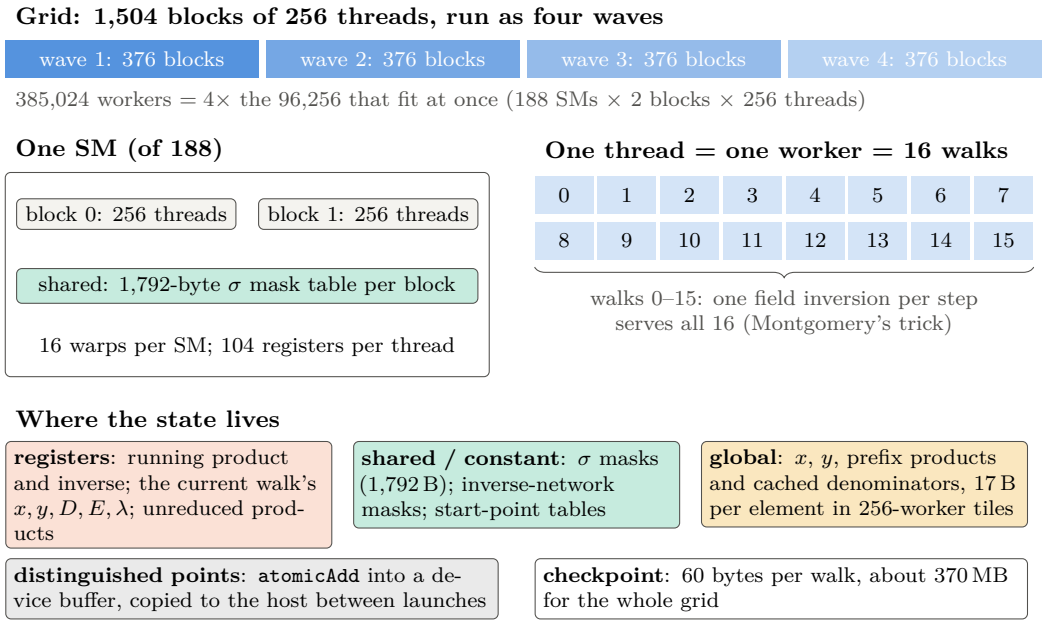


Figure 4: Anatomy of the audited σ-walk kernel on the RTX PRO 6000. Each thread owns 16 walks that share one field inversion per step; two 256-thread blocks fit per SM, and the 385,024-worker grid runs in four waves. The 1,024-step loop runs inside the kernel; every step streams each walk's compact state through global memory, and the distinguished-point buffer is drained between launches.

that an Ada SM would retire integer logic at half the per-clock rate of `sm_120`. The L40S fits that prediction: with the software product it reached 34.4 million iterations per second per SM, 0.427 of the RTX PRO 6000's 80.4 on its shipping build, and `CLMAD` took it to 62.2 (0.774) [E22]. Because the walk is bound by integer logic and the carry-less unit, not by memory bandwidth or tensor cores, the part that wins is the one with the most SMs at the highest clock for the money, which on today's price lists is the RTX PRO 6000 rather than the B200 (Section 8.8).

8 GPU implementation

On the GPU the exponent is fixed, so the only question is throughput. As everywhere in this paper, rates are in billions of iterations per second on one card. An iteration is one complete update of one walk, including its share of the batched inversion, the distinguished-point test and the restart path. Every row here is engineering: the walk and its collision structure stay the same, and the ratio to the generic-group floor does not move. What does move is the ratio to a second boundary, the GPU's carry-less unit, which we derived in Section 8.1 before tuning the kernel toward it.

8.1 Performance floor and roofline

Any walk that keeps the rho collision structure performs one affine addition per iteration. With Montgomery's batched inversion [40] across W walks, that addition costs a fixed number of field products plus $1/W$ of an inversion. In the packed representation a 131×131 product is a two-way Karatsuba over 64×64 carry-less multiplies, which is six `CLMAD` instructions. At batch $W = 16$ the shipping kernel issues 77 products per 16 slots, or 4.81 per update. Table 6 prices one update on the RTX PRO 6000 (188 SMs) [E10, §1].

The two pipes are independent, and each sets its own floor. On the carry-less unit, 28.9 `CLMAD` at 1.65 lanes per SM-clock is 17.5 SM-clocks per update, a ceiling of 24.7–26.1 billion

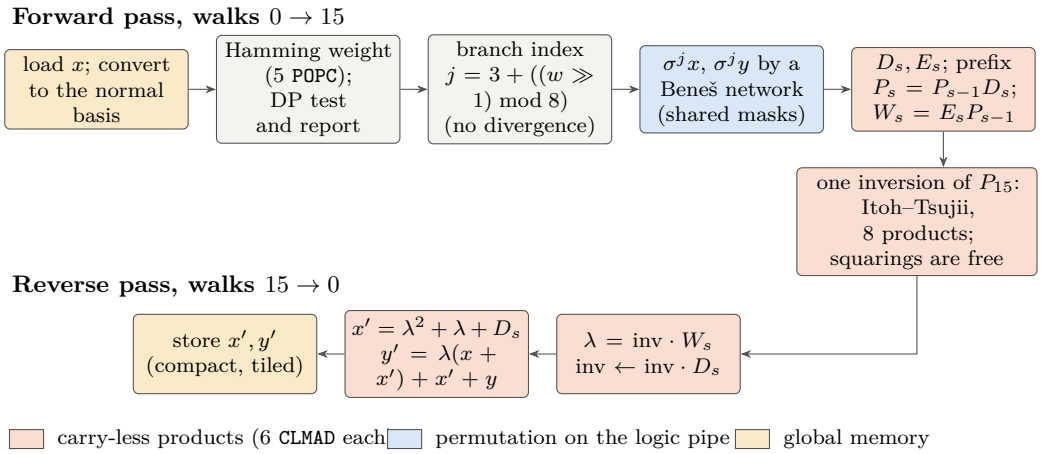


Figure 5: One step of the σ-walk for the 16 walks of a thread. The forward pass computes each walk's denominator and a running prefix product; one inversion of the final product then serves all 16; the reverse pass peels off each walk's inverse and finishes the affine addition. Per walk that is 4.81 products of six `CLMAD` each plus a sixteenth of the inversion; everything else, including every Frobenius power, runs on the logic pipe.

Table 6: Arithmetic floor of one affine-addition update on the RTX PRO 6000, and the budgets three throughput targets allow at 100% pipe utilisation. The carry-less unit issues 1.65 lane operations per SM-clock (1/37.7 of the logic pipe's 62.1, measured on an RTX PRO 4500 with the same SM and later confirmed on the 6000 at 1/38.0); sustained clock 2.30 GHz, maximum 2.43 GHz. Both floors sit above the 28 billion iterations/s budget.

component	`CLMAD`/upd.	ALU slots/upd.
4.81 products (6 `CLMAD` each)	28.9	≈ 830
squarings	5.0	≈ 80
batched inversion (/16)	4.6	≈ 270
floor	38.5 (28.9, squarings on ALU)	1,090–1,180
budget at 14.64 billion iterations/s	48.7–51.4	1,834–1,938
budget at 20 billion iterations/s	35.6–37.6	1,343–1,418
budget at 28 billion iterations/s	25.4–26.9	959–1,013

iterations/s even if everything else were free. On the logic pipe, about 1,090 ALU slots already exceed the 959–1,013 that 28 billion iterations/s would allow. At 90% utilisation of either pipe the ceiling is 22.2–23.4 billion iterations/s. The note was written to decide whether a different iteration function could take one card from the audited 14.64 to 28 billion iterations/s. The answer came before any kernel was built: it cannot, because 28 billion iterations/s is below the floor of any walk that performs one affine addition per iteration.

Our estimate of the carry-less rate went up as the kernels got closer to it. A kernel-mix measurement on the 6000 at 1.62 lane-`CLMAD`s per SM-clock priced the 20 billion iterations/s table-walk kernel at 33.1 `CLMAD` per update (28.9 products, 3.0 inversion, 1.25 normal-basis squarings) and gave a floor of 22.3 billion iterations/s [E24, §1]. A screened (not yet verified) three-limb Karatsuba arm then implied at least 1.654 lane-`CLMAD`s per SM-clock, which refuted that figure. The roofline of the 20.078 billion iterations/s build (Table 7) puts the carry-less speed of light at 27.4 billion iterations/s with the unit 2.00 lanes wide, while the logic pipe, at 1,319.6 lane-instructions per update, is 91.2% busy and runs out first [E19]. Thirty billion iterations/s would need at most 30.3 `CLMAD` and 968 ALU slots per update, a 27% cut in integer work that the note calls "a two-card number". The 28 billion iterations/s verdict survives the

Table 7: Roofline of the 20.078 billion iterations/s table-walk build on the RTX PRO 6000: 188 SMs at 2.415 GHz, 22.61 SM-clocks per update. Work per update is measured lane-instructions; the ceiling is the rate at which that unit alone would saturate.

unit	peak lanes/SM-clk	work/update	ceiling (billion it/s)	busy
ALU (logic)	64	1,319.6	22.0	91.2%
`CLMAD`	2.00	33.1	27.4	73.2%
issue	128	1,698.9	34.2	58.7%
load/store	16	80.0	90.8	22.1%
FMA	64	188.6	154	13.0%

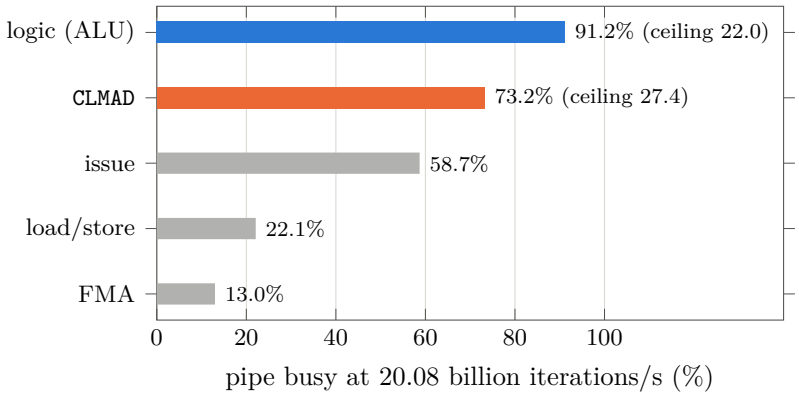


Figure 6: The roofline of Table 7 as a picture: the logic pipe is the binding constraint for the table walk's arithmetic on the RTX PRO 6000, with the carry-less unit second; memory and the FMA pipe are far from busy. In brackets, the rate at which each unit alone would saturate, in billions of iterations per second.

revision because it never rested on the carry-less unit alone.

The same note measures the carry-less unit on other dies: 37.8 logic slots per `CLMAD.lo` on the 6000, 3.78 on the B200 and 2.00 on the H100. On the data-centre parts, then, the logic pipe binds by a wide margin, and on the B200 the carry-less floor alone would be 255 billion iterations/s [E24, §6.1–6.2].

We had to correct the static accounting behind Table 6 once. An earlier SASS counter reported about 1,570 ALU slots and 29 `CLMAD` per update; its replacement counts 2,271 ALU and 45 `CLMAD` statically, or 2,324 once `POPC` and `FLO` are priced at their measured quarter rate [E10, §3]. The floor argument does not depend on which count is used, since the budgets are below both.

8.2 Acceptance criteria

With two exceptions named in Section 8.9, every step in this section is a paired comparison on one GPU allocation: the candidate and control are interleaved, three to six repetitions each, and we report the candidate's ratio to the control for every pair. We adopt a change only if it clears a rule declared before the run. For most of the ledger that rule is a gain of at least 1% on both the benchmark and the DP34 collection workload, with every pair favouring the candidate. For the later table-walk work it is a geometric-mean ratio at or above a stated threshold, with every pair above a stated minimum [E27; E28]. Every adopted kernel re-walks every device report on the host reference (300 of 300 in the later notes) and recovers planted logarithms on $\mathrm{GF}(2^{41})$. A candidate that fails the rule stays in the tree behind a flag, with its ratio on record; Section 8.9 collects them.

Table 8: Packed-backend rungs on one RTX PRO 6000. Each gain is the note's own paired comparison against the adjacent control, not a ratio of non-adjacent headlines. Blocks/SM is the resident-block bound; "conf." marks a comparison that also changed occupancy, compiler or walk population (the packed row's control ran 49 M walks once and its candidate 1.5 M walks three times). Class: engineering.

rung	control	candidate	gain
bitsliced control	—	0.852294	baseline
packed normal basis, five words (conf.)	0.852294	2.957961	3.47×
single polynomial product, 2 blocks/SM	2.956981	3.366572	+13.85%
denominator cache, 4 blocks/SM	3.506509	4.024103	+14.76%
same, 6 blocks/SM (conf.)	3.506509	4.094367	+16.76%
Frobenius permutation networks	4.115575	5.811421	+41.21%
polynomial-basis batch chain (conf.)	5.832342	6.054107	+3.80%
paired products (conf.)	6.123614	6.215430	+1.50%
polynomial-basis state	6.301859	6.547850	+3.90%
direct reduction	6.826154	6.906059	+1.17%
generated straight-line product (conf.)	6.860749	6.924275	+0.93%
256-worker tiles	6.852888	6.994745	+2.07%
native `clmad` (CUDA 13.3)	7.110440	8.703518	+22.40%
batch 16, 385,024 workers	8.673447	13.206088	+52.26%
weighted prefixes, paired Frobenius	13.110335	13.323276	+1.62%
compact 17-byte state	13.548376	14.403112	+6.31%
shared-memory mask table	14.296584	14.411102	+0.80%
public audit of that build	—	14.637530	DP34 14.106673

8.3 Optimisation ledger

The original client was bitsliced, with 32 walks in the bit lanes of a word and every field operation a straight-line program of `LOP3` instructions (Section 5.1). When we audited the packed replacement, that control measured 0.852294 billion iterations/s on an RTX PRO 6000 [E16]. Table 8 traces the path from there to the audited 14.637530 billion iterations/s (14.106673 billion iterations/s while collecting at weight 34). Each step is the corresponding note's own paired comparison. They do not form a single chain at constant occupancy and compiler, and where a note changed more than one thing at once, the table says so.

Sources, one per rung: packed basis [E16]; single product [E29]; denominator cache [E30]; Frobenius networks [E17]; polynomial chain [E31]; paired products [E32]; polynomial state [E33]; direct reduction [E34]; generated product [E35]; tiles [E36]; native carry-less [E18]; batch tuning [E37]; weighted prefixes [E38]; compact state [E39]; shared masks [E40].

Three steps account for most of the factor, and each is a change of representation rather than a tuning knob. The Frobenius permutation networks (+41%) replace the repeated squarings that computed σ^j with fixed Beneš permutation networks over the permuted normal basis, checked on 3,120 GPU vectors and on 256 basis vectors under all 131 powers. Native carry-less multiplication (+22%) replaces an emulated 32×32 carry-less product with the `clmad.lo/hi` instructions that CUDA 13.3 exposes (Section 7.1). That change in turn let the batch drop from 32 to 16 walks per inversion with 385,024 workers, four times the 96,256 that fit on the card at once, so the grid runs in four waves (+52%), at the same two-block occupancy and the same 122 registers per thread. The note suggests a cache-capacity explanation (the hot state shrinks from 245 to 122 MiB) but did not measure it [E37]. Between the Frobenius-network confirmation (5.81 billion iterations/s) and the shipping walk (14.64 billion iterations/s), the gain is those two changes plus layout work. The table gives each its own pair, and we do not quote any single "remaining factor" as if it had been measured.

The σ-walk did not stop at 14.64. A fused single-pass variant measured 15.436677 against

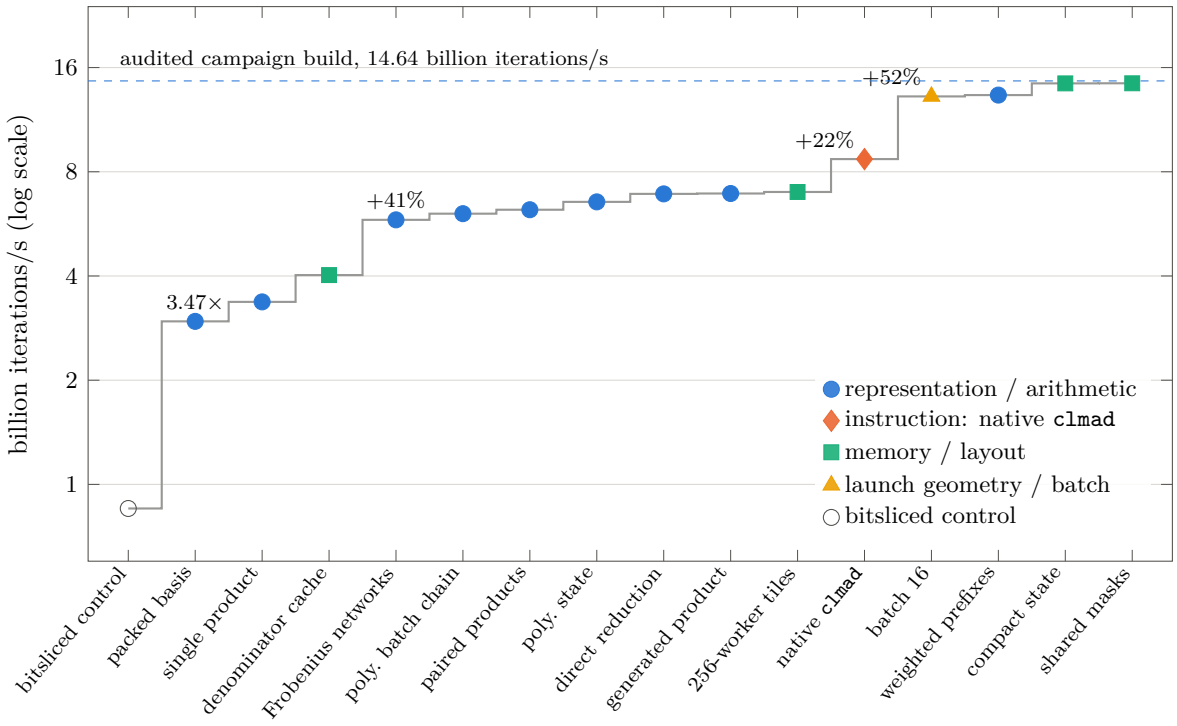


Figure 7: The packed-backend ledger on one RTX PRO 6000 (Table 8), each step's candidate rate, with the marker showing what kind of change it was. No step changed the walk: the iteration function, the step count and the distinguished points are the same throughout. Steps are separate paired comparisons, not one chain, so the line joins candidates measured in different sessions.

15.015004 billion iterations/s (1.028×, five pairs). A σ step through nibble tables in shared memory (68,352 bytes, 512×1 blocks) measured 15.884369 against 15.513653 billion iterations/s, a paired median of 1.023930 with every pair at or above 1.0238 and an A/A drift of 0.099%. That clears a pre-registered gate of geometric mean ≥ 1.015 with every pair ≥ 1.005 [E9; E28]. These are the current σ-walk headlines. The campaign configuration of Section 13 was frozen before they existed and still ships the audited 14.64 build.

8.4 Sources of the speedup

None of the sixteen steps changed the algorithm. The iteration function $R \mapsto \sigma^j(R) + R$, the branch rule, the distinguished-point rule, the seeds and therefore the expected number of iterations are those of Bailey et al. at every step; from the direct reduction onward each adopted build produced the byte-identical sorted corpus of 5,149 distinguished points on the same workload, and the earlier steps matched their controls' checkpoints byte for byte [E39; E40]. What changed is how a step is computed, and Figure 8 splits that by kind. Representation and arithmetic (the packed basis, one product instead of two, the Frobenius networks, the polynomial-basis state and its reduction) account for 6.3×; launch geometry and batch size for 1.5×; memory layout (the denominator cache, tiles, compact state and shared masks) for 1.3×; and the one new hardware feature, `clmad`, for 1.2× directly, plus the batch change it enabled. Figures 4 through 9 show the resulting kernel. The changes that do alter the walk, the table walk and the RTX PRO 4500's bridge map, are kept separate and labelled as iteration-function changes, because they change the walk constant as well as the rate (Section 4.5).

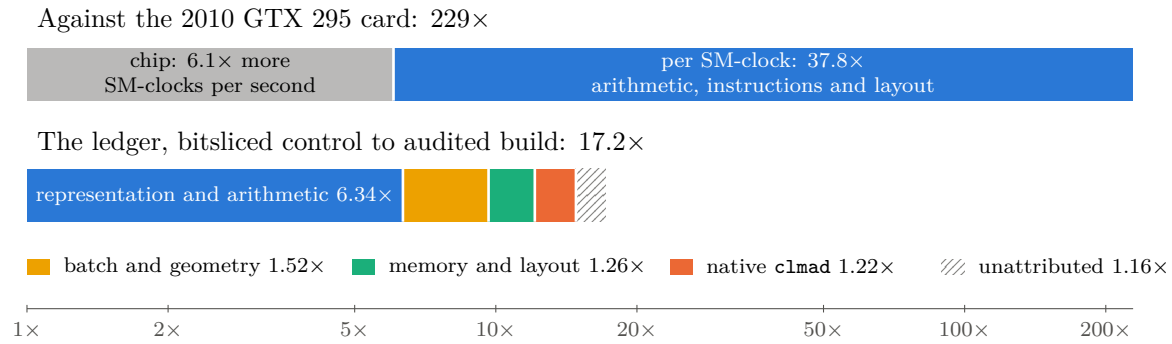


Figure 8: Where the speedup came from, on a log scale so that segment widths multiply. Top: against the 2010 GTX 295 card, about 6× is a bigger, faster chip and about 38× is per-SM-clock work. Bottom: the sixteen paired ledger steps grouped by kind (Figure 7); their gains multiply to 14.8× against the 17.2× from the bitsliced control to the audited build, and the remaining 1.16× is steps taken outside the ledger (launch shape, a compiler upgrade) and session-to-session drift. None of it changed the rho algorithm.

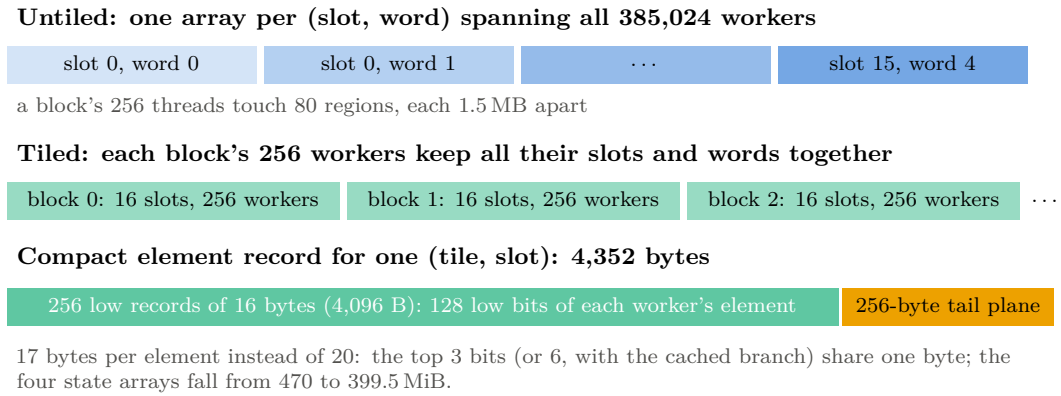


Figure 9: How walk state is laid out in global memory. Tiling puts all of a block's state in one contiguous region while each warp still reads adjacent words (+2.1%); the compact record then packs each 131-bit element into 17 bytes (+6.3%). Neither changes a single bit of the walk.

8.5 Table-walk arithmetic

This subsection measures how cheap the table walk's step can be made. Every rate in it predates the exact cycle rule v3, so it describes the arithmetic rather than a walk we would run today; Section 8.6 explains why.

The σ-walk spends part of every step on σ^j and the basis conversions around it. A redesigned r-adding table walk (Section 4.2) removes both. Each step adds a precomputed table point $\pm\sigma^k(T_h)$ whose Frobenius power is a shared-memory lookup, and the static ALU count drops from 2,324 to 1,831 slots per update. On the same host and automatic geometry it measured 16.560337 billion iterations/s against 14.412634 billion iterations/s for the shipping walk (+14.9%, six alternating repetitions, every paired ratio between 1.146 and 1.169). In the audited 385,024-worker geometry it measured 16.351936 against 14.975448 (+9.2%). All 300 device reports re-walked correctly, and all 64 planted logarithms were recovered on GF(2^{41}) [E10, §6.2–6.3; E41].

Before the build, the note had set itself a target of 17.0 billion iterations/s (§4.4). Both medians missed it, by 2.6% and 3.8%, so under its own rule the table walk was not made the campaign default; it remains available behind `WALK_TABLE=1`. This is the acceptance rule doing its job, even when the result is a real speedup that falls short of the bar we set.

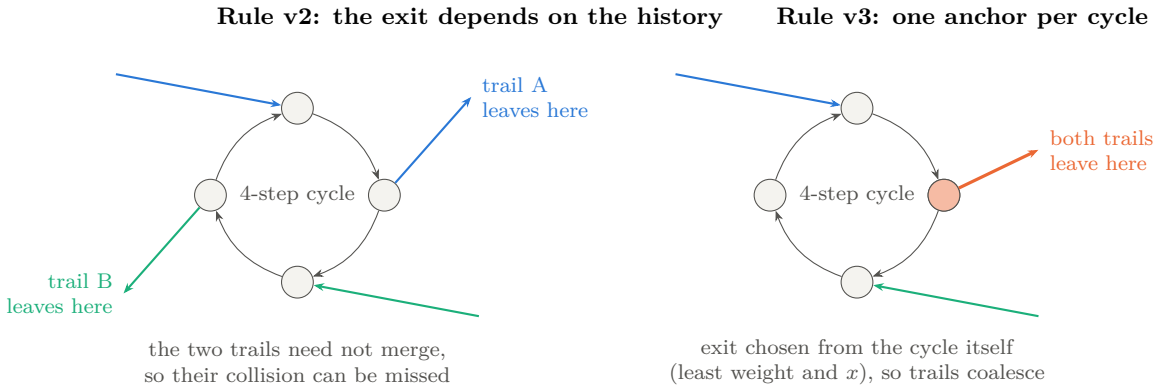


Figure 10: Why the table walk's 20.08 does not count. A table walk on Frobenius classes has short fruitless cycles that a trail must detect and leave. Under rule v2 the exit depended on how a trail arrived, so two trails entering the same cycle from different directions could leave by different edges and so fail to merge. Rule v3 confirms the cycle by walking it (at most eight raw steps) and leaves at one anchor chosen from the cycle's own points, so every trail entering it leaves the same way. The σ-walk has no fruitless cycles at all.

As first built, the table walk also changes the walk constant for the worse. Section 4.4 measures its cycle trapping at dpWeight 32 and prices the as-built kernel at 4.85–6.26 times the σ-walk's cost per solve; a revised cycle rule is projected, but not yet measured, at 0.81–0.86. The throughput rows here are therefore rates, not costs per solve.

With a different geometry the same walk goes further. A one-block layout (512 threads and one resident block per SM, which gives each block 64 KB of L1 instead of 28 KB), denominators rebuilt from the step tag instead of stored, inlined products and a pipelined selection pass took a 17.406 billion iterations/s reference to 20.078 billion iterations/s (+15.3%, five samples in 20.077–20.080). That is 0.73 of the 27.4 billion iterations/s carry-less speed of light (0.90 of the since-refuted 22.3). The geometry alone was worth +9.4%, tag denominators +1.5% and inlining +2.3%. Collecting at weight 34 it holds 19.04 billion iterations/s against 16.70. All 300 reports re-walked correctly, and the distinguished-point count matched the reference's 472,776 [E42]. This is the `gpu-rtx-pro6000-20b` build, and it ran cycle rule v2.

8.6 σ-walk versus table walk

Readers who skim the tables will see 20.08 next to 15.88 and conclude that the table walk is the faster walk on Blackwell. Today it is not, and the reason is correctness rather than arithmetic.

The table walk adds a precomputed point instead of a Frobenius image of itself, which removes the basis conversions around σ^j and makes each step cheaper. The price is fruitless cycles: short loops in which a trail returns to a class it has already visited and makes no progress (Section 4.2). The σ-walk has none at all. Every table-walk rule is a way of noticing such a loop and leaving it, and the rules have had three versions:

- *v1*, as first built, let some cycle patterns through. The kernel ran at 16.56 billion iterations/s on the RTX PRO 6000 and 5.55 on the RTX PRO 4500, but at campaign trail lengths it is modelled to trap about half of all trails, so its cost per solve is 4.85–6.26 times the σ-walk's.
- *v2* closed those patterns, and the one-block build ran it at 20.08 on the RTX PRO 6000; the B200 sweep reached 19.40 on a rule older than v3. A review then found that v2's exit from a cycle depended on the history a trail arrived with. Two trails entering the same cycle from different directions could leave it by different edges, which undermines the collision structure that rho relies on.

- *v3* fixes that. On a hint it follows at most eight raw steps, confirms an exact cycle, and leaves at one anchor chosen from the cycle itself, so every trail entering a cycle leaves it the same way. The probe sits on the critical path, and the exact v3 kernel currently runs at 5.10 billion iterations/s on the RTX PRO 6000 [E13; E1, "The earlier 20.078 B/s table result"].

So the honest summary is this. The σ-walk is the fastest *correct* walk we have on Blackwell, at 15.88 billion iterations/s on one RTX PRO 6000 and 14.64 in the audited campaign build. The table walk is the faster *arithmetic*: its v2 rate shows what the step costs without the cycle probe, and that is the target for a faster exact rule. Even a fast exact table walk only pays if its cost per solve, which also depends on its walk constant, beats the σ-walk's; under v2 that was projected at 0.81–0.86× and has never been measured. Section 17.1 lists the routes we see.

8.7 RTX PRO 4500

The campaign's second GPU family is the RTX PRO 4500 (82 SMs, 165 W) in the AWS `g7.2xlarge`. We keep its ledger separate because on this card the power cap binds, not the pipe. The shipping walk measures 5.107 billion iterations/s here at 1.95 GHz, within about 1% of what the 6000's per-SM-clock rate predicts (Section 13.3). The card's own best preset before the map change sat near 6.2 billion iterations/s. Candidates such as the larger split batches were slower here because they raised memory activity, and the power cap answered by lowering the clock [E10, §6.3; E1, "larger split-batch experiment"]. We had declared a goal of 12 billion iterations/s for this card (Section 13.3), and it was reached by changing the map rather than the kernel. The complete two-bridge x-only map of Section 4.3 measured a median of 12.499203 billion iterations/s against 11.218437 billion iterations/s for its matched modulus-32 control, a paired geometric-mean ratio of 1.121079 with 95% interval [1.105668, 1.136704]. A fresh three-run confirmation gave 12.567029, 12.457221 and 12.366415 billion iterations/s, and the production build ran at 12.632704 billion iterations/s over 2^{36} updates [E1, "ECC2K-130 and ECC2K-95"; E11]. We have not measured its DP34 collection rate on this card, so we do not give one.

8.8 Cross-GPU comparison

Table 9 puts every part we measured side by side, with the σ-walk and the table walk in separate columns. The σ-walk column is one survey run of the same packed preset on each Modal SKU (automatic workers, three valid `-bench` repeats, unpaired); the next column gives the best paired result on that part, where one exists. The table-walk rates were all measured under cycle rules older than the current v3, for the reason given in Section 8.6 [E43; E1, "Table-walk correctness update"].

Three things stand out. First, nothing beats the RTX PRO 6000 on this walk, per card, per SM or per dollar. It has the most SMs, a high clock and Blackwell's unified integer lanes, and the walk is bound by integer logic and the carry-less unit rather than by memory or tensor cores, so the usual rankings of GPUs for machine-learning work do not carry over: the B200 runs the σ-walk at 0.585× the 6000 and the H100 at half. Second, per SM, the Blackwell workstation die leads the data-centre parts even though its carry-less unit is the slowest in the table (Section 7.2), because the shipping walk was tuned for exactly that balance. Third, the data-centre dies reward a different kernel. With a table walk tuned for its fast carry-less unit the B200 reaches 19.40 (under a pre-v3 cycle rule), more than twice its σ-walk rate.

On the B200, an automatic sweep over twenty single-knob arms (launch geometry, product form, scheduling, memory residency) later lifted the table-walk build from 15.365 to 19.402 billion iterations/s (1.2627, every pair in 1.2623–1.2632). The winning combination of the fused pass, batch 32, three-limb Karatsuba and 512-thread blocks was adopted as `gpu-b200-19b`. The same sweep on the 6000 found a 640×1 arm at 19.993 against 19.951, with two of five pairs

Table 9: Rates in billions of iterations per second on one GPU. "σ survey": the shipping packed preset, one unpaired Modal survey run (the RTX PRO 4500 row is its AWS measurement). "Best paired": the best paired result on that part. "Table, pre-v3": table-walk rates under cycle rules v1 or v2, which are superseded (Section 8.6); the exact v3 table walk measures 5.10 on the RTX PRO 6000. "M/SM": million iterations per second per SM, our quotient of the survey rate by the SM count. "T/$": trillions of iterations per dollar at Modal list price, 2026-09-19.

GPU	arch.	SMs	σ survey	best paired	table (pre-v3)	M/SM	T/$
RTX PRO 6000	Blackwell `sm_120`	188	15.116	15.884 (σ, nibble table)	20.078 (v2)	80.4	17.95
RTX PRO 4500	Blackwell `sm_120`	82	5.107	12.499 (x-only bridge)	5.552 (v1)	—	—
B200	Blackwell `sm_100`	148	8.840	—	19.402	59.7	5.09
B300	Blackwell `sm_103`	148	7.575	—	—	51.2	3.84
H200	Hopper `sm_90`	132	7.632	—	—	57.8	6.05
H100	Hopper `sm_90`	132	7.540	—	—	57.1	6.87
L40S	Ada `sm_89`	142	8.696	—	—	61.2	16.04
L4	Ada `sm_89`	58	2.561	—	—	44.2	11.54
A100-80GB	Ampere `sm_80`	108	4.632	—	—	42.9	6.67
A100-40GB	Ampere `sm_80`	108	4.618	—	—	42.8	7.92
A10	Ampere `sm_86`	72	2.423	—	—	33.7	7.92
T4 (no `clmad`)	Turing `sm_75`	40	0.541	—	—	13.5	3.30

below one, and adopted nothing [E26]. The trades that change sign from one die to the other are exactly those that move work from the logic pipe onto the carry-less unit. The top-bit correction on `CLMAD` costs 15% on the 6000 but gains 13.4% on the B200, and stacked there with the normal-basis inversion and λ^2 on `CLMAD` it reaches 1.379× (15.372 billion iterations/s), just as the per-die `CLMAD` rates of Section 8.1 predict. A two-chain kernel that splits each warp's Montgomery chain in two lost on both dies (0.826× on the B200, 0.696× on the 6000), which points to the register file rather than to any unit's rate [E24, §5, §6.2 and §6.4].

8.9 Rejected optimisations

Table 10 lists every GPU candidate that we built, measured against the rule, and did not adopt. We record them so that anyone who has the same idea later finds a measurement rather than folklore. Every row passed its correctness checks; what fell short was the rate.

Sources: stream Karatsuba [E44]; add-combine [E45]; top-`CLMAD` [E25]; batch screen [E39]; two chains [E24]; λ-projective [E46]; the 4500 rows [E1, "ECC2K-130 and ECC2K-95"], each with its own benchmark record.

Two rows need a word of explanation. The λ-projective candidate was never built. Priced in the unit of Table 6, it came to 60.5 `CLMAD` per update against 38.1 for affine at batch 16, a ceiling of 11.8–12.5 billion iterations/s, below the 14.41 we had already measured. The condition we had declared for building it (a negative cost difference at $W = 16$) could not be met, so we did not start. The top-`CLMAD` and stream-Karatsuba rows are the two comparisons in this section that were not interleaved on one allocation (the stream-Karatsuba pair also differs in source digest). We report both with that caveat; neither would have cleared the rule in any case.

9 CPU implementations

The CPU client matters for three reasons. Every GPU report is verified on it, it runs the ECC2K-95 end-to-end control, and the campaign's spot fleet included CPU workers whose points cannot be told apart from the GPUs'. It is also where the instruction-set story of Section 6 can be measured most directly, because the same source compiles to scalar 64-bit words, AVX2, AVX-512 and NEON.

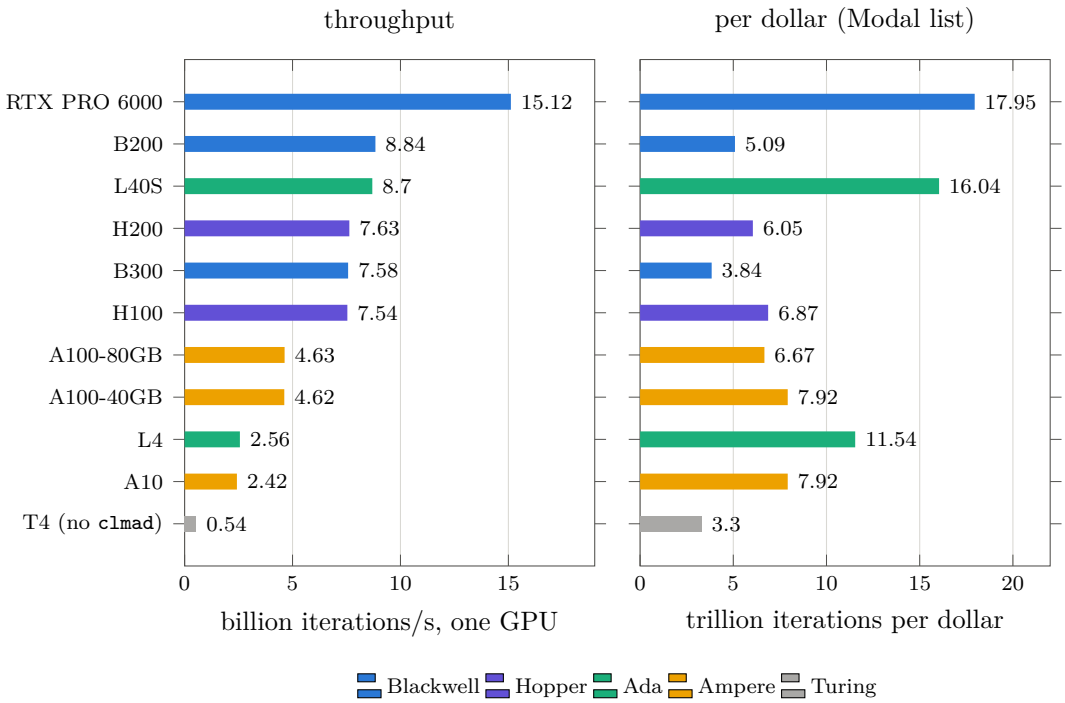


Figure 11: The campaign's σ-walk on eleven GPUs, one survey run each with the same preset (Table 9), by throughput and by iterations per dollar at Modal list price. The workstation Blackwell part leads on both; the data-centre Blackwell and Hopper parts run this integer-bound walk at about half its rate and a third of its value.

9.1 Bitsliced vector implementations

Table 11 collects the measurements [E1, "Throughput"], and Figure 12 draws them. A bitsliced client gains from a CPU's vector extensions in two ways: wider words carry more walks per instruction, and three-input logic fuses the XOR that follows every AND. On x86 the step from 64-bit words to AVX2 is 2.4× and AVX-512 adds another 1.18×; AVX-512's `vpternlogd` is the CPU counterpart of the GPU's `LOP3`. On Arm the gain splits cleanly. On the M4 Pro the 128-bit NEON word with `BSL` alone is worth 1.76× and `EOR3`, from the SHA3 extension, a further 1.24×; a core without `FEAT_SHA3` (Graviton2) still gets 1.81× from the wider word and `BSL`. On this code the compiler matters as much as the instruction set: gcc emits 9,372 instructions for the multiply leaf, 59% of them loads and stores of spilled temporaries, where clang emits 1,902 plus five outlined calls. We enable the SHA3 path only when the CPU reports the feature: an ungated probe on a Graviton2 picks a `+sha3` spelling under both compilers, and the binary dies with `SIGILL` on its 8,907 `eor3` instructions [E1, "Word width"].

The AVX-512 build's limit is data movement, not arithmetic. It spends about half of each multiply on moves (4,069 logic instructions against 4,287 `vmovdqa32`), because the Karatsuba leaf keeps 254 values live with only 32 vector registers. A scheduling pass over the same DAG recovered moves only in the conversion (4,287 to 3,903) and missed its declared target; repeated single-thread runs in a shared container ranged from 15 to 18.6 million iterations per second, too noisy for a throughput claim [E47].

9.2 Carry-less multiplication on CPUs

Every current CPU family also has a carry-less multiply: PCLMULQDQ and its 512-bit form VPCLMULQDQ on x86, PMULL on Arm. The bitsliced client deliberately does not use it. Its multiply leaf costs about 11 instructions per walk amortised over 512 lanes, while a per-walk

Table 10: GPU candidates measured and declined. Ratio is candidate over the paired control; intervals are 95% paired bootstrap where the note reports one. "4500" rows are on the RTX PRO 4500 against the selected runtime of the day; the others are on the RTX PRO 6000.

candidate	control	candidate	ratio
stream Karatsuba compile option	0.857	0.748	0.873
integer add in place of OR in `clmul32`	6.8615	6.8080	0.992
top three-bit correction moved onto `CLMAD`	15.116	12.859	0.851
batch 12 / 10 against batch 16	14.26	13.75 / 13.70	$<$ screen
batch 18 against batch 16 (larger population)	14.47	14.29	$<$ screen
table walk against pre-declared 17.0 billion iterations/s	14.41	16.56	1.149, missed
two Montgomery chains per warp	19.981	13.903	0.696
λ-projective coordinates (priced, not built)	—	ceiling 11.8–12.5	< 14.41
4500: hybrid reduction	6.2459	5.3459	0.852 $[0.845, 0.860]$
4500: half-native product tail	6.1898	6.0303	0.977 $[0.971, 0.982]$
4500: quotient fan-in	6.1739	6.0139	0.977 $[0.972, 0.982]$
4500: register-retained inverse, 3 / 4 blocks	6.1724	6.1487 / 5.6922	0.997 / 0.924
4500: paired inverse leaves, 3 / 4 blocks	6.1643	6.1146 / 5.6501	0.994 / 0.918
4500: logical-pair inversion, 3 / 4 blocks	6.1675	6.1037 / 5.6723	0.960 / 0.921
4500: seven-node inversion, 3 / 4 blocks	6.1826	6.1388 / 5.6904	0.995 / 0.923
4500: seven-node shared-X cache, six slots	6.2624	6.3741	1.010 $[0.991, 1.029]$
4500: split batch 24 / 32	6.2742	6.1218 / 5.9950	0.972 / 0.948
4500: direct polynomial δ-3/4 routing	6.2724	4.3635	0.690 $[0.676, 0.704]$
4500: 65/66-bit, 33-bit products; byte masks (four)	—	—	all < 1

131-bit carry-less product needs about nine `vpclmulqdq` plus a reduction for a single walk [E47, §7]. A packed client is the other design, the CPU twin of the GPU client: one walk per lane of 64-bit limbs, the product on the carry-less multiplier. The `cryptanalysis` repository has one, using PMULL on Arm and PCLMULQDQ on x86. On one M4 Pro core it runs an iteration in 53.5 ns, 18.6 million iterations per second, against 153 ns with the software product in the same engine, a 2.9× gain from the instruction; all 14 cores give 139–159 million. It walks the table walk, not the campaign's σ-walk, and has not been measured on x86 at all [E86]. On the same M4 Pro the integrated GPU, through Metal and a software product, reaches 389–390 million.

Two other CPU instructions were priced or measured on smaller binary fields. GFNI's affine transform (`vgf2p8affineqb`) speeds up a basis change 3–5× for fields up to 2^{63} in the repository's index-calculus code, but for ECC2K-130's bitsliced conversions it was priced as a loss, since it acts on bytes inside a register where the bitsliced maps act on whole words [E87]. PCLMULQDQ made a 41-bit Koblitz rho 1.76× faster end to end [E88].

9.3 Prime-field arithmetic on CPUs

The repository's prime-field CPU solver is a reference implementation, not a fast one: it uses heap-allocated big integers, not four 64-bit limbs with `MULX`/`ADX` carry chains or AVX-512 IFMA, and manages about 32,000 affine additions per second per thread on P-256 [E52]. That is the CPU counterpart of the gap the GPU closed with explicit carry chains (Section 10), and it is listed as an open question in Section 17.6.

9.4 CPU contribution to the campaign

In the campaign a CPU worker was a whole multi-core instance, and the twelve CPU slots self-reported a median of 2.3 billion iterations per second on 2026-09-17, about a sixth of an

Table 11: Bitsliced CPU client, GF(2^{131}), one core, median of five; M it/s is million iterations per second. Speedups are against the g++ 64-lane build on the same host. The Xeon's model is not recorded beyond its clock, and the record does not say whether the AVX2 row was built with AVX-512VL's ternary logic on 256-bit words. The 2009 qhasm implementation of Bailey et al. reached 533 cycles per iteration on a Core 2 with 128-bit vectors.

host	word	lanes	M it/s	note
Xeon 2.1 GHz	`uint64_t`	64	4.42	475 cycles/iteration
Xeon 2.1 GHz	AVX2	256	10.7	196 cycles/iteration
Xeon 2.1 GHz	AVX-512	512	12.6	167 cycles/it.; 4 threads 40 M
M4 Pro, g++ 16.1	NEON	128	11.342	2.19× over 64 lanes (5.185)
M4 Pro, Apple clang 17	NEON	128	15.673	3.02× over g++/64
Graviton3 (V1, SHA3), clang	NEON	128	6.214	2.38× over g++/64
Graviton2 (N1, no SHA3), clang	NEON	128	4.220	2.09× over g++/64
Modal container, 4 threads	AVX-512	512	12.4	49.4 M on 4 threads

RTX PRO 6000. On Modal, a 32-thread (16-core) sidecar beside a GPU adds about 25% to the container's cost for 2.7% more iterations, so it is off by default [E48; E1, "Walking on the container's CPUs too"]. Per dollar, a CPU walker is roughly ten times worse than the GPU.

10 Application to NIST P-256 and P-384

Most of what made the ECC2K-130 client fast has nothing to do with binary fields. Sharing one inversion across many walks, using the hardware's own carry instructions instead of emulating them, choosing the launch shape by measurement, and refusing occupancy trades that look good on paper are all properties of an affine-addition rho walk on a GPU, not of $\mathbb{F}_{2^{131}}$. So once the ECC2K-130 ladder was written down, the natural question was how much of it carries over to a prime-field walk. We tried it on NIST P-256 and P-384, on the same RTX PRO 6000, and this section reports what happened.

Two things should be clear before the numbers. First, these are throughput measurements on a walk that nobody expects to finish: at the rates below a P-256 collision is still about $2 \cdot 10^{29}$ seconds away (Section 16.3), and nothing here says anything about the security of the NIST curves. Second, every row is engineering in the sense of Section 1. The walk computes the same function on the same curve at every rung, with one step per counted step, so $S = \text{steps}/\sqrt{n}$ does not move; only the cost of a step does [E49].

10.1 Walk and field arithmetic

The library holds a P-256 element in eight 32-bit limbs and a P-384 element in twelve, one element per thread, with CIOS Montgomery multiplication. Both primes have low limb $2^{32} - 1$, so the Montgomery constant is $n' = 1$, and every limb of either modulus in radix 2^{32} is 0, 1, $2^{32} - 1$ or $2^{32} - 2$; the reduction row therefore needs carries but no multiplications. A generalised-Mersenne (Solinas) field is kept alongside as a matched competitor, so that the device rather than an argument decides between the two [E50]. Both representations are checked on 20,000 random pairs per curve against Boost.Multiprecision, and the Jacobian doubling and mixed addition against independent affine arithmetic. The benchmark protocol was pre-registered before any device time: workloads, seeds, the threads-per-block and register-cap sweep, and a 5% threshold below which the simpler configuration wins by rule [E51].

The rho walk on top of the field mirrors the structure of Section 4, minus the Frobenius. It is an r-adding walk on the low bits of x with the negation map as the only fold (fold 2, a $\sqrt{2}$ saving; $j \neq 0$ on both curves, so ± 1 is the whole automorphism group), distinguished points by a

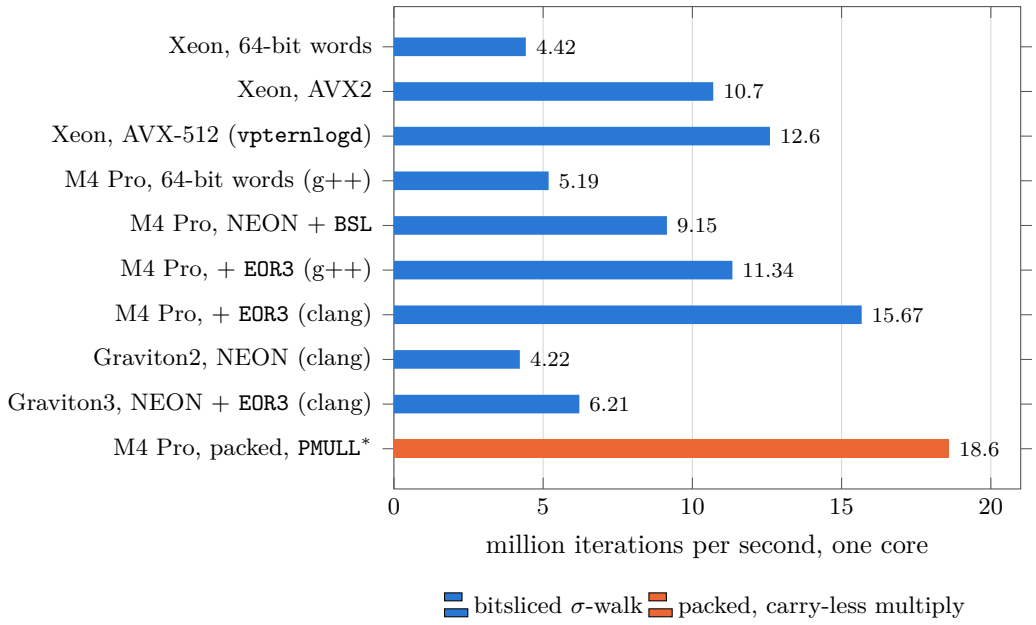


Figure 12: CPU rates per core by instruction set. Bitsliced rows are the campaign walk (Table 11); each step down a group adds one extension. *The packed client walks the table walk, so its row is not a paired comparison with the others.

mask on the next byte at probability 2^{-24}, fruitless two-cycles detected and escaped, and walks batched per thread so that one inversion serves all of them through Montgomery's trick [E50]. The inversion is Fermat's a^{p-2} with a four-bit fixed window, which for P-256 is on the order of 256 squarings and 64 window multiplications (our count from the exponent length, not a measured figure). That is the important difference from the binary field. On ECC2K-130 an inversion costs eight multiplications because squaring is free (Section 5); here it costs a few hundred, so the batch width W matters far more.

The host test suite runs the exact kernel body on the CPU: group-law identities, determinism of a single step, the negation fold as a class function, and an end-to-end logarithm on a toy curve of order about 2^{20}, where the folded walk needs the expected $\sqrt{2}$ fewer steps than the unfolded one [E50, "Correctness"].

The CPU solver across all fifteen curves. The GPU walk has a CPU counterpart: a host solver that takes any of the fifteen FIPS 186-4 curves (P-192 to P-521, K-163 to K-571, B-163 to B-571), audits it for the structural precondition of each known attack (trace one, composite order, embedding degree at most 100, composite extension degree), and runs the branch the curve calls for: Smart–Semaev–Satoh–Araki, Pohlig–Hellman, interval baby-step giant-step and kangaroo, or whole-group rho. The audit certifies the absence of every precondition on all fifteen. The rho folds the negation map everywhere and the Frobenius on the Koblitz curves, the class of size $2m$ carried exactly through the eigenvalue λ of τ on the subgroup, which is the binary-curve cousin of the σ-class walk of Section 4. On one host we measured interval solves of width 2^{20} to 2^{32} on P-256, K-283 and B-409 in 43 ms to 8.7 s with the planted scalar recovered every time; Frobenius folding on toy Koblitz curves cutting additions by the predicted $\sqrt{2m}$ (6.8× at $m = 23$, 9× at $m = 41$) at a cost of m squarings per step; and the Smart attack breaking CM-constructed anomalous curves of 129, 257 and 522 bits in 3, 10 and 33 ms. No full-width scalar has been recovered on any NIST curve, and none of the fifteen has a structural weakness; the documentation says both [E52].

Table 12: ECC2K-130 levers applied to the NIST P-256 and P-384 rho walk on one RTX PRO 6000. Rates are in billions of iterations per second. Each NIST ratio compares adjacent cells of the same sweep; the cells are single runs, not the interleaved repeated pairs of Section 8. Rows marked * come from the round-2 sweep. Class: engineering.

NIST rung	ECC2K-130 analogue	cells compared	P-256	P-384
first working build	bitsliced control	portable, 128 thr., $W = 8$	1.075	0.210
explicit PTX carry chains	native `clmad` (+22.4%)	same cell, PTX build	1.19×	1.38×
compile-time W, $W = 8 \to 16$	batch tuning (+52%)	256 thr., PTX	1.08×	1.37×
128 → 256 threads/block	launch geometry (+9.4%)	$W = 16$, PTX	1.03×	1.01×
$W = 32$, register-resident inversion*	batch tuning	256 thr., $W = 16 \to 32$	1.22×	1.31×
register caps 96/128/160*	two chains, blocks/SM (declined)	256 thr., $W = 32$	0.97×	0.82–0.83×
best cell*		PTX, 256 thr., $W = 32$	**1.632**	**0.548**
cumulative*		from the first working build	**1.52**×	**2.61**×

10.2 Baseline measurement

The kernel had been written and verified entirely on hosts without a GPU (Section 11.1), and the first device run measured nothing at all. The batched-inversion routine was marked host-only, and `nvcc`, after a warning, silently dropped the call from device code. The walk kernel compiled to a load and an exit: four registers, no stack, microseconds per launch and zero distinguished points. The CPU tests could not have caught it, because the host compiler happily compiles host-only code. A second bug, an unguarded `atomicAdd` on a null cycle counter, faulted on the first fruitless two-cycle, which arrives about once every 512 steps. Both are fixed and now covered by tests, and the device benchmark checks every launch for errors [E49, "What this round fixed"].

This is the same lesson that Section 5.1 drew from code size, in a different form: a static count or a host emulation tells you what the device ought to do, and only the device tells you what it did. Our correctness check on hardware is therefore the distinguished-point count. Because the walk is deterministic and points are distinguished with probability 2^{-24}, every cell must report about walks × iterations/2^{24} of them. Every cell did, for example 11,969 observed against 12,085 expected on the first P-256 run and 12,154 against 12,096 in the best cell [E49, "Correctness evidence on hardware"]. The CPU suite also checks the specialised steppers below bit for bit against the generic one, for every batch width that was compiled.

10.3 Optimisation steps

Table 12 lines the NIST rungs up against the ECC2K-130 rungs they borrow from. The starting point is the portable build at 128 threads per block and $W = 8$: 1.075 billion iterations/s on P-256 and 0.210 on P-384, both on one RTX PRO 6000 Blackwell (188 SMs, CUDA 13.3, `sm_120`).

Carry chains. On ECC2K-130 the single largest instruction-level gain was letting the hardware do the carry-less product itself. The prime-field counterpart is the carry: a portable C++

multi-precision product hides its carries in compares and adds, and Section 11.2 already showed that the 256-bit product is carry-bound rather than multiply-bound. Writing the product rows and the sparse reduction with explicit `mad.lo.cc`/`madc.hi.cc` chains lifted the walk by 1.19× on P-256 and 1.38× on P-384 at the same launch shape (1.282 and 0.289 billion iterations/s). The larger gain on P-384 is what one would expect from twelve limbs instead of eight: there are more carries per product to save. At the field level the same change took a dependent Montgomery multiply from 54.9 to 73.4 billion multiplications/s on P-256 and from 20.3 to 32.2 billion multiplications/s on P-384. Under the pre-registered protocol the Solinas competitor did not win either. On P-256 the two multiplications are within 5% (70.4 against 73.4 billion multiplications/s), a tie that the protocol awards to the simpler configuration. On P-384 the Solinas multiplication is 14% faster (36.7 against 32.2) but the Montgomery doubling is 4% faster (1,712 against 1,647 million doublings/s). These field rows are a first run at one block size, not the protocol's three repeats, so the question stays open [E49, "Field arithmetic"]. The walk is built on the Montgomery field for both curves; a Solinas walk has not been compiled or measured [E50].

Batch width. This is where the two fields differ most, and the reason is the inversion. On ECC2K-130 the batch went *down*, from 32 walks per inversion to 16, because an inversion was already cheap and a smaller hot state fitted the cache better. On the NIST curves an inversion costs hundreds of multiplications, so every extra walk sharing it pays. Templating the stepper on W at compile time let the per-walk arrays live in registers instead of 64-wide local arrays, and the rate climbed with W on every geometry: at 256 threads per block from 0.466 at $W = 2$ to 1.340 at $W = 16$ on P-256, and from 0.096 to 0.438 on P-384 (Figure 13) [E49, "Rho walk"]. A second round passed the inversion's scratch arrays by reference and compiled a $W = 32$ instance. That cell reached 1.632 and 0.548 billion iterations/s, 1.22× and 1.31× over $W = 16$ at the same geometry [E49, round 2]. We cannot cleanly split that gain between the larger batch and the by-reference inversion, because round 1 never compiled $W = 32$. What we can say is that at $W \leq 16$ the two rounds agree within noise (1.306/0.434 against 1.310/0.408 at 128 threads, $W = 16$), which suggests, without proving it, that most of the gain comes from the batch.

Launch shape. Doubling threads per block from 128 to 256 was worth 2.6% on P-256 and under 1% on P-384 at $W = 16$. That is small next to the one-block geometry's 9.4% on ECC2K-130, and it makes sense: the binary kernel was shaped by a shared-memory table and an L1 budget, while the prime-field walk is limited by registers.

Rejected changes. At $W = 32$ both instantiations sit at the 255-register ceiling. The standard counter-measure is a register cap, trading spills for occupancy, and the round-2 sweep tried caps of 96, 128 and 160 registers. All of them lost to the unrestricted build: P-256 fell to 1.589–1.595 and P-384 to 0.447–0.455. That matches the ECC2K-130 experience with the two-chain kernel and the four-block configurations of Table 10: on this card, occupancy bought with spilled state has not paid on either field.

10.4 Transfer from ECC2K-130

Three of the ECC2K-130 levers transferred directly. Native carry handling, batch width and the refusal of register caps all moved the NIST walk in the same direction they moved the binary one, and the combined effect was 1.52× on P-256 and 2.61× on P-384 over the first working build. The size of each lever did not transfer. The batch width, a modest knob on ECC2K-130, was the largest single factor here because a prime-field inversion takes some forty times as many multiplications as a normal-basis one. Launch geometry, a real gain on the binary walk, was

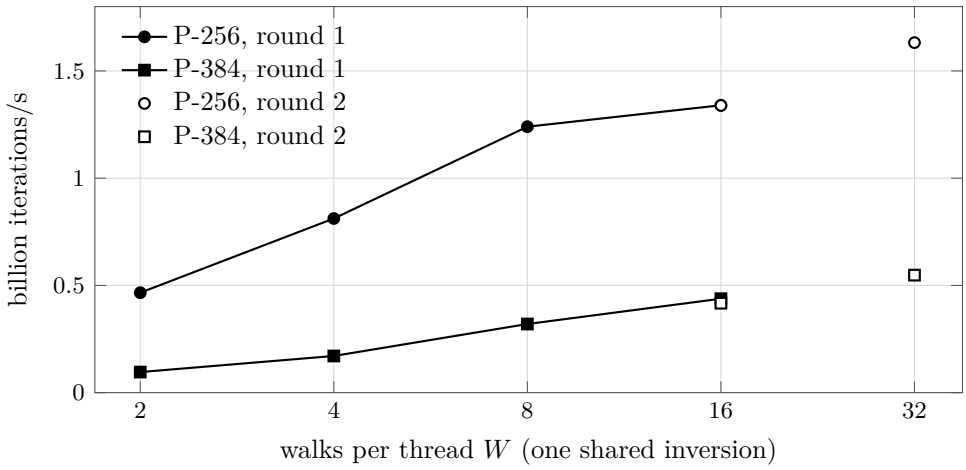


Figure 13: The NIST rho walk on one RTX PRO 6000 at 256 threads per block, PTX build, unrestricted registers, against the batch width W. Filled markers are the round-1 sweep, open markers the round-2 sweep with the by-reference inversion and the $W = 32$ instance. Each point is one cell of 8,192 iterations; distinguished-point counts matched the 2^{-24} rate in every cell.

nearly flat. In other words the ladder carried over, but its rungs were spaced differently, and the only way to find the spacing was to measure it.

The comparison also puts the two walks on one scale. On the same card the ECC2K-130 campaign build runs 14.64 billion complete iterations per second and the best P-256 walk 1.63 billion iterations, a factor of about nine. A 131-bit binary product is six `CLMAD` instructions; a 256-bit prime product is two hundred or so carry-chained integer instructions (Table 14). That factor of nine is the carry-less unit's advantage made concrete.

The next candidate steps are the ones the round-2 record names: a $W = 64$ instance (which will spill at the register ceiling and probably lose, but is worth one cell to confirm), reseeding off the critical path, and a second walk per step from the same table read. A repeated, interleaved pairing of the best cell against its control, in the style of Section 8, would also turn these single-run ratios into intervals. None of that changes the exponent, and none of it is meant to.

11 Other GPU kernels

The ECC2K-130 client is one of several GPU arithmetic libraries in the two repositories. The others target the 256-bit prime field of secp256k1 (a Koblitz curve in the SEC sense, with $j = 0$ and the order-3 endomorphism), the NIST primes, the polynomial-basis field of ECC2K-95, and the linear algebra and polynomial-system solvers that index calculus needs. They follow the same discipline as the ECC2K-130 client: a Python or Rust oracle, a host emulation of the identical device source, bit-identical steppers, planted logarithms, and static instruction accounting from the assembler. What most of them still lack is a measurement on a device. We start by saying which is which, because a static instruction count and a measured rate are different kinds of evidence, and the rest of the paper keeps them apart.

11.1 Implementation status

Table 13 is the inventory. Every "never" repeats a sentence in that family's own README, and the reason is the same each time: the code was written and verified in containers with no GPU or no complete CUDA toolkit, and the device runs were planned for rented hardware that the campaign's account hold (Section 13.5) or missing credentials then put out of reach. The NIST

Table 13: GPU arithmetic in the two repositories by evidence class. "Emulated" means the device source was executed on the host and checked against an oracle; "static" means instruction, register and stack counts from `ptxas` with no execution.

kernel family	field	on a device	otherwise
ECC2K-130 client (Sections 8, 13)	$\mathbb{F}_{2^{131}}$, ONB	measured, audited	emulated, static
F4 Gröbner elimination (`cryptanalysis`)	$\mathbb{F}_2$ Macaulay matrices	measured (RTX 5090, H100)	emulated
Macaulay reduced echelon	$\mathbb{F}_p$, $p = 2083$	one scheduling run (RTX PRO 4500)	emulated, static
secp256k1 / Curve25519 rho, baby-step giant-step	$\mathbb{F}_p$, two 256-bit primes	never	emulated, static
secp256k1 kangaroo	$\mathbb{F}_p$, 256-bit	never	emulated, static
P-256 / P-384 field and rho walk (Section 10)	$\mathbb{F}_p$, 256- and 384-bit	measured (RTX PRO 6000, 2026-10-12)	host-verified; protocol pre-registered
ECC2K-95 kernel	$\mathbb{F}_{2^{97}}$, polynomial basis	never	emulated, static
Semaev S_4 sweep; FES	$\mathbb{F}_{2^n}$; $\mathbb{F}_2$	never	host-timed; emulated
sparse matrix–vector	$\mathbb{Z}/m$, $m < 2^{63}$	never	CPU-rechecked
64-bit prime-field rho (`cryptanalysis`)	$\mathbb{F}_p$, $p < 2^{64}$	never	emulated, static

Table 14: Static PTX instruction counts for one 256-bit field multiplication (clang, `sm_90`, `-O3`). The hard floor for the product is 128 instructions: 64 limb products at two instructions each. Class: engineering; nothing here has executed on a device.

	portable C++	inline PTX, before	inline PTX, after
512-bit product	236	171	137
secp256k1 reduction	146	130	77
full multiply	472	328	**210**
Montgomery CIOS multiply	475	—	352

kernels were in the same position until they finally ran, which is the subject of Section 10 [E53; E54; E5; E55; E56, "How this is tested"]. The numbers below are therefore of three kinds, and we label each one: a static count, an emulated operation count or walk constant, or (in Section 11.7) a device measurement.

11.2 256-bit prime-field arithmetic for secp256k1

The field layer holds an element in eight 32-bit limbs, because the integer datapath is 32 bits wide and the carry-chain instructions `mad.lo.cc`, `madc.hi.cc` and `addc` exist only at that granularity. For secp256k1 the 512-bit product is folded with $2^{256} \equiv 2^{32} + 977 \pmod{p}$, about 40% fewer multiply-adds than Montgomery; for any other odd modulus below 2^{256} a Montgomery CIOS path is used, and the scalar ring modulo n reuses it on the host. Points are Jacobian with the standard $a = 0$ doubling ($2M + 5S$) and mixed addition ($7M + 4S$) formulas. The library is a throughput library, not a signing library, and is not constant-time [E53; E53]. A second special-reduction path, added in October 2026, serves Curve25519 (Ed25519's prime-order group in short-Weierstrass form, $p = 2^{255} - 19$, folded with $2^{256} \equiv 38$) through the same walk, baby-step giant-step and test suites; it is host-verified against the RFC 7748 constants and has not run on a device [E53].

Table 14 is the instruction ledger. The portable path emits no carry instructions at all: of

the 444 instructions in a portable squaring, 178 are adds, 102 shifts, 100 ANDs and only 46 multiplies, so the multiply is carry-bound, not multiply-bound, and the inline-PTX rewrite's 55% cut comes from the carry chains. The seven assembly blocks are checked by an interpreter against the portable path on 20,000 random and boundary inputs each, and that check is mutation-tested; it does not cover register allocation or device semantics, which is why `FP_PTX` is off by default [E57, §2; E53].

In the unit of Section 5, a secp256k1 inversion is 270 multiplications (255 squarings and 15 multiplications along the libsecp256k1 addition chain), a rho step with batch W is $6 + 270/W$ multiplications (3 for the affine addition, 3 for Montgomery's trick), $7 + 270/W$ with the $j = 0$ fold: 39.8 at $W = 8$, 14.4 at $W = 32$, 8.1 at $W = 128$, against 276 unbatched [E57, §1]. Three things were tried and declined on the static count: a dedicated squaring routine (-2.7% portable, $+23.5\%$ on the PTX path; reverted), lazy reduction to $[0, 2^{256})$ (worth about 10% of a multiply, but it breaks the canonical form that equality, the distinguished-point test, the cycle hash and the fold comparison rely on), and Karatsuba (judged to lose on a carry-bound multiply). Tensor cores were considered and not reached for [E57, §2 and §4.4]. Marking the inversion and ladder routines `__noinline__` and replacing two 16-entry tables by a small double-scalar ladder took the walk kernel's stack from 4,664 to 1,616 B (the low-memory variant from 4,008 to 792 B) and the compile from over ten minutes to 46 s, the same code-size lesson as Section 5.1 [E57, §3].

11.3 Automorphism folding and baby-step giant-step

The rho walk is r-adding on the low bits of x, with distinguished points by a mask on the next byte and three folds: identity, the negation map, and on $j = 0$ curves the full $\langle \beta, -1 \rangle$ orbit of order six, chosen by minimising x over $\{x, \beta x, \beta^2 x\}$ at the cost of one multiplication ($\beta^2 x = -(x + \beta x)$) and two compares, about 10% of a batched step for $\sqrt{3}$ fewer steps. Only an automorphism code is kept per step; a report is $\{x, \text{walk}, \text{restart}, \text{steps}\}$, 48 bytes, and the host replays the two colliding walks from their seeds, multiplying (a, b) by $(-1)^{\text{neg}} \lambda^k$ at each step. Fruitless cycles of length 2 to 6 are detected with a five-deep ring and escaped by doubling the member with the smallest hash; their measured rates at $R = 8$ agree with the prediction to about 10% [E53, "The walk"].

On a 36-bit $j = 0$ toy curve in host emulation (8 solves per fold, 16 walks, $R = 256$, period 2^7), the mean steps were 305,740, 199,042 and 113,522 for folds 1, 2 and 6 against predictions of 249,454, 176,391 and 101,839 (1.23×, 1.13×, 1.11×; ±0.18× on a mean of eight), all 8/8 correct, so fold 6 buys 1.75× over the negation map (predicted 1.73). In the unit $S = \text{ops}/\sqrt{n}$ that is 1.54, 1.00 and 0.57. The three steppers (batched, low-memory, one-inversion-per-walk) are bit-identical over hundreds of iterations with the same distinguished-point multiset, and every point replays on the host to $aP + bQ$, which is what proves the λ bookkeeping on secp256k1 itself [E53, "How this is tested"].

Baby-step giant-step on the same arithmetic keeps an x-keyed table of 16-byte entries in lock-free open addressing (one 64-bit `atomicCAS` per probe, about 2.5 slots per probe at half load, a false-candidate rate of load$/2^{32}$ and none observed), and walks W giant chains per thread sharing one inversion, 94 registers and no spills at $W = 8$. With the negation map the layout is $m = \sqrt{w}/2$ babies and $2m - 1$ giants, the Galbraith–Wang–Zhang row at $1.00\sqrt{w}$ expected against the textbook $1.41\sqrt{w}$. A 2^{66}-wide interval is 2^{32} entries and 64 GB, the top of one 80 GB device; beyond that the method is out of memory, not out of time [E53, "Baby-step giant-step"]. Table 15 is the measured cost on a 40-bit toy curve, in emulation, every phase charged.

The classification is the same as every row in Section 8: the batched stepper is engineering, the layouts are the published ones, and nothing moves the ratio to the generic floor. A Rust port of the same baby-step giant-step on the CPU runs at 37–49 M steps/s on four threads at $S = 1.03$ cold and 0.55 amortised, 6–11× the big-integer implementation it replaced [E53,

Table 15: Rho and baby-step giant-step on a 40-bit toy curve ($n = 649{,}523{,}094{,}257$), host emulation of the device source, 128 chains, mean over 8 random targets. S = group additions/$\sqrt{n}$ with table construction charged. The rho rows predate the order-six fold and were single solves ($\pm 0.5\times$). Twelve targets on a 2^{26} interval cost $S = 0.560$ per target against 2.131 for one cold solve.

row	table/$\sqrt{n}$	mean S	predicted	correct
rho, negation map (reference)	0	0.85	—	1/1
rho, no negation map	0	1.42	—	1/1
BSGS textbook, reference stepper	0.707	1.44	1.41	21/21
BSGS textbook, batched $W = 16$	0.707	1.44 (bit-identical)	1.41	21/21
BSGS negation layout, $W = 16$	0.500	1.07	1.00	21/21
giant phase only	0.500	0.56	0.50	8/8

"Cost, measured"]. The emulation itself runs at about 380 k steps/s per core, a figure that says nothing about a device and is quoted only so that no one mistakes it for one.

11.4 Pollard kangaroo for interval discrete logarithms

The interval solver reuses the verified field and point code for $Q = kG$ with $k \in [a, a+w)$ at about $2\sqrt{w}$ steps: two herds that alternate by thread parity so every warp is half tame and half wild, a jump table of 2^6 pseudo-random scalars with mean jump $\sqrt{w}/2$ and 25-word entries (odd, so the shared-memory reads are bank-conflict-free), and the target centred at $Q' = Q - (a + w/2)G$. The invariant that makes it testable is that after every step a tame kangaroo at distance d is at dG and a wild one at $Q' + dG$, which pins the jump table, the distance accumulator and the point arithmetic together [E54, "How the walk works"].

Its measured constant is the useful number. In emulation on a 2^{31} interval with 120 trials per configuration, steps are $C\sqrt{w}$ with $C = 2.79 \pm 0.13$ for the default (median 2.65); the idealised two-herd bound is 2.0, Pollard's classic lambda about 3.3, and published parallel implementations about 2.1, so this one is about a third above the best published. A jump-parameter sweep at 20 trials per cell had shown C from 2.50 to 3.95, an apparent 36% win; rerun at 120 trials the best cell and the default were within half a standard error. The defaults sit on a flat optimum and the sweep was noise, which is recorded so that it is not rediscovered. Restarting at a distinguished point or not was likewise within noise [E54, "Measured cost"]. A multi-device split (one host thread per device, disjoint even index bases so herd parity stays global, one host table) is tested by reproducing the combined herd exactly from two contexts and recovering a 28-bit key across them; multi-machine is not done [E54, "Multi-GPU"]. The tool ships no real puzzle keys (no trustworthy offline source), does no address scanning, and has not solved any puzzle; the synthetic 20-, 28- and 32-bit registry entries are regression fixtures.

11.5 P-256 and P-384

The NIST library has graduated from this inventory: it now has device measurements, and Section 10 reports them as the ECC2K-130 ladder carried over to prime fields. One historical note belongs here. The library's first device attempt was an infrastructure non-result, because the cloud credentials were not exposed to the CI runner and the GPU steps were skipped. The attempt record says it "must not be cited as performance evidence", and we do not cite it [E55].

11.6 Binary fields beyond ECC2K-130

The ECC2K-95 kernel. Section 3.3 used this kernel as the end-to-end control; its arithmetic is the packed, polynomial-basis counterpart of Section 5.2. An element is four 32-bit words; a

32-bit carry-less product is sixteen widening integer multiplies with the interleaved-mask trick (operands split by bit index modulo 4 so no partial sum exceeds 8 and nothing carries); the 1-bit top word at $m = 97$ is split off so that a three-way Karatsuba over three words (six leaves) plus seven single-bit masks replaces nine leaves; reduction by $t^{97} + t^6 + 1$ is two folds, the second an extract and two XORs; squaring is a bit-spread kept in 64-bit form (78 instructions against 104 for two 32-bit spreads); inversion is Itoh–Tsujii at about 7 multiplications and 96 squarings. The static count is 304 instructions per multiplication, down from 405, with the six software leaves 96 of the 304; the native `clmad` that gave the ECC2K-130 client +22% would do the same product in six instructions and is not yet ported because the pip-wheel toolchain stops at `ptxas` 12.9 [E5, "What the field arithmetic actually costs"; E5].

Per step the cost model is one inversion split W ways plus two multiplications, one squaring and the three multiplications of Montgomery's trick, with the two Frobenius chains of the walk paid in squarings: 3,813 instructions at $W = 8$, 44% multiplies, 27% the Frobenius chains, 25% the inversion's squarings. Because τ^k is $\mathbb{F}_2$-linear it can be applied as 25 four-bit-window table reads at any k, 296 instructions against 78 per squaring; tables for $\tau^{6,12,24,48}$ cut the inversion's 3,744 squaring instructions at $k = 48$ to 296 and the step to 3,083 (-19%), and $W = 4$ with tables beats $W = 8$ without at 656 against 1,048 bytes of local memory. The table traffic, 400 words per inversion, is unmeasured, and the tables are off by default [E5, "τ^k is not k squarings"]. The low-memory stepper at $W = 8$ compiles to 128 registers and 464 B of stack with four blocks per SM, 512 threads per SM on `sm_90`, `sm_100` and `sm_120`; the prime-field walk of Section 11.2 needs 226 registers and 792 B at 256 threads per SM, so the binary walk gets 1.5–2× the occupancy before either has run [E5, "Occupancy"]. A 40-bit end-to-end test of eight logarithms on one CPU core fell from 2.1–2.7 s to 1.02–1.05 s across the rewrite at the same 191,840 steps, which the README itself files as engineering with S flat.

The pair table. For Koblitz index calculus the same directory builds the meet-in-the-middle table of all $P_i + P_j$ over a factor base, one thread per row with the row's denominators batch-inverted, folded to signed Frobenius orbits by the least cyclic rotation of x in a normal basis (one representative times an orbit-sorted suffix, $2n$ times fewer sums). Stored words shrink 43.4× at $n = 23$ (6,256 against 271,216) and 77.2× at $n = 41$ (11,152 against 861,328). Verification runs each emulated thread on its own CPU thread, requires identical tables at several thread counts and chunkings, round-trips every stored pair and 19,999 generator multiples into the Rust table, keeps a control that must lose pairs, and runs under ThreadSanitizer with a deliberately racy control that must be reported; the bug that motivated all this (the base size passed as the field degree) fails 960 of 1,176 keys when restored [E5, "The $|F|^2$ pair table"; E5]. The fold does not reach $n = 97$ (limited to $n \le 62$) and the device round trip is planned, not run.

Polynomial systems. Two more kernels serve the decomposition step of Section 14.1. A Gray-code fast exhaustive search over quadratic Boolean systems (one thread per sub-cube, $n \le 40$, ≤ 64 equations, shards checked against brute force) exists alongside the production Rust path and has not run on a device [E58]. A Semaev S_4 pairs-and-solve sweep (fix X_1, X_2, reduce the quartic in X_3 against the linearised polynomial of the factor-base subspace, $O(l)$ field operations per pair instead of 2^l) has a `clmad` path and a software fallback, is tested against exhaustive triple enumeration, and on one host core went from 5.07 to 0.44 ms per 2^l-row sweep at $n = 21$. Its README's conclusion is the one that matters: relation collection stays $\Theta(2^n)$ with any oracle polynomial in the base, "the oracle got roughly a thousand times faster and the attack's exponent did not move"; a GPU is another constant. The same README records and retracts an earlier claim that NVIDIA has no carry-less multiply [E59].

Linear algebra. A batched dense reduced-echelon solver over $\mathbb{F}_p$ ($p = 2083$, 226×286 matrices of rank 222) serves the $E(\mathbb{F}_{p^3})$ index calculus whose Macaulay step is 57% of each solve. One

archived pipeline experiment did run on an RTX PRO 4500: 64 matrices per invocation, 36 runs, 2,340 matrices verified against the serial reduction, device reduction about 0.045 s per batch inside a 0.32 s cold invocation dominated by setup, and a two-slot overlap 5.1% faster on held-out seeds, which the record declines to call a speedup for want of an exclusive host and an interval. Its own conclusion is that a GPU changes the crossover with rho "by not a single bit"; it would only enable larger p [E60; E61]. (The directory README's statement that no kernel has run on a GPU predates that experiment.) A sparse matrix–vector worker modulo any $m < 2^{63}$ is correctness-first, every product rechecked on the CPU, with no rate [E62].

11.7 F4 elimination on cloud GPUs

The `cryptanalysis` library's F4 Gröbner-basis path for Weil-descent systems is the one piece of index-calculus arithmetic with device measurements. On a rented RTX 5090 (`sm_120`, 32 GB) the degree-6 Macaulay matrices are built and eliminated 25 to 75 times faster than on the Xeon host; the $168{,}363 \times 188{,}203$ matrix went from 426 s to 5.7 s, the degree-of-regularity sweep from 18.8 to 8.2 s, and the batched per-node decider was correct on the device and 4.4× slower than the host's default engine [E63]. On an H100 a tiled update to the elimination kernel was kept at 1.11–1.49× and a merged-basis panel reverted at 1.5–5.3× slower; the largest matrix fell from 4,781 to 3,586 ms [E63]. The records call these what they are, stage diagnostics of one oracle matrix, not an ECDLP cost, and a later crossover harness keeps its qualified speedup null until a matched device-and-host resource receipt exists [E63].

The `cryptanalysis` library's own 64-bit prime-field CUDA rho (one limb, Montgomery with $R = 2^{64}$, eight walks per thread sharing a Fermat inversion so that a step is 18 multiplications instead of 99) is in the same state as the kernels above: 77,000 device-arithmetic assertions against the host library, whole-solver emulation on four groups, a cubin for `sm_70` through `sm_90` with 64–72 registers and no spills, and no execution on a device; its emulator rows sit at $S \approx 1.3$–2.6 alongside the CPU rho and show only that S does not grow with the walk count [E56; E56, "GPU rho kernel"]. Its efficiency scorer, built for the ECC2K-130 walk, is the one place the two threads meet: charging the packed walk's 4,328 lane-instructions per iteration to the measured dual-issue rate of the RTX PRO 6000 (78.3 per SM-clock, 188 SMs, 2,307 MHz) gives an attainable 7.85 billion iterations/s per card against an audited 6.905 for that build, 88% of the issue roof, and the same method on the wide multiply reproduces the "31 billion iterations/s if the walk did nothing else" of the earlier ceiling note [E64; E64].

11.8 Summary

Two lessons hold across the fields. The first is that, on this hardware, the carry-less unit is the whole difference between the binary and prime-field kernels. A 131-bit binary product is six `CLMAD`, while a 256-bit prime-field product is at best 210 carry-chained integer instructions. The binary walk is therefore limited by a narrow unit at 1.6–2.0 lanes per SM-clock, the prime-field walk by carries on the wide integer pipe, and the two want different occupancies, batch sizes and arithmetic representations. Section 10 confirms this on a device: the same levers helped both walks, but by very different amounts. The second lesson is that each of these libraries reached a verified, bit-identical, statically accounted state without ever touching a device, and that state is worth exactly what Section 8.1 says a pre-tuning floor is worth. It bounds what a device could do; it does not tell you what one did. The NIST walk is a reminder of how wide that gap can be: its first device run measured an empty kernel. Only the ECC2K-130 client and, more recently, the NIST walk have crossed to the measured side, which is why they get sections of their own and the rest share this one.

12 FPGA implementation

In some ways this field suits an FPGA better than a GPU. The GPU has one narrow carry-less unit per SM, while an FPGA can be filled with carry-less multipliers, and σ^k is just wiring. We designed a VHDL engine for the AWS F2 (Xilinx VU47P), estimated its rate before any board time, and then measured it.

12.1 Pre-implementation estimate

Before synthesis, we mapped the client's own straight-line multiplier to LUT6s (about 5,563 at 32 levels, known to be about 40% high for ignoring dual-output packing) and calibrated against the 3,071 and 3,620 LUTs that Bernstein et al. report for GF(2^{113}) and GF(2^{127}) multipliers [17], extrapolated to 3,781 at $m = 131$. With multipliers at 79% of a core and 5.33 multipliers per core, a core is 25.5 KLUT at literature packing or 37.5 KLUT with the repository's unpacked circuit; 18% shell overhead on 1,303,680 CLB LUTs at 70–80% utilisation gives 29–34 or 20–23 cores at an estimated 250–350 MHz. The estimate came to 5–12 billion iterations/s per VU47P, with a central value of 8 billion iterations/s: slower per chip than the 14.1 billion iterations/s GPU, roughly level on cost per solve, and 2–5× better per watt [E65].

12.2 Architecture

Multiplier. `gf131_mul` follows the Bernstein–Lange route in hardware: `prep` converts each operand to the polynomial basis (991 XOR2 per operand), a recursive Karatsuba `gf2_kmul` splits $131 \to 66 \to 33 \to 17$ bits into 27 leaf products of 17×17, and `to_onb` converts the 261-coefficient product back (3,167 XOR2). There is no reduction step. At three levels the product is 7,803 AND and 9,404 XOR2, 14,553 XOR2 per multiply including conversions, against 17,161 AND and 22,049 XOR2 for schoolbook. Out of context in Vivado 2025.2 at a 4.0 ns clock, three levels map to 4,855 LUTs and 4,647 FFs with +2.68 ns of slack [E4].

A DSP leaf computes a 17-bit GF(2) product as six 25×16 integer products with coefficients spaced three bits apart (split $9 + 8$ against $6 + 6 + 5$) so that no column count exceeds six and nothing carries; the parities are XORed into place with 22 LUTs. Eleven of the 27 leaves go to DSPs, 66 DSP48E2 per multiplier, which takes the multiplier to 4,004 LUTs at a latency of 14 clocks including a 261-FF copy register (11 clocks with LUT leaves). The pblock holds 7,992 DSPs, so 120 engines can take 11 DSP leaves and 128 engines ten [E4; E4].

Step and batch. An unbatched step is ten multiplies: $d = x + \sigma^j x$, $\lambda = (y + \sigma^j y)/d$ with the eight-multiply Itoh–Tsujii inversion, $x_3 = \lambda^2 + \lambda + d$, $y_3 = \lambda(x + x_3) + x_3 + y$. `ec2k_batch_pipe` shares one inversion across W walks through a product tree held in UltraRAM (four URAM288 per engine): $W - 1$ forward multiplies, 8 for the inversion, $2(W - 1)$ backward, W for λ and W for y_3, $5W + 5$ per batch, $5 + 5/W$ per step: 5.31 at $W = 16$, 5.16 at $W = 32$, 5.08 at $W = 64$. A batch is a chain of $2 \log_2 W + 10$ bursts, and several batches are kept in flight to hide the multiplier latency. Table 16 is the simulated rate [E4; E4].

Walker and host port. Walks live in a block-RAM FIFO of $(\mathrm{id}, \mathrm{steps}, x, y, \mathrm{dp})$, 285 bits by 512, read through a four-entry prefetch. Only the low 13 bits of the step count travel with the walk; the high 19 sit in a LUTRAM table bumped once every 8,192 steps. A step whose x has weight at most `DP_WEIGHT` is flagged, moves into a report register at the FIFO head and is held until acknowledged, so no report is ever lost and the `DROPPED` counter always reads zero. `ec2k_axil` is an AXI4-Lite slave in front of `NENG` walkers with a credit-based register spine (about 870 FFs and 300 LUTs per engine), introduced after the first flat 48-engine build placed at −2.5 ns. The host program is a drop-in for the CUDA client under the same worker script:

Table 16: Simulated clocks per step of the batched step unit at multiplier latency 10, against the arithmetic bound $5 + 5/W$. The default geometry is 64×8 (512 walks per engine), which runs at 5.16 clocks per step with either multiplier.

W	batches in flight	200 steps	6,144 steps	bound
8	8	8.02	6.62	5.63
16	4	9.86	8.38	5.31
16	8	6.98	5.67	5.31
16	16	6.46	5.34	5.31
32	8	7.20	5.20	5.16

Table 17: F2 images on one VU47P (`f2.6xlarge`). All runs had zero dropped reports; for every image except the 112- and 120-engine ones, 32 or 64 sampled distinguished points were re-walked on the host against the client reference from the same seed and matched. The 128-engine image is the promoted AFI.

engines	MHz	geometry	routed WNS (ns)	million it/s	clk/step
48	333.3	16×16, tree in BRAM	+0.022	3,010.7	5.31
64	333.3	same	+0.040	4,013.7	5.32
80	333.3	17 tiles/engine	+0.027	5,015.8	5.32
96	333.3	same	+0.002	6,018.0	5.32
112	375	32×8, tree in URAM	+0.049	8,123	5.17
120	333.3	32×8, tree in URAM	+0.063	7,730	5.17
128	**333.3**	32×8, tree in URAM, LUT multiplier	+0.048	**8,249**	**5.17**

it writes the same 32-byte records and can re-walk the first N reports from their seeds on the CPU (`-verify N`); production workers leave verification to the campaign merge [E4; E4].

Verification. The reference script imports the client's field model and emits 256 multiplier vectors (including 0, the all-ones 1, single basis elements and the extreme coordinates, each also checked against an independent direct convolution), 200 steps (twelve full batches of 16 plus a partial batch, so the flush path runs every time) and 64 walks of random order-ℓ points on the challenge curve. The batch testbench passes for every W from 2 to 32 with 2 to 16 batches in flight; the walker testbench checks every report's id, step count and point against the reference trajectory (64 distinguished points from 797 steps in 19,194 clocks) [E4; E4].

12.3 Measurements on AWS F2

Table 17 lists every image we built, placed, loaded and timed, and Figure 14 plots them [E66]. The 48-engine run read back its geometry (48 engines of 512 walks, weight 34, 333.3 MHz), seeded 24,576 walks over AXI-Lite in 7.3 s, held 3,010 million iterations/s from ten seconds on, and completed 103 billion iterations in a 40 s run; simulation had predicted 5.29 clocks per step against the measured 5.31. The promoted 128-engine image holds 8.25 billion iterations/s. A design whose engines share nothing but the host port should scale linearly with engine count, and the measured images sit on that line. Builds that missed timing (128 engines at 375 MHz, 112 at 400) and the 144-engine build that failed block-RAM placement (3,744 RAMB18 sites needed against 3,576) are recorded alongside; the projection at 5.16 clocks per step is 8.8 billion iterations/s at 136 engines and 9.3 at 144 or at 128 engines and 375 MHz, neither yet met (128 at 375 MHz routed at -0.21 ns; 144 failed placement) [E4].

Per engine the current revision is 6,790–6,940 LUTs, 9,371 FFs, 12 RAMB36, 4 URAM and 66 DSPs; the 96-engine image used 62% of the LUTs and 81% of the block RAM, and block RAM, not logic, is what bounds the engine count on this part [E4].

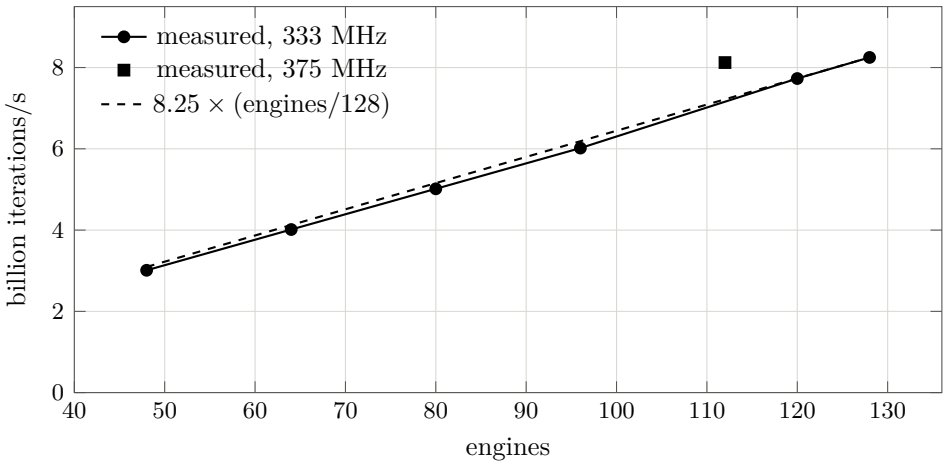


Figure 14: ECC2K-130 FPGA engine on AWS F2. The 128-engine image at 333 MHz is 8.25 billion iterations/s; the pre-board estimate was 5–12 billion iterations/s with a central 8. The measurement landed on the central estimate; it is a device measurement, not the estimate restated.

12.4 Comparison with GPUs

At the prices observed on 2026-09-11, an F2 instance at $1.980 on demand or $0.709 spot against an RTX PRO 6000 at $4.14 and $1.31 per GPU-hour, the expected $2^{60.9}$ iterations cost about 72,400 F2-hours at the measured 8.25 billion iterations/s, roughly $51k spot, against 42,400 GPU-hours and $56k spot at the campaign's 14.11 billion iterations/s collection rate.[1] So the two platforms are level on cost per solve, within the noise of spot pricing, and the FPGA is about 2–5× better per watt on the note's estimated power envelopes (75–150 W against a 600 W board). FPGA workers follow the same client contract as the GPUs and write the same records, so a collision between an FPGA trail and a GPU trail is merged like any other.

13 The ECC2K-130 campaign

At the campaign's collection rate, $2^{60.9}$ iterations is about five years on a single GPU. The campaign is therefore not a single solve but a collection effort: a corpus of distinguished points, each re-walkable from its seed, grown on whatever hardware is cheapest at the time and merged as it arrives. This section covers its configuration, its calibration, its storage protocol and where it stands now.

13.1 Configuration

The configuration is one file in the repository; the copy in the storage bucket is what workers read [E8; E8]. Walk: σ (Section 4.1). `dpWeight`: 32, uniform campaign-wide, and a campaign run refuses any other. Batch 16, 256-thread blocks, two resident blocks per SM, 385,024 workers = 6,160,384 logical walks per GPU, 1,024 steps per launch, checkpoint and upload every 600 s, restart every 48 h. The iteration guard `maxIters` in the repository file is 2^{32}, twelve trail lengths at weight 32; it was 2^{30}, three trail lengths, which a model on the measured rates says cut 4.9% of walks and discarded 15.6% of the fleet's steps until an audit found it. Whether the bucket copy was raised before the fleet stopped is not recorded [E67]. The fields that define the corpus (workers, batch, geometry, curve, cutoff, walk and the state-layout knobs) freeze once any slot

[1] Derived here from the measured rate and the prices recorded in E65; the note's own table priced the hypothetical 8 billion iterations/s image at $53k spot.

Table 18: Iterations per distinguished point at cutoff 32, frozen 2026-09-17 18:47 UTC. Per-worker spread 0.04 in the exponent. The last column is the median self-reported rate of the family's workers on the same day.

family	workers	iterations	points	it/point	median billion it/s
RTX PRO 6000 (`g7e`)	38	$5.606 \cdot 10^{15}$	15,724,672	$2^{28.41}$	14.66
L40S (`g6e`)	34	$2.723 \cdot 10^{15}$	7,643,600	$2^{28.41}$	7.47
RTX PRO 4500 (`g7`)	48	$3.081 \cdot 10^{15}$	8,640,973	$2^{28.41}$	4.87
CPU	12	$2.804 \cdot 10^{14}$	785,410	$2^{28.41}$	2.29
all	132	$1.169 \cdot 10^{16}$	32,794,655	$\mathbf{2^{28.41}}$	$\approx$ 1,068 total

exists; `WITNESS=0` is pinned. The preset is the audited RTX PRO 6000 build at CUDA 13.3.1 and `sm_120` [E8]. A checkpoint is about 370 MB (six million walks at 60 bytes).

13.2 Distinguished-point cutoff

The planning numbers follow from the cutoff [E8, "The numbers that decide the plan"]. At weight 34 the expected corpus is $2^{35.6}$ records, about 1.7 TB, one per $2^{25.27}$ iterations; at weight 32 one report per $2^{28.41}$ iterations, $2^{3.14} = 8.8\times$ rarer, so about $2^{32.5}$ records and 0.19 TB at the expected work, about 40 points/s, 3.4 M points/day and 0.11 GB/day from a GPU collecting at 14.11 billion iterations/s. The price of a rarer point is the work left in flight when a collision lands: 43 GPU-hours per GPU at weight 32 (6.16 M walks times $2^{28.41}$ steps), 13% of the expected work at 128 GPUs and more than the whole expected work at 1,024, against 4.9 GPU-hours per GPU at weight 34. The campaign's fleet size never approached the range where that trade reverses.

We measured the interval twice, and the first measurement was wrong. A fleet-wide ratio of $7.93 \cdot 10^{15}$ checkpointed iterations to 31.6 M uploaded points on 2026-09-15 had been quoted as $2^{27.9}$ iterations per point. Neither counter was clean: the checkpoint sum multiplied every slot's per-walk base by the Blackwell grid constant, inflating the auto-sized Ada slots by the ratio of the grids (about 26× on an L4-sized grid), and the point total mixed in two retired slots that had walked at weight 34. The replacement reads from each live worker the client's own two counters, iterations this run on the grid the client built itself and points reported over the same interval, frozen at 2026-09-17 18:47 UTC across 132 workers (Table 18) [E48]. The result, $2^{28.41} = 3.56 \cdot 10^8$, is the same on every hardware family to 0.04 in the exponent, and sits $2^{0.60}$ below the naive binomial $P(\mathrm{HW} \le 32) = 2^{-29.01}$, as Bailey et al.'s $2^{25.27}$ sits $2^{0.57}$ below the binomial at weight 34. We classify the correction as accounting: the walk did not change, only the number did, and we retire the earlier figure rather than averaging it in.

13.3 Fleet economics

The total cost depends on the work, not on the size of the fleet: at the 2026-09-11 spot price of $1.31 per RTX PRO 6000 hour the expected 42,400 GPU-hours are about $56k whether 8 GPUs take 221 days or 512 take 3.4; on demand the figure is $142k–$175k [E8]. What fleet size changes is the calendar, and the in-flight loss described above. The campaign ran mixed families because iterations per dollar, not iterations per card, is the objective: on 2026-09-18 the best-value quote was an L4 in `eu-north-1` at $56 \cdot 10^{12}$ iterations per dollar against $38 \cdot 10^{12}$ for the RTX PRO 6000 and $28 \cdot 10^{12}$ for the RTX PRO 4500, and at that quote a full reference walk costs about $38k [E68]. The per-SKU rates behind those ratios are measured, not list: L40S 8.84 billion iterations/s with `clmad`, L4 2.49, T4 0.53 without it [E22; E23].

The RTX PRO 4500 is the most instructive case. Its measured 5.107 billion iterations/s on the shipping walk is exactly what the 6000's rate predicts from SM count and clock (14.64 ×

$82/188 \times 1.95/2.44 = 5.10$), and at that rate the 6000 wins per dollar by 1.2–1.3×; the break-even against the 6000's shipping rate is 6.2–6.6 billion iterations/s in the two cheapest regions and 8.8 in the third, figures the note says must be redone with both cards on their best builds [E69]. That is why a 12 billion iterations/s goal was set for the card and why the x-only bridge map of Section 4.3 was pursued there rather than on the 6000: a power-capped die rewards fewer instructions per step more than a pipe-bound one does. The separate goal of 28 billion iterations/s on one 6000 was priced below the pipe (Section 8.1) and its grid and occupancy screens, run on an RTX PRO 4500 because no 6000 could be allocated, found nothing faster than the shipping configuration [E70].

A power survey closes the loop: on the 6000 the 20 billion iterations/s build drew 583 W for 34.4 M updates per joule and a fused-pass variant 544 W for 37.0 M/J in the same session, 1.072× per joule in five paired repetitions against a 1.03 target; per joule, not per second, is the figure that transfers to a 165 W card, and the model, calibrated on the 4500's 5.107 and 5.552 billion iterations/s receipts (ratio error +2.6%), predicts 7.2–7.4 billion iterations/s for the fused build there [E71]. That prediction still needs a measurement; the bridge map got there first.

13.4 Storage, seeding and certification

A record is 32 bytes: a little-endian 64-bit seed and the canonical normal-basis x of the endpoint, three 64-bit words; no scalar coefficients are stored, they are rebuilt by replay. The seed is `run id (16)` ‖ `walk index (32)` ‖ `restart counter (16)` with run id = slot +1; a splitmix64 finaliser expands it to the 128-bit c of the start point. Two contributors who pick different run ids never start the same walk (independently seeded walks can still meet, which is the point), and a permanent seed registry does compare-and-set across providers (Modal GPU runs are ids 8000–8999, CPU sidecars 9000–9999); a race test with eight claimants acquired exactly one [E72; E72; E67, "Seeds"; E8].

Uploads are ordered so that no pointer ever names data that is not there: payloads before their commit manifests, both before checkpoint publication. Strict manifests bind curve, cutoff, guard, seed and key semantics, worker geometry, source hash, worker binary hash and replay binary hash, and every immutable payload carries its SHA-256. Slot ownership is one S3 object per slot claimed with a conditional put on the ETag, a 60 s heartbeat and a 180 s lease; checkpoints are content-addressed and the owner-checked heartbeat publishes the pointer, so a stale worker can leave orphaned blobs but cannot commit. The note is explicit that this is integrity against failure, not authentication against an actor able to rewrite both data and manifests [E72; E8, "Slots"].

Merging buckets records by a hash of the key limbs (the first layout used the key's low bits) into 4,096 files, sorts, drops exact duplicates (the client store, the database primary key and the merge are three layers; 9.95 M re-reports had been dropped by 2026-09-25) and reports equal keys with different seeds. Such a pair goes into a two-record file and to the CPU client, which re-walks both trails, aligns Frobenius and negation, and verifies $[k]P = Q$; a $\mathrm{GF}(2^{41})$ pair with known seeds that recovers $k = 369250562913$ is the local acceptance fixture. An exhaustive check of 76,904,684 step multisets confirms that no multiset of at most 32 σ-walk steps closes an orbit, so no short fruitless cycle exists; a trail crossing itself at random is estimated at about 2^{-65} per trail, about 10^{-10} over the campaign, and the iteration guard releases such a walk [E72; E67; E8].

The public dashboard publishes aggregates only: not point keys, coefficients, seeds or database locations [E73; E73].

Table 19: Campaign state at the audited snapshot of 2026-09-25 05:05 UTC and at the published feed of 2026-10-10 15:42 UTC. The pooled iterations-per-point ratio is below $2^{28.41}$, partly because the corpus still holds weight-34 records from the first two slots; the mean length of walks that ended by 2026-09-25 is $2^{28.34}$. Sustained production peaked at about 842,000 points per hour on 2026-09-23, about 83 billion iterations/s.

	2026-09-25	2026-10-10
distinguished points stored	367,123,410	485,528,850
iterations (checkpointed)	$2^{56.38}$	$2^{56.68}$
share of expected work	4.4%	5.4%
collisions	0	0
duplicate re-reports dropped	9,953,494	14,153,833
slots / walking / off-weight	195 / 2 / 7	227 / 0 / 7
points in the last 24 h	741,048	0
walk rate over the last hour	3.65 billion iterations/s	0
last point stored	—	2026-10-07 14:41

13.5 Current state

The campaign was created on 2026-09-11. Its fleet on 2026-09-17 was about 133 workers (the 132 with enough points to calibrate are in Table 18), producing about 2,600 points/s by the ingest's count; the ingest path was re-engineered that week from one object at a time (2,450 records/s, below production) to a threaded loader measured at 137,753 rows/s, and a five-and-a-half-hour blind spot during a key layout change was recovered in 23 minutes (4,321 objects, 63,351,803 rows, zero failed), taking the store from 67.6 M to 124.3 M points [E8]. On 2026-09-18 the spot market reclaimed every worker and the cloud account was placed on an account-verification (risk) hold that has blocked new instance launches since; collection continued at a small scale on another provider. Its first four GPU runs reused the run ids of the first four slots by mistake and re-walked them (813k of one run's 912k records byte-identical to the slot they replayed, all dropped as re-reports, and an inadvertent end-to-end replay check); later runs use the reserved ids 8000–8999 [E74; E1, "Collecting for the campaign from Modal"].

Table 19 gives two snapshots: the audited one of 2026-09-25, while a small fleet was still walking, and the published feed of 2026-10-10, by which time the last point had landed (2026-10-07) and no slot was walking; the sustained peak and the mean trail length in the caption are from the audit [E67; E75; E67]. Having no collision yet is exactly what we should expect: $2^{56.68}$ iterations is 5.4% of $2^{60.9}$, and the probability of no collision after a fraction c of the expected work is about $\exp(-\pi c^2/4)$, here 0.998. Collection is paused, not finished: every slot's checkpoint and the corpus are intact, and a worker that resumes a slot continues its walks from the checkpoint.

14 Structural alternatives to rho

Everything above is engineering against a fixed exponent. Three lines of work asked, in the same accounting, whether the exponent itself could move for this curve. We report each with its measurement, so that anyone who has the same idea later finds a number instead of folklore. None of them changed the campaign.

14.1 Index calculus on the Koblitz curve

Index calculus on $E(\mathbb{F}_{2^n})$ decomposes a random point into m factor-base points through Semaev's summation polynomials [22, 26, 27, 29, 46, 50]; on a Koblitz curve the factor base can be taken Frobenius-stable so that columns are orbits, saving a factor n in relations and

Table 20: Four-summand compact-orbit index calculus on the ECC2K-130 curve family, single thread, $K = 600$ columns, precompute fully charged (left), and the extrapolation to $n = 131$ ($r = 6.8 \cdot 10^{38}$) at $1.85 \cdot 10^6$ probes/s (right). The minimum base for one expected four-tuple per point is $B > r^{1/4} = 5.1 \cdot 10^9$ points, $K \geq 2 \cdot 10^7$.

n	r (bits)	rank wall	probes/relation
61	47.2	13.1 s	$\sim 4 \cdot 10^4$
71	52.3	1,485 s	$\sim 4.6 \cdot 10^6$
73	56.3	18,398 s	$5.7 \cdot 10^7$

K	probes/relation	total probes	core-years
600	$1.1 \cdot 10^{29}$	$6.6 \cdot 10^{31}$	$1.1 \cdot 10^{18}$
10^6	$4.0 \cdot 10^{22}$	$4.0 \cdot 10^{28}$	$6.8 \cdot 10^{14}$
10^9	$4.0 \cdot 10^{16}$	$4.0 \cdot 10^{25}$	$6.8 \cdot 10^{11}$
10^{12}	$4.0 \cdot 10^{10}$	$4.0 \cdot 10^{22}$	$6.8 \cdot 10^8$

n^2 in linear algebra against rho's $\sqrt{2n}$. Two structural facts about $n = 131$ come first. In a normal basis the Frobenius is a coordinate permutation, so a Frobenius-stable subspace is an $\mathbb{F}_2[t]/(t^n - 1)$-submodule, a binary cyclic code of length n, and at $n = 131$ (and $n = 163$) 2 is primitive modulo n, so $t^n - 1$ has exactly two factors and the only stable subspaces have dimension $0, 1, 130, 131$. The usual subspace construction is empty at exactly the interesting sizes; forced onto the 130-dimensional subspace, the orbit trick is a net loss of 57–96 bits for $m \in \{2, \ldots, 5\}$ [E76; E77]. The Couveignes–Lercier elliptic-period alternative needs an auxiliary curve over $\mathbb{F}_2$ with $n \mid \#E'(\mathbb{F}_2)$, and Hasse gives at most 5 [E76]. What survives is a weight-bounded base $\{R : \mathrm{HW}(x_R) \leq w\}$ in the normal basis, which is Frobenius-stable for every n, is the predicate the walk already computes, and is not a low-degree algebraic condition, so only a solver that does not need linearity can use it.

SAT decomposition. Chaining Semaev's S_3 through the intermediate sums, with squaring free, encodes a k-point decomposition as a CNF whose multiplications cost the generated Karatsuba circuit's gates (15,588 at $n = 131$ against 2.2 million for the definition). With $k = 3$, $w = 3$ and CryptoMiniSat the median solve is 0.11 s at $n = 9$ and 2.64 s at $n = 11$; at $n = 23$ no instance finished, one running 25 minutes without returning: roughly $2^{2.3}$ per field bit over that range, from three points [E1, "Index calculus: SAT point decomposition"].

The compact-orbit ladder. A separate line of work solves rungs of the same curve family ($a = 0$, the ECC2K-130 equation) with a four-summand decomposition, at $n = 41$ and 53 over 85- and 220-column bases and at $n = 61, 71, 73$ over a 600-column base, fully charged, and fits the rank cost. Table 20 is the measured cost and its extrapolation to $n = 131$ under the fit probes/relation $\approx C\, r/(n^2 K^2)$ with $C = 1$ [E78]. The fitted C (0.33 at $n = 61$, 1.51 at $n = 71$, 1.27 at $n = 73$, recomputed here from the exact subgroup orders) disagrees with the structural model's 0.25 by a factor of 1.3 to 6, and the note records the gap as open; the extrapolation uses $C = 1$, between the structural 0.25 and the fitted 1.3–1.5 of the two largest rungs.

Put in the same unit as the rest of the paper, the verdict is clear. Rho at $2^{60.9}$ iterations on one RTX PRO 6000 is about five GPU-years at the campaign's 14.11 billion iterations/s. Four-summand index calculus has total work about $r/(n^2 K)$ probes: at $K = 10^9$ that is $4 \cdot 10^{25}$, some $2 \cdot 10^7$ times rho's iteration count with one probe counted as one step (uncalibrated), and the base at which the precompute alone equals rho is $K \approx 1.8 \cdot 10^{16}$ columns, beyond any linear algebra contemplated. The rungs that beat rho do so only in the online class after a precompute that is excluded from the clock (at $n = 73$, 18,400 s of precompute against 46 s of rho online, a 400× total-work deficit traded for a 185× exploratory online wall ratio on one frozen target, on a shared host), and with the precompute charged the method spends 26–1,200× more probes than one rho solve takes steps on every rung, again uncalibrated. Beating rho on total work needs a relation-shape change (five- or six-summand relations, whose per-point yield scales as $B^5/(120r)$), not engineering, and that oracle is unmeasured [E78, §3–4]. In this repository's boundary ledger the Koblitz rows at a 39-bit subgroup run from $S = \mathrm{ops}/\sqrt{r} = 58.8$

(300× signed-Frobenius rho) for the plain three-summand meet-in-the-middle oracle down to 5.12 (26×; 5.92 and 30× on re-measurement) for a balanced two-summand base; the secondary complete-CPU panel on disjoint public points at $n = 37, 41, 53$ has rho ahead in every cell [E79; E79].

14.2 Point decomposition

The `cryptanalysis` library prices the same question from the other end: given rho's $2^{60.81}$ orbit-walk iterations, it derives the speed an m-summand decomposition must reach for index calculus to fit inside that budget, and compares it with the best measured decomposition. With a base of 2^ℓ points, $m = 4$ at $\ell = 29$ needs $2^{48.6}$ decompositions and $2^{60.0}$ of linear algebra, leaving 2^{12} rho-iterations per decomposition; $m = 5$ at $\ell = 28$ leaves 2^{33}; $m = 6$ at $\ell = 24$ leaves 2^{37}; $m = 3$ has no budget at all because the linear algebra alone exceeds rho. The best measured decomposition is the non-algebraic meet-in-the-middle at exponent $\lceil m/2 \rceil$ in ℓ, and extrapolated to $n = 131$ it is 2^{58}, 2^{84} and 2^{72} for $m = 4, 5, 6$: 35–51 bits over the budget. Algebraic solvers land far higher: SAT fits at 2^4 ($m = 3$) to 2^7 ($m = 4$) per unit of ℓ, msolve grows faster still, and WDSat at about 2^3, the exhaustive-search exponent. An attack needs a decomposition about 2^{35}–2^{50} times faster than meet-in-the-middle at $\ell \approx 25$–30, which is the obstruction that stops Wagner's k-tree from applying to elliptic curves; the trace-homomorphism clause that halves UNSAT conflicts is a constant, not an exponent [E80]. A Riemann–Roch (Nagao) oracle on the real curve was measured at factor-base dimensions 4–8 and its derived product law (relations × targets × oracle $= m \cdot 2^{131}$) leaves even a free oracle above rho at $m = 3$ ($2^{68.6}$) and below it only from $m = 4$ ($2^{56.4}$, leaving $2^{4.4}$ of headroom for the oracle itself), where no measured oracle comes close [E81].

14.3 Discussion

We carried each alternative through to a measurement, and each landed above rho by a margin that is a matter of exponent, not constant. For ECC2K-130 the honest summary is that the generic-group floor, $\sqrt{\pi\ell/(2 \cdot 262)}$ class-walk steps, is still what the campaign is walking toward, and the only thing this work has managed to move is the cost of one step.

15 Performance upper bounds

Everything in this paper is engineering against a fixed exponent, so the natural question is how much engineering is left. We answer it per platform with ceilings derived from measured pipe rates: each assumes the named unit runs at its full measured rate and that every other cost is zero. They are upper bounds, not forecasts. Figure 15 draws them against today's measurements.

15.1 RTX PRO 6000

The ceiling for any walk that performs one affine addition per step is set by two pipes (Section 8.1). On the carry-less unit, 28.9 `CLMAD` per update at 1.65 lanes per SM-clock caps the card at 24.7–26.1 billion iterations per second with everything else free; on the logic pipe, the arithmetic alone already needs more slots than 28 allows [E10, §1]. The roofline of the table walk's build puts its own speed of light between 22.7 and 27.4, depending on how fast `CLMAD` sustains in that kernel's instruction mix, and the logic pipe binds first, at 22.0 [E19, §2 and §4]. For the σ-walk with its current carry-less arithmetic (38.125 `CLMAD` per update at two lanes), the same calculation gives about 24.0 with every other cost free [E85].

So the realistic headroom on one card is about 1.5×: from 15.88 for the best σ-walk to the low or middle twenties, and only by cutting integer work. Thirty is out of reach for any

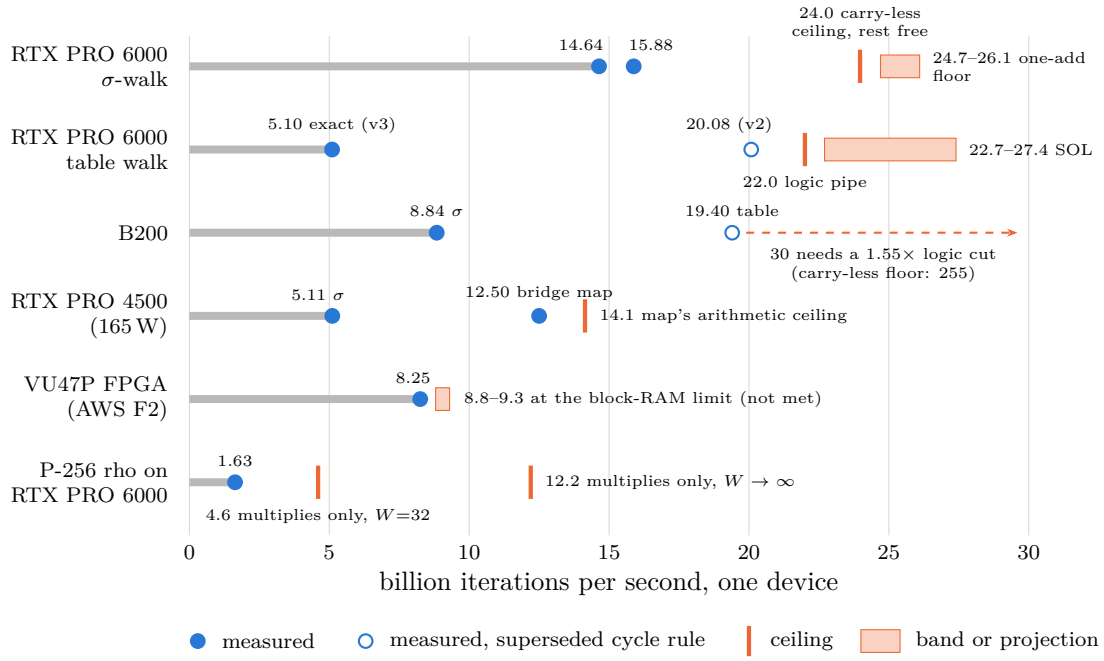


Figure 15: How far each platform can go, against what it does today. Ceilings are derived from measured pipe rates with every other cost set to zero, so they are upper bounds, not targets: the RTX PRO 6000's are its carry-less and logic pipes for any walk with one affine addition per step (Section 8.1); the NIST bounds count only field multiplications at the measured multiply rate (Section 15.4). None of these moves the $2^{60.9}$ iteration count.

one-addition walk on this card; it would need integer work cut from about 1,320 to 968 lane-instructions per update with the carry-less unit fully busy, which the note calls a two-card number [E19, §5]. Two cards at today's rates already exceed it.

15.2 Data-centre parts and the RTX PRO 4500

On the B200 the carry-less unit is so fast that its floor, 255 billion iterations per second, is irrelevant; the logic pipe is the only ceiling. Thirty would need at most 618 logic slots per update against 955 after the sweep, a 1.55× cut, and the candidate offloads (reduction, basis conversions and branch selection as carry-less operations) add up to roughly that, though none has been built [E24, §6.2 and §6.5; E26, §3.4]. The H100 would need at most 521 slots for the same rate [E24, §6.5]. On the power-capped RTX PRO 4500, the bridge map's own arithmetic ceiling is 14.1, measured with the rest of the map stripped out, so 15 is not available on that map either [E11].

15.3 FPGA

The F2 engine scales linearly with engine count and is bounded by block RAM, not logic. At 5.16 clocks per step the projection is 8.8 billion iterations per second at 136 engines and 9.3 at 144 engines or at 128 engines and 375 MHz; the 144-engine build failed placement and the 375 MHz builds missed timing, so neither is met [E4]. About 13% above today's 8.25 is all this part offers without a new design.

15.4 NIST walk

The prime-field walk has more room, because it is further from any ceiling. A step costs about $6 + 320/W$ field multiplications: three for the affine addition, three for Montgomery's

trick, and a share of a Fermat inversion with a four-bit window (about 256 squarings and 64 multiplications; our count). At the measured 73.4 billion dependent multiplications per second on P-256, that caps the walk at about 4.6 billion iterations per second at $W = 32$ and 12.2 as W grows without limit. The walk runs at 1.632, the cost of about 45 multiplications per step against 16 counted, so roughly two thirds of each step is still outside the multiplier: selection, distinguished-point checks, loads and the serial latency of each thread's inversion. These bounds are ours, derived from the measured multiply rate, and that rate is itself a single dependent chain, so a throughput-bound multiplier would raise them [E49].

15.5 Implications for solution cost

Cost per solve scales inversely with the rate. At the campaign's collection rate of 14.11 billion iterations per second the expected $2^{60.9}$ iterations are 42,400 RTX PRO 6000 hours, about \$56k at the 2026-09-11 spot price [E8]. If the σ-walk reached 24 the same work would be about 24,900 GPU-hours and \$33k, and at 26, about 23,000 GPU-hours and \$30k (our scaling of the measured figures). Even the ceiling leaves the solve a five-figure rental problem. No engineering in this paper or in the bounds above changes the $2^{60.9}$; only an algorithmic advance would, and Section 14 found none. Section 16 turns these rates into calendar time for fleets of each kind of device.

16 Time and cost to solution

The rates in this paper translate directly into calendar time and money, and this section does that translation for every device we measured. Nothing here is a new measurement: each estimate divides the expected work by a rate recorded earlier in the paper and states what else it assumes. The question for ECC2K-130 is no longer whether a solve is feasible but how many cards, for how long, at what price. For the NIST curves, the same arithmetic gives an answer that no plausible amount of hardware changes.

16.1 Model

Three quantities set the time to a collision.

Expected work. The σ-walk needs $2^{60.9} \approx 2.15 \cdot 10^{18}$ iterations in expectation (Section 4.4). One device running at R billion iterations per second therefore needs $2.15 \cdot 10^{18}/R$ nanoseconds, or $68.2/R$ years (Table 2), and a fleet of N such devices needs $1/N$ of that.

Variance. Rho's running time is random. The probability of no collision after a fraction c of the expected work is about $\exp(-\pi c^2/4)$ (Section 13.5). Half of all solves finish by $c = 0.94$, nine in ten by $c = 1.71$ and 99 in 100 by $c = 2.42$ (our arithmetic). So a plan that must finish with 90% probability should budget 1.7 times the expected work.

Detection delay. A collision is only seen when the second of the two merged trails reaches a distinguished point. At cutoff 32 that is on average another $2^{28.41}$ steps of one walk (Section 13.2). Each walk advances at R divided by the number of walks on the card. With 6.16 million walks on an RTX PRO 6000, the delay is about 42 hours at 14.64 and 38 hours at 15.88, whatever the fleet size (our arithmetic; the campaign note gives 43 GPU-hours per GPU at 14.11 [E8]). Every card keeps walking during the delay, so a large fleet pays for it in GPU-hours as well as calendar time. At 1,000 cards it adds about as much work as the solve itself. Cutoff 34 cuts the delay to about 4.4 hours at 15.88, at the cost of an 8.8× larger corpus (about 1.7 TB instead of 0.19 TB).

Table 21: Expected time to walk $2^{60.9}$ iterations of ECC2K-130 at each recorded rate, our arithmetic. "Years, one": device-years on a single device. "Days, 100" and "Days, 1,000": calendar days for a fleet of that size, excluding the detection delay D (about 1.6–1.7 days on the RTX PRO 6000 at cutoff 32). The median is 0.94× these figures and the 90% point 1.71×. "Cost": the expected work at the Modal list price implied by Table 9's iterations per dollar (2026-09-19). Spot prices are much lower (text). Table-walk rates are absent because no table-walk build has a measured cost per solve (Section 8.6).

device	build	billion it/s	years, one	days, 100	days, 1,000	cost
σ-walk (merges with the campaign corpus)						
RTX PRO 6000	campaign preset	14.64	4.66	17.0	1.70	$124k
RTX PRO 6000	nibble-table σ	15.88	4.29	15.7	1.57	$114k
L40S	survey	8.70	7.84	28.6	2.86	$134k
B200	survey	8.84	7.71	28.2	2.82	$423k
B300	survey	7.58	9.00	32.9	3.29	$560k
H200	survey	7.63	8.93	32.6	3.26	$356k
H100	survey	7.54	9.04	33.0	3.30	$313k
A100-80GB	survey	4.63	14.7	53.8	5.38	$323k
L4	survey	2.56	26.6	97.2	9.72	$186k
A10	survey	2.42	28.1	102.8	10.3	$272k
T4	no `clmad`	0.541	126	460	46.0	$652k
other walks and platforms						
RTX PRO 4500	x-only bridge map	12.50	5.46	19.9	1.99	—
RTX PRO 4500	σ-walk	5.11	13.3	48.8	4.88	—
VU47P (AWS F2)	128 engines	8.25	8.26	30.2	3.02	—
Xeon, one core	AVX-512 bitsliced	0.0126	5,410	—	—	—
bound and history						
RTX PRO 6000	ceiling (Section 15.1)	24–26	2.6–2.8	9.6–10.4	0.96–1.04	—
GTX 295 card	2010 kernel	0.064	1,065	3,891	389	—

A fleet of N devices at R therefore expects a collision after

$$T(N) \approx \frac{2^{60.9}}{N\,R} + D,$$

where D is the detection delay. The first term is multiplied by 0.94 for the median and by 1.71 for a 90% plan.

Existing corpus. The campaign has already walked $2^{56.68}$ iterations without a collision (Table 19). A new fleet that runs the same σ-walk at cutoff 32 and merges with that corpus inherits the credit. Given no collision so far, the expected remaining work is 0.949 of the total (our arithmetic). That is a saving of 5%; it changes the plan, not the scale.

16.2 Estimates by device and fleet size

Table 21 applies the first term of the model to every rate in this paper. The σ-walk rows are the survey and paired rates of Table 9. Points from these rows can merge with the campaign corpus. The other rows use other walks or other platforms.

On the best card, a solve is about four device-years. A fleet of five RTX PRO 6000s finishes in about a year. With 56 the expected time is a month, including the detection delay. With 290 it is a week. A thousand cards bring the expected time down to about 3.2 days, 1.6 of walking and 1.6 of waiting for the last trail to reach a distinguished point. Nine times in ten, a thousand cards finish within 4.3 days. Beyond a few hundred cards the delay dominates, which is where Figure 16's dashed line bends away from the solid one. A larger fleet buys little calendar time unless it also lowers the cutoff.

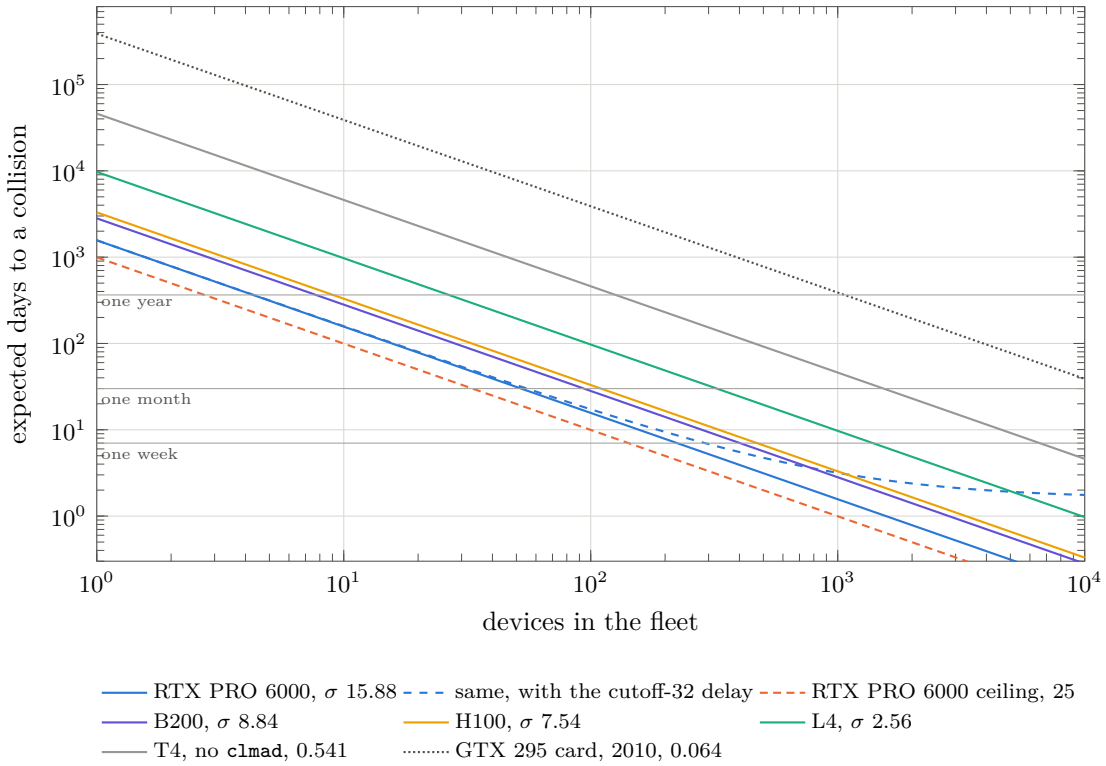


Figure 16: Expected calendar time to walk $2^{60.9}$ iterations of ECC2K-130 against fleet size, at measured one-device rates in billions of iterations per second (Table 21), with the 2010 GTX 295 for comparison. Solid lines count walking time only. The dashed line adds the RTX PRO 6000's detection delay at cutoff 32 (about 1.6 days), which sets a floor that more cards cannot lower. The median solve is 0.94× the plotted time and the 90% point 1.71×.

The new GPUs moved this by more than two orders of magnitude. With the 2010 kernel on a GTX 295, a one-year solve needed about 1,066 cards and a one-week solve more than 55,000. Today the same one-year and one-week targets need five cards and 290. If the RTX PRO 6000 reached its ceiling, the solve would be about 2.7 device-years. Even the campaign's own peak, about 83 billion iterations/s from a mixed spot fleet (Table 19), would finish the remaining expected work in about 285 days.

Cost. The cost of the work does not depend on the size of the fleet. At Modal list prices the cheapest per solve are the RTX PRO 6000 and the L40S, about $114k–$134k. The data-centre Blackwell and Hopper parts cost two and a half to five times as much for the same work, because this integer-bound walk does not use what they are priced for (Section 8.8). Spot capacity is cheaper again. At the 2026-09-11 spot price of $1.31 per RTX PRO 6000 hour the expected work is about $56k at the campaign's collection rate of 14.11. An AWS F2 at its spot price comes to about $51k (Section 12.4). The best quote seen, an L4 in `eu-north-1` on 2026-09-18, priced a full walk at about $38k [E8; E68]. Those spot figures hold only while spot capacity lasts. The spot market reclaimed the campaign's whole fleet on 2026-09-18 (Section 13.5), and a solve on spot capacity has to plan for that.

Assumptions. Every rate in Table 21 is sustained for the whole solve. The survey rows are single unpaired runs, and the campaign's own fleet measured its RTX PRO 6000 workers at a median of 14.66 (Table 18), so the survey rates are close to what a fleet sees but are not guaranteed. The bridge map on the RTX PRO 4500 walks a different function, so its points do not merge with the σ-walk corpus. Its frozen study measured 1.007× the work of the map

it replaced (Section 4.3). The FPGA row is one card of an `f2.6xlarge`. The CPU row is one core and assumes the per-core rate holds across a server. The costs exclude storage, egress and the merge.

16.3 NIST P-256 and P-384

The same model applies to the NIST walk of Section 10, and it shows why that section's rates say nothing about the security of those curves. With the negation map, the expected work for a group of prime order n is about $\sqrt{\pi n/4}$ iterations. For P-256 that is $2^{127.8} \approx 3.0 \cdot 10^{38}$ iterations, and for P-384 $2^{191.8} \approx 5.6 \cdot 10^{57}$ (our arithmetic from the standard group orders; any walk constant above one only adds to it).

- **P-256.** At the best measured 1.632 billion iterations/s on one RTX PRO 6000, the expected time is $1.9 \cdot 10^{29}$ seconds, about $5.9 \cdot 10^{21}$ card-years. A billion cards would need $5.9 \cdot 10^{12}$ years, more than 400 times the age of the universe. At the multiplies-only bound of 12.2 (Section 15.4) it is still $7.8 \cdot 10^{20}$ card-years.
- **P-384.** At 0.548 the expected time is $1.0 \cdot 10^{49}$ seconds, about $3.2 \cdot 10^{41}$ card-years.

The engineering of Section 10 multiplied the P-256 rate by 1.52 and the P-384 rate by 2.61, and divided these times by the same factors. P-256 needs about 2^{67} times the work of ECC2K-130, so a speedup of any engineering size leaves it where it was. The levers that bring ECC2K-130 within a fleet's reach shorten a P-256 attack by a factor that disappears against the exponent.

17 Open problems and future work

This section collects what we would try next, and what would settle each question. Some items carry an estimate from the note that proposed them; we quote those as estimates, not results. None of them moves the exponent. What they could move is the constant, and on Blackwell the constant is still being found.

17.1 A fast exact table walk

The biggest open question is whether a correct table walk can be made faster than the σ-walk again. The 20.08 billion iterations/s build ran cycle rule v2, whose exits from a fruitless cycle depended on how a trail had entered it. That was found in review and fixed by rule v3, which picks one exit for each detected cycle regardless of history (Section 4.5). The exact v3 kernel runs at 5.10 billion iterations/s today, against 15.88 for the best σ-walk [E1, "The earlier 20.078 B/s table result"; E13]. Almost all of that gap is the cost of probing for cycles inside the critical path, and several ideas attack it:

- *Deferred cycle detection (a rule v4).* Detect cycles after the fact rather than on every hinted step. The note estimates this recovers "most of the gap, 15–19" billion iterations/s, but it is a new rule identity, so the reference model, the corpus header and the replay and walk-constant tools all have to move with it [E82, §3.1].
- *A cheaper probe step* in the polynomial basis (estimated at 30–45% of the probe's critical path, implemented and passing the host probe, not yet re-measured on a device), and *early termination* of the probe through the formal $\mathbb{Z}[\tau]$ sum (estimated −35% probe work) [E82, §3.2–3.3].
- *Moving the probe off the critical path* onto a concurrent resolver, at an estimated 4.5% of ALU work with its latency hidden [E82, §3.4–3.5].
- *The paired cost-per-solve run.* Even a fast exact table walk only pays if its cost per solve beats the σ-walk's. The v2 projection was 0.81–0.86×, which held only while the table walk kept at least 0.932–0.943 of the σ-walk's rate, and only for the first 15–22% of a campaign.

The paired run that would test it has never been executed; its job script is all that exists [E6, §11.1 and §11.5].

17.2 Remaining σ-walk optimisations

The σ-walk is at 15.88 billion iterations/s on one RTX PRO 6000, and the repository's one-card goal of 26 is unmet. The roofline says where the slack is. The polynomial reduction is the largest single item, 29% of the integer work, and the path to 30 billion iterations/s needs integer work cut from about 1,320 to 968 lane-instructions per update with the carry-less unit fully busy, which the note calls a two-card number [E19, §2 and §5]. Open items:

- A shared-memory square table for the σ-walk compiles and fits on `sm_120`; its device occupancy and paired rate are pending [E83].
- Toom-3 over $\mathbb{F}_2$ is not implemented. A Karatsuba-family search found nothing below 8,859 bitsliced instructions, but the packed kernel's product has never been tried in Toom form [E1, "What is not done"].
- On the RTX PRO 4500, the bridge map's collection rate at cutoff 34 is unmeasured, and a pre-registered rare-phase skip has not been run [E11].

17.3 Blackwell pipeline behaviour

Three questions about the hardware itself would sharpen every estimate above.

- *Is the integer-multiply pipe separate from the logic pipe?* If it is, moving shifts onto `IMAD.SHL` is the next integer lever; if the two share a datapath, the kernel is already at 96%. A small probe settles it: a 1:1 `LOP3`/`IMAD` mix running at 1.5× or more says separate, under 1.2× says shared. It has not been run [E19, §4.1 and §7].
- *What does `CLMAD` sustain in the walk's own instruction mix?* The carry-less speed of light for the table walk lies between 22.7 and 27.4 billion iterations/s depending on whether paired `CLMAD`s beside a 40-`LOP3` chain reach 1.90 lanes per SM-clock or stop at 1.70. A screened three-limb Karatsuba arm implies at least 1.654; a verified run would pin it [E19, §3, §4.2 and §7].
- *Hardware counters.* Nsight Compute failed to profile on the rented RTX PRO 4500 and cannot read counters inside Modal containers, so no pipe-utilisation figure in this paper comes from a hardware counter. A host with working counters would replace our probe-based estimates with measurements [E70, "Nsight findings"].

17.4 Data-centre parts

Kernel tuning does not transfer between dies. The B200's three winning knobs (fused pass, batch 32, three-limb Karatsuba) are among the RTX PRO 6000's largest losses, so every new part needs its own sweep [E26, §4]. On the B200 the logic pipe binds hard, and 30 billion iterations/s would need at most 618 ALU slots per update against 955 after the sweep. The candidate offloads all move work onto the carry-less unit: the reduction as products against the modulus (−290 ALU, +19 `CLMAD`), the basis conversions and Frobenius networks as carry-less linear maps (about 300 slots), and the branch selection as polynomial evaluation (about 175). None has been built [E24, §6.2 and §6.5; E26, §3.4]. The B300 is an open case: NVIDIA cut its FP64 rate to 1/64 of FP32 [10], and if `CLMAD` runs on the FP64 pipe there as it does elsewhere, the B300 would price carry-less work like the workstation die rather than like the B200. Its σ-walk is already slower per SM than the B200's (51.2 against 59.7 million iterations per second), but the probe of Table 5 has not been run on it. The H100 has been probed but never run the

kernel. On the power-capped RTX PRO 4500, the fused build is predicted at 7.2–7.4 billion iterations/s on the shipping walk from an energy model calibrated on one pair, and a single run would confirm or break the model [E71, §4–5]. Multi-GPU instances are not implemented: one process drives one device.

17.5 NIST walk

The prime-field walk has had two rounds of tuning, so its open questions are nearer the surface:

- $W = 64$. Both curves now sit at the 255-register ceiling at $W = 32$, so $W = 64$ will probably spill and lose; one cell would confirm it.
- *Separating the round-2 gain* between the larger batch and the by-reference inversion, with a $W = 32$ cell built without the new inversion.
- *A Solinas-field walk.* The field benchmark favours Solinas for the P-384 multiplication; the walk has never been built on it.
- *Independent chains per thread, reseeding off the critical path, and a second walk per table read*, each named as a next rung in the results file but not yet priced [E49; E50, "Measurement rule"].
- *A larger jump table.* The walk uses 2^8 entries, so a fruitless two-cycle starts about once every 512 steps. The secp256k1 notes estimate that going to 2^{11} cuts the escape work eightfold; the same argument applies here, untested [E57, §4.1–4.3].
- *Repeated, interleaved pairs* for the best cell against its control, which would turn the single-run ratios of Table 12 into intervals.

17.6 CPUs

The CPU side has had the least attention and has the clearest untested levers.

- *A packed σ-walk on x86 with VPCLMULQDQ.* The packed client's carry-less path is measured only with PMULL on an M4 Pro and only for the table walk. AVX-512's `vpclmulqdq` does four 64×64 products per instruction, and AMD's Zen 5 and Intel's current server cores run AVX-512 at full width; nobody has measured either [E86].
- *Arm's newer vector extensions.* SVE2's `PMULLB`/`PMULLT` and its three-input `EOR3`/`BCAX` would serve both the packed and the bitsliced client on Neoverse V-class cores; none of the code uses SVE today.
- *Prime fields with real carry chains.* A four-limb Montgomery multiply with `MULX`/`ADX`, or 52-bit limbs on AVX-512 IFMA, would replace the big-integer reference solver; this is the CPU version of the GPU's PTX carry-chain step and has not been tried for rho [E52].
- *GFNI after a layout change.* GFNI wins on small-field basis changes but was priced as a loss for ECC2K-130's bitsliced conversions because it acts on bytes within a register; a byte-oriented layout could change that verdict [E87].

17.7 Methodology

Two lessons from this work are worth keeping as rules. Static instruction counts are not throughput: two candidates predicted from their static counts at 0.887 and 0.976 measured 0.986 and 0.905 on the device [E27; E84]. And a rate is not a cost per solve: every engineering gain in this paper is a same-walk scheduling result until a paired run on the walk constant says otherwise. Tensor cores, finally, remain a negative result for this walk: they need one operand shared across eight or sixteen lanes, and no product in the walk has one [E24, §1].

18 Related work

The 2009 attempt. Bailey et al. [12] fixed the iteration function this campaign walks, $R \mapsto \sigma^{j}(R) + R$ with j from the weight of x, the weight-34 distinguished-point rule, and the $2^{60.9}$ budget, and reported per-platform rates on Core 2 (533 cycles per iteration with 128-bit vectors), Cell, GPUs, FPGAs [25] and ASIC estimates; their SHARCS paper [11] priced the whole Certicom ECC2 list. Bos, Kleinjung, Niederhagen and Schwabe's Cell implementation [19] is the bitsliced lineage this client's CPU and first GPU backends belong to. The GPU line ends with Bernstein et al.'s GTX 295 kernel [15], which reached 1,164 cycles per iteration per multiprocessor and about 64 million iterations per second per card; Niederhagen's thesis [41] collects the final Cell and GPU figures. We found no later published ECC2K-130 rate for any GPU generation between Fermi and Blackwell, which is part of why Section 2 compares directly against 2010. The type-II optimal-polynomial-basis multiplication of Bernstein and Lange [14] is used verbatim; the conversion networks generated here match their hand-designed chains to within a few instructions (Section 5.1). Bernstein's bit operation counts for binary multiplication [13] are the reference the three-input-logic finding is measured against. Bernstein et al.'s later FPGA designs [17], which set the current size record for a completed binary-curve computation by solving a $2^{117.35}$-order instance on sect113r2 and were extrapolated to a 127-bit field, are the calibration for the pre-board estimate of Section 12, which the measurement then met. Between those two, Wenger and Wolfger solved a 113-bit Koblitz-curve logarithm on a Virtex-6 cluster at one iteration per clock [54] and then a 113-bit binary curve on Kintex-7 parts [55]. Harley's group solved ECC2K-95 in 1998 and ECC2K-108 in April 2000, the latter with about 1,300 volunteers' machines over four months [32, 33]; those two are the only Koblitz challenges completed, and their walks are the ancestors of the one in Section 4.1.

Rho on automorphism classes. The $\sqrt{2m}$ saving from Frobenius and negation on Koblitz curves is due to Wiener and Zuccherato [56] and Gallant, Lambert and Vanstone [28], generalised by Duursma, Gaudry and Morain [24]. Teske's r-adding walks [51] and the Bernstein–Lange–Schwabe treatment of fruitless cycles under the negation map [16] are the background to Section 4.2; the cycle classes measured there are the τ-relation analogue, and the measured walk constants of Table 3 are, to our knowledge, the first reported for this branch distribution on this family. The distinguished-point method is van Oorschot and Wiener's [52]. The 112-bit prime-field record of Bos et al. [20] is the nearest completed computation in scale.

Index calculus on binary curves. Semaev's summation polynomials [50], Gaudry's and Diem's index calculus [22, 29], the Faugère–Perret–Petit–Renault and Petit–Quisquater analyses of Weil-descent systems [26, 46] and Galbraith and Gebregiyorgis's characteristic-two algorithms [27] are the methods Section 14 prices on this curve. The structural closure at $n = 131$ (no Frobenius-stable subspace of intermediate dimension) is elementary once the normal basis is in view, and the measured ladder confirms that the surviving weight-bounded bases do not reach the budget.

Measurement discipline. The companion paper [7] states the boundary-unit-table-class rule and applies it across rho, baby-step giant-step, kangaroo and prime-field index calculus; this paper is, in effect, its ECC2K-130 chapter at full length.

Prime-field rho on GPUs. Bos et al.'s 112-bit record on PlayStation 3 consoles [20] is the nearest completed prime-field computation in scale. Our NIST walk (Section 10) reuses the standard ingredients, an r-adding walk [51], the negation map with fruitless-cycle handling [16] and Montgomery's simultaneous inversion [40]; what it adds is a measured account of which GPU optimisations from the binary campaign carry over to it.

19 Conclusion

We summarise the state of Pollard rho on current NVIDIA hardware as measured in this work.

Per-card throughput. In 2010 the best GTX 295 kernel ran ECC2K-130 at 0.064 billion iterations per second. On one RTX PRO 6000 Blackwell the same walk runs at 14.64 in the audited campaign build and 15.88 in the best later build, and the whole $2^{60.9}$ budget is about five GPU-years. Roughly a sixth of that factor is a bigger, faster chip; the rest is arithmetic. Sixteen paired steps took the client from a 0.85 bitsliced control to 14.64, and three representation changes did most of the work: Frobenius permutation networks, the native carry-less product, and the smaller batch that the native product made possible.

Time to solution. At 15.88 billion iterations per second, five RTX PRO 6000s expect to find the collision in about a year, 56 in a month, 290 in a week, and a thousand in about three days, half of them spent waiting for the last trail to reach a distinguished point at cutoff 32. The work costs about $114k at list price and $38k–$56k on the spot quotes we saw, whatever the fleet size. The same rates leave P-256 about $6 \cdot 10^{21}$ card-years away (Section 16).

Carry-less multiplication across dies. PTX 9.3 gives every part from `sm_80` onward a carry-less multiply, but not at the same rate. On the RTX PRO 6000 it issues at about 1.7 lanes per SM-clock, the cost of 38 logic instructions; on the B200 and H100 it is ten to eighteen times faster. The workstation part still wins on this walk, per card, per SM and per dollar, because the walk is bound by integer logic and the RTX PRO 6000 has the most SMs at the highest clock for the money. On the data-centre parts the logic pipe binds instead, the right kernel moves work onto the carry-less unit, and kernel tuning does not travel between them. For any walk with one affine addition per step, 28 billion iterations per second on one RTX PRO 6000 is out of reach on both pipes.

Choice of iteration function. The table walk does the arithmetic of a step more cheaply, and reached 20.08 under cycle rule v2. But v2's exits from fruitless cycles depended on how a trail arrived, and the exact v3 rule that fixes this runs at 5.10 today. Closing that gap, with a deferred or off-critical-path cycle check, is the largest open optimisation in this paper, and even then a table walk must win on cost per solve, which nobody has measured.

Transfer to prime fields. On NIST P-256 and P-384 on the same card, explicit carry chains, a wider shared inversion and the same refusal of register-cap trades gave 1.52× and 2.61×, to 1.632 and 0.548 billion iterations per second. The batch width, a modest knob on the binary walk, was the biggest lever on the prime one, because a Fermat inversion costs some forty times as many multiplications as a normal-basis one. A P-256 step remains about nine times as expensive as an ECC2K-130 step on the same card, which is the carry-less unit's advantage in one number.

Remaining headroom. None of the speedup came from changing the algorithm: every step kept Bailey et al.'s walk and its distinguished points bit for bit, and the gains split into representation and arithmetic (6.3×), batch and geometry (1.5×), memory layout (1.3×) and the new instruction (1.2×). The same pipe measurements bound what is left. Any walk with one affine addition per step tops out at about 24–26 billion iterations per second on one RTX PRO 6000, so the best σ-walk has roughly 1.5× to go, all of it in integer work; a B200 would need a 1.55× cut in logic instructions to reach 30; the F2 FPGA stops near 9.3. On CPUs the instruction sets matter as much as on GPUs: three-input logic and wider words give 2.4–3× to the bitsliced

client, and a packed client on the CPU's own carry-less multiply runs 2.9× faster than with a software product.

Asymptotic cost and the campaign. The campaign's walk needs $2^{60.91}$–$2^{60.92}$ iterations, which agrees with the 2009 estimate. Index calculus on the curve costs 10^7 times as much as rho, and the best measured decomposition is 35–51 bits short of its budget. The campaign stored 486 million points and $2^{56.68}$ iterations, 5.4% of the expected work, with no collision, every record re-walkable from its seed; its fleet was reclaimed on 2026-09-18 and collection is paused. Whether the remaining 94.6% is walked is a question of money, not mathematics, and at five GPU-years it is a much smaller question than it was in 2009.

A Reproduction

The companion repository [8] will carry the ECC2K-130 client (CUDA and CPU), the NIST P-256/P-384 rho kernel, the paper source, the benchmark data behind every figure, the evidence records of Appendix B, and a Docker image.

Status. The repository is assembled but not yet public. The CPU client, its validation suite, a planted-logarithm solve and every mode of the image's entry point have been run from its copy of the source; the Docker image itself has not yet been built, and its GPU modes have not been run. The commands below describe the intended workflow and will be checked before the repository is published.

Docker. The image compiles the client with CUDA 13.3 for Ampere, Ada, Hopper and both Blackwell families, with the audited RTX PRO 6000 preset (the σ-walk, batch 16, 256-thread blocks, native `clmad`, compact tiled state and shared Frobenius masks), and runs the client's validation suite during the build. By default a container walks ECC2K-130 with the campaign's parameters (cutoff 32, a 2^{32} iteration guard, checkpoints every ten minutes) on the GPU if one is visible and on the CPU otherwise, writing 32-byte distinguished-point records and a resumable checkpoint to its data volume:

```
docker build -t rho-blackwell -f docker/Dockerfile .
docker run --gpus all -v rho-data:/data rho-blackwell   # walk ECC2K-130
docker run --gpus all rho-blackwell bench                # throughput only
docker run rho-blackwell test                              # validation suite
docker run rho-blackwell planted                           # small planted logs
docker run --gpus all rho-blackwell ecc2k95                # ECC2K-95, answer known
docker run --gpus all rho-blackwell nist                   # P-256/P-384 rho bench
```

A container without `RUN_ID` picks a random run identifier from a range the campaign never used (20000–65535), keeps it with its data and warns that it did so. Two containers with the same identifier walk the same trails, and with random choices that happens with probability about 0.1% for 10 containers, 10% for 100 and 63% for 300, so any run of more than one container must assign each a distinct `RUN_ID`. A collision is found by merging the corpora of all containers that walked with the same cutoff; a single container is unlikely to find one alone.

Building without Docker. The client needs a C++17 compiler and, for the GPU build, CUDA 13.3 or newer (the native carry-less product requires PTX 9.3). Its make targets build the CPU client, the GPU client with any preset of this paper, and the test suites; the repository's README lists the targets for each measured build, including the fused and nibble-table σ-walks and the B200 and RTX PRO 4500 presets.

Checking a number. Every measured number cites an evidence record [En]. Each record in the companion repository will hold the notes, data or source that report the measurement, so a reader can check a number without the development history.

B Evidence records

Every measured number in this paper cites one of the records below as [En], optionally with a section of that record. Each record is a measurement report, dataset, protocol or source file kept in a public repository named after its title. The companion repository being prepared for this paper (Appendix A) will hold a copy of every record under its number.

E1 ECC2K-130 client reference: field, walk, validation and throughput. *design and measurement notes*; crypto.

E2 Independent C golden model of $\mathbb{F}_{2^{131}}$ and the curve. *source and tests*; cryptanalysis.

E3 Field model and code generator for $\mathbb{F}_{2^{131}}$. *source*; crypto.

E4 FPGA engine: VHDL design, reference model and test vectors. *design and source*; crypto.

E5 ECC2K-95 GPU kernel and Koblitz pair table. *design notes and source*; crypto.

E6 Walk constants and fruitless-cycle analysis on toy Koblitz curves. *measurement report*; crypto.

E7 Witness counters and their cost. *measurement report*; crypto.

E8 Campaign operations, configuration and planning numbers. *operations notes*; crypto.

E9 Fused single-pass σ-walk. *paired benchmark*; crypto.

E10 Iteration functions compared: σ-walk, table walk and their cost model. *measurement report*; crypto.

E11 x-only doubling-and-bridge map on the RTX PRO 4500. *paired benchmark*; crypto.

E12 Iteration-guard study. *analysis*; crypto.

E13 Table-walk cycle escape, rule v3. *design and measurement notes*; crypto.

E14 Packed $\mathbb{F}_{2^{131}}$ arithmetic for CUDA. *source*; crypto.

E15 Instruction-issue rates and the throughput ceiling. *measurement report*; crypto.

E16 Ledger: packed normal-basis backend and its audit. *paired benchmark*; crypto.

E17 Ledger: Frobenius permutation networks. *paired benchmark*; crypto.

E18 Ledger: native carry-less multiplication (`clmad`). *paired benchmark*; crypto.

E19 Roofline of the table-walk build on the RTX PRO 6000. *measurement report*; crypto.

E20 Host reference implementation of the step. *source*; crypto.

E21 ECC2K-95 kernel: `clmad` port. *measurement report*; crypto.

E22 Ada L4 and L40S with and without `clmad`. *measurement report*; crypto.

E23 Turing T4 without `clmad`. *measurement report*; crypto.

E24 Two-chain kernel and `CLMAD` rates on three dies. *measurement report*; crypto.

E25 Declined: top-bit correction on `CLMAD`. *paired benchmark*; crypto.

E26 Automatic single-knob sweep on the B200 and RTX PRO 6000. *measurement report*; crypto.

E27 Native-instruction candidates and the acceptance rule. *protocol*; crypto.

E28 σ step through shared-memory nibble tables. *paired benchmark*; crypto.

E29 Ledger: single polynomial product. *paired benchmark*; crypto.

E30 Ledger: denominator cache. *paired benchmark*; crypto.

E31 Ledger: polynomial-basis batch chain. *paired benchmark*; crypto.

E32 Ledger: paired products. *paired benchmark*; crypto.

E33 Ledger: polynomial-basis state. *paired benchmark*; crypto.

E34 Ledger: direct reduction. *paired benchmark*; crypto.

E35 Ledger: generated straight-line product. *paired benchmark*; crypto.

E36 Ledger: 256-worker tiles. *paired benchmark*; crypto.

E37 Ledger: batch 16 with 385,024 workers. *paired benchmark*; crypto.

E38 Ledger: weighted prefixes and paired Frobenius. *paired benchmark*; crypto.

E39 Ledger: compact 17-byte state and batch screens. *paired benchmark*; crypto.

E40 Ledger: shared-memory mask table. *paired benchmark*; crypto.

E41 Table walk against σ-walk: paired comparison data. *dataset*; crypto.

E42 One-block table-walk geometry (the 20.08 build). *paired benchmark*; crypto.

E43 Eleven-GPU survey on Modal. *measurement report*; crypto.

E44 Declined: stream-Karatsuba compile option. *benchmark*; crypto.

E45 Declined: integer add in the carry-less product. *paired benchmark*; crypto.

E46 Declined: λ-projective coordinates, priced. *analysis*; crypto.

E47 AVX-512 host scheduling pass. *measurement report*; crypto.

E48 Distinguished-point interval calibration at cutoff 32. *measurement report and raw counters*; crypto.

E49 NIST P-256/P-384 rho on the RTX PRO 6000: device results, rounds 1 and 2 (round 2: the $W = 32$ and register-cap cells, merged in pull request 1778). *measurement report*; crypto.

E50 NIST P-256/P-384 GPU library: field, walk and tests. *design notes and source*; crypto.

E51 Pre-registered NIST GPU arithmetic protocol. *protocol*; crypto.

E52 Fifteen-curve NIST ECDLP solver and audit. *design and measurement notes*; crypto.

E53 secp256k1 GPU library: field, walk, folds and baby-step giant-step. *design notes and source*; crypto.

E54 GPU kangaroo interval solver. *design and measurement notes*; crypto.

E55 First NIST GPU attempt (infrastructure non-result). *record*; crypto.

E56 64-bit prime-field GPU rho. *design and measurement notes*; cryptanalysis.

E57 secp256k1 GPU arithmetic: Blackwell optimisation notes. *analysis*; crypto.

E58 Fast exhaustive search for quadratic Boolean systems. *design notes*; crypto.

E59 Semaev S_4 pairs-and-solve sweep. *design and measurement notes*; crypto.

E60 Batched Macaulay reduced-echelon solver. *design notes*; crypto.

E61 Macaulay reduction pipeline on an RTX PRO 4500. *measurement report*; crypto.

E62 Sparse matrix–vector worker. *design notes*; crypto.

E63 F4 elimination on rented RTX 5090 and H100 GPUs. *measurement reports*; cryptanalysis.

E64 GPU efficiency scorer and the packed-walk profile. *tool and profile*; cryptanalysis.

E65 FPGA pre-board estimate. *analysis*; crypto.

E66 FPGA engine on AWS F2: builds and measurements. *measurement report*; crypto.

E67 Campaign audit and live snapshot, 2026-09-25. *audit report and dataset*; crypto.

E68 Fleet price quotes, 2026-09-18. *operations record*; crypto.

E69 RTX PRO 4500: rates and per-dollar break-even. *measurement report*; crypto.

E70 The 28-billion goal: grid and occupancy screens. *measurement report*; crypto.

E71 Power-bound model and updates per joule. *measurement report*; crypto.

E72 Storage, seeds and certification protocol. *protocol*; crypto.

E73 Public campaign dashboard. *documentation*; crypto.

E74 Cloud account hold, 2026-09-18. *operations record*; crypto.

E75 Published campaign feed, 2026-10-10. *dataset*; crypto.

E76 Index calculus on Koblitz curves: structural analysis. *analysis*; crypto.

E77 Evidence record: Frobenius-stable factor bases at $n = 131$. *evidence record*; crypto-autoresearcher.

E78 Index-calculus feasibility for ECC2K-130. *analysis*; crypto.

E79 Index-calculus boundary ledger. *measurement ledger*; crypto.

E80 Point decomposition against the rho budget. *measurement report*; cryptanalysis.

E81 Evidence record: Riemann–Roch oracle product law. *evidence record*; crypto-autoresearcher.

E82 Cycle-probe latency and the routes to a faster exact rule. *analysis*; crypto.

E83 σ-walk shared-memory square table: compile and resource gate. *gate record*; crypto.

E84 Native-instruction candidates on the RTX PRO 4500. *paired benchmark*; crypto.

E85 Sparse-basis and compound σ maps: modelled ceilings. *analysis*; crypto.

E86 Packed CPU client on PMULL/PCLMULQDQ and the Metal client. *design and measurement notes*; cryptanalysis.

E87 GFNI for basis changes: measured on small fields, priced for ECC2K-130. *measurement report and analysis*; crypto.

E88 Carry-less reduction in small binary fields (PCLMULQDQ). *measurement report*; crypto.

References

[1] NVIDIA Ampere GA102 GPU architecture. Whitepaper V2.0, NVIDIA Corporation, 2021.

[2] NVIDIA H100 tensor core GPU architecture. Whitepaper V1.02, NVIDIA Corporation, 2022.

[3] NVIDIA Ada GPU architecture. Whitepaper V2.02, NVIDIA Corporation, 2023.

[4] 5th gen AMD EPYC processor architecture. White paper, Advanced Micro Devices, 2025.

[5] NVIDIA RTX Blackwell GPU architecture. Whitepaper V1.1, NVIDIA Corporation, 2025.

[6] *VMware vSphere Tuning Guide for AMD EPYC 9004 Processors*. Advanced Micro Devices, 2023.

[7] Anonymous. The constant in front of $\sqrt{n}$: measured Pollard rho, ECC2K-130 arithmetic, and index calculus over prime fields. Companion draft, https://github.com/aburan28/cryptanalysis/tree/main/paper, 2026.

[8] Anonymous. rho-blackwell: code, data and evidence records for "Pollard rho on NVIDIA Blackwell". https://github.com/aburan28/rho-blackwell, 2026. In preparation; not yet public.

[9] *Arm A64 Instruction Set for A-profile Architecture*. Arm Limited, 2026. ISA XML release 2026-09.

[10] Kyle Aubrey and Nick Stam. Inside NVIDIA Blackwell Ultra: The chip powering the AI factory era. NVIDIA Technical Blog, 2025.

[11] Daniel V. Bailey, Brian Baldwin, Lejla Batina, Daniel J. Bernstein, Peter Birkner, Joppe W. Bos, Gauthier van Damme, Giacomo de Meulenaer, Junfeng Fan, Tim Güneysu, Frank Gürkaynak, Thorsten Kleinjung, Tanja Lange, Nele Mentens, Christof Paar, Francesco Regazzoni, Peter Schwabe, and Leif Uhsadel. The Certicom challenges ECC2-X. In *Workshop on Special-purpose Hardware for Attacking Cryptographic Systems – SHARCS 2009*, pages 51–82, 2009.

[12] Daniel V. Bailey, Lejla Batina, Daniel J. Bernstein, Peter Birkner, Joppe W. Bos, Hsieh-Chung Chen, Chen-Mou Cheng, Gauthier van Damme, Giacomo de Meulenaer, Junfeng Fan, Tim Güneysu, Frank Gürkaynak, Thorsten Kleinjung, Tanja Lange, Nele Mentens, Ruben Niederhagen, Christof Paar, Francesco Regazzoni, Peter Schwabe, Leif Uhsadel, Anthony Van Herrewege, and Bo-Yin Yang. Breaking ECC2K-130. IACR Cryptology ePrint Archive, Report 2009/541, 2009.

[13] Daniel J. Bernstein. Batch binary Edwards. In *CRYPTO 2009*, volume 5677 of *LNCS*, pages 317–336. Springer, 2009.

[14] Daniel J. Bernstein and Tanja Lange. Type-II optimal polynomial bases. In *Arithmetic of Finite Fields – WAIFI 2010*, volume 6087 of *LNCS*, pages 41–61. Springer, 2010.

[15] Daniel J. Bernstein, Hsieh-Chung Chen, Chen-Mou Cheng, Tanja Lange, Ruben Niederhagen, Peter Schwabe, and Bo-Yin Yang. ECC2K-130 on NVIDIA GPUs. In *Progress in Cryptology – INDOCRYPT 2010*, volume 6498 of *LNCS*, pages 328–346. Springer, 2010. doi: 10.1007/978-3-642-17401-8_23. Updated version: IACR Cryptology ePrint Archive, Report 2012/002.

[16] Daniel J. Bernstein, Tanja Lange, and Peter Schwabe. On the correct use of the negation map in the Pollard rho method. In *Public Key Cryptography – PKC 2011*, volume 6571 of *LNCS*, pages 128–146. Springer, 2011.

[17] Daniel J. Bernstein, Susanne Engels, Tanja Lange, Ruben Niederhagen, Christof Paar, Peter Schwabe, and Ralf Zimmermann. Faster elliptic-curve discrete logarithms on FPGAs. IACR Cryptology ePrint Archive, Report 2016/382, 2016.

[18] Eli Biham. A fast new DES implementation in software. In *Fast Software Encryption – FSE 1997*, pages 260–272. Springer, 1997.

[19] Joppe W. Bos, Thorsten Kleinjung, Ruben Niederhagen, and Peter Schwabe. ECC2K-130 on Cell CPUs. In *Progress in Cryptology – AFRICACRYPT 2010*, volume 6055 of *LNCS*, pages 225–242. Springer, 2010.

[20] Joppe W. Bos, Marcelo E. Kaihara, Thorsten Kleinjung, Arjen K. Lenstra, and Peter L. Montgomery. Solving a 112-bit prime elliptic curve discrete logarithm problem on game consoles using sloppy reduction. *International Journal of Applied Cryptography*, 2(3):212–228, 2012.

[21] Certicom Research. The Certicom ECC challenge. https://www.certicom.com/content/certicom/en/the-certicom-ecc-challenge.html, 1997.

[22] Claus Diem. On the discrete logarithm problem in elliptic curves. *Compositio Mathematica*, 147(1):75–104, 2011.

[23] Nir Drucker, Shay Gueron, and Vlad Krasnov. Fast multiplication of binary polynomials with the forthcoming vectorized VPCLMULQDQ instruction. In *2018 IEEE 25th Symposium on Computer Arithmetic (ARITH)*, pages 115–119, 2018. doi: 10.1109/ARITH.2018.8464777.

[24] Iwan Duursma, Pierrick Gaudry, and François Morain. Speeding up the discrete log computation on curves with automorphisms. In *ASIACRYPT 1999*, pages 103–121. Springer, 1999.

[25] Junfeng Fan, Daniel V. Bailey, Lejla Batina, Tim Güneysu, Christof Paar, and Ingrid Verbauwhede. Breaking elliptic curve cryptosystems using reconfigurable hardware. In *International Conference on Field Programmable Logic and Applications – FPL 2010*, pages 133–138. IEEE, 2010.

[26] Jean-Charles Faugère, Ludovic Perret, Christophe Petit, and Guénaël Renault. Improving the complexity of index calculus algorithms in elliptic curves over binary fields. In *EUROCRYPT 2012*, pages 27–44. Springer, 2012.

[27] Steven D. Galbraith and Shishay W. Gebregiyorgis. Summation polynomial algorithms for elliptic curves in characteristic two. In *INDOCRYPT 2014*, pages 409–427. Springer, 2014.

[28] Robert P. Gallant, Robert J. Lambert, and Scott A. Vanstone. Improving the parallelized Pollard lambda search on anomalous binary curves. *Mathematics of Computation*, 69(232): 1699–1705, 2000.

[29] Pierrick Gaudry. Index calculus for abelian varieties of small dimension and the elliptic curve discrete logarithm problem. *Journal of Symbolic Computation*, 44(12):1690–1702, 2009.

[30] Shay Gueron and Michael E. Kounavis. Intel carry-less multiplication instruction and its usage for computing the GCM mode. White Paper 323640-002, Intel Corporation, 2014.

[31] Shay Gueron and Vlad Krasnov. Accelerating big integer arithmetic using Intel IFMA extensions. In *2016 IEEE 23rd Symposium on Computer Arithmetic (ARITH)*, pages 32–38, 2016. doi: 10.1109/ARITH.2016.22.

[32] Robert Harley. Elliptic curve discrete logarithms: ECC2K-95. Announcement of the solution of the Certicom ECC2K-95 challenge, 1998.

[33] Robert Harley, Damien Doligez, Daniel de Rauglaudre, and Xavier Leroy. Elliptic curve discrete logarithms: ECC2K-108. INRIA announcement of the solution of the Certicom ECC2K-108 challenge, 13 April 2000, 2000.

[34] *Intel Advanced Performance Extensions (APX) Architecture Specification.* Intel Corporation, 2026.

[35] *Intel Advanced Vector Extensions 10.2 Architecture Specification.* Intel Corporation, 2026.

[36] *Intel 64 and IA-32 Architectures Software Developer's Manual, Volume 1.* Intel Corporation, 2026.

[37] Toshiya Itoh and Shigeo Tsujii. A fast algorithm for computing multiplicative inverses in GF(2^m) using normal bases. *Information and Computation*, 78(3):171–177, 1988.

[38] Anatolii Karatsuba and Yuri Ofman. Multiplication of many-digital numbers by automatic computers. *Doklady Akademii Nauk SSSR*, 145:293–294, 1962.

[39] Neal Koblitz. CM-curves with good cryptographic properties. In *CRYPTO 1991*, pages 279–287. Springer, 1992.

[40] Peter L. Montgomery. Speeding the Pollard and elliptic curve methods of factorization. *Mathematics of Computation*, 48(177):243–264, 1987.

[41] Ruben Niederhagen. *Parallel Cryptanalysis.* PhD thesis, Eindhoven University of Technology, 2012.

[42] *CUDA Binary Utilities: Instruction Set Reference*. NVIDIA Corporation, 2026. URL https://docs.nvidia.com/cuda/cuda-binary-utilities/index.html.

[43] *CUDA C++ Best Practices Guide, Release 13.4*. NVIDIA Corporation, 2026. URL https://docs.nvidia.com/cuda/cuda-c-best-practices-guide/index.html.

[44] *Parallel Thread Execution ISA, Release 9.3*. NVIDIA Corporation, 2026. URL https://docs.nvidia.com/cuda/archive/13.3.0/pdf/ptx_isa_9.3.pdf. Shipped with CUDA 13.3; `clmad` in §9.7.1.5.

[45] Erdinc Ozturk, James Guilford, Vinodh Gopal, and Wajdi Feghali. New instructions supporting large integer arithmetic on Intel architecture processors. White Paper 327831-001, Intel Corporation, 2012.

[46] Christophe Petit and Jean-Jacques Quisquater. On polynomial systems arising from a Weil descent. In *ASIACRYPT 2012*, pages 451–466. Springer, 2012.

[47] J. M. Pollard. Monte Carlo methods for index computation (mod p). *Mathematics of Computation*, 32(143):918–924, 1978.

[48] *RISC-V Bit-Manipulation ISA-extensions, Version 1.0.0*. RISC-V International, 2021.

[49] *RISC-V Cryptography Extensions Volume II: Vector Instructions, Version 1.0.0*. RISC-V International, 2023.

[50] Igor Semaev. Summation polynomials and the discrete logarithm problem on elliptic curves. IACR Cryptology ePrint Archive, Report 2004/031, 2004.

[51] Edlyn Teske. On random walks for Pollard's rho method. *Mathematics of Computation*, 70(234):809–825, 2001.

[52] Paul C. van Oorschot and Michael J. Wiener. Parallel collision search with cryptanalytic applications. *Journal of Cryptology*, 12(1):1–28, 1999.

[53] Jean-Luc Watson, Charles Gouert, Michael B. Sullivan, and G. Edward Suh. Building faster cryptography with carryless multiplication in NVIDIA CUDA 13.3. NVIDIA Technical Blog, 2026. URL https://developer.nvidia.com/blog/building-faster-cryptography-with-carryless-multiplication-in-nvidia-cuda-13-3/.

[54] Erich Wenger and Paul Wolfger. Solving the discrete logarithm of a 113-bit Koblitz curve with an FPGA cluster. In *Selected Areas in Cryptography – SAC 2014*, volume 8781 of *LNCS*, pages 363–379. Springer, 2014. doi: 10.1007/978-3-319-13051-4_22.

[55] Erich Wenger and Paul Wolfger. Harder, better, faster, stronger: elliptic curve discrete logarithm computations on FPGAs. *Journal of Cryptographic Engineering*, 6(4):287–297, 2016. doi: 10.1007/s13389-015-0108-z.

[56] Michael J. Wiener and Robert J. Zuccherato. Faster attacks on elliptic curve cryptosystems. In *Selected Areas in Cryptography – SAC 1998*, pages 190–200. Springer, 1999.